

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

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FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
AC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level access control policy that a. Addresses purpose	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5		AC-01
AC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level access control policy that a. Addresses purpose	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRCP), including policies,	5		AC-01
AC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level access control policy that a. Addresses purpose	Functional	Subset Of	Identity & Access Management (IAM)	IAC-01	Mechanisms exist to facilitate the implementation of identification and access management controls.	10		AC-01
AC-02	Account Management	a. Define and document the types of accounts allowed and specifically prohibited for use within the system;b. Assign account managers;c. Require [Assignment: organization-defined prerequisites and criteria for group and role membership]	Functional	Intersects With	Termination of Employment	IAC-07.2	Mechanisms exist to revoke user access rights in a timely manner, upon termination of employment or contract	5		AC-02
AC-02	Account Management	a. Define and document the types of accounts allowed and specifically prohibited for use within the system;b. Assign account managers;c. Require [Assignment: organization-defined prerequisites and criteria for group and role membership]	Functional	Intersects With	Account Management	IAC-15	Mechanisms exist to: (1) Define authorized system account types; (2) Define prohibited system account	5		AC-02
AC-02	Account Management	a. Define and document the types of accounts allowed and specifically prohibited for use within the system;b. Assign account managers;c. Require [Assignment: organization-defined prerequisites and criteria for group and role membership]	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5		AC-02
AC-02	Account Management	a. Define and document the types of accounts allowed and specifically prohibited for use within the system;b. Assign account managers;c. Require [Assignment: organization-defined prerequisites and criteria for group and role membership]	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5		AC-02
AC-02(01)	Account Management Automated System Account Management	Support the management of system accounts using [Assignment: organization-defined automated mechanisms].	Functional	Intersects With	Automated System Account Management (Directory Services)	IAC-15.1	Automated mechanisms exist to support the management of system accounts (e.g., directory services).	5		
AC-02(02)	Account Management Automated Temporary and Emergency Account	Automatically [Selection (one): remove; disable] temporary and emergency accounts after [Assignment: organization-defined time period for each type of account].	Functional	Equal	Removal of Temporary / Emergency Accounts	IAC-15.2	Automated mechanisms exist to disable or remove temporary and emergency accounts after an organization-defined time period for	10		
AC-02(03)	Account Management Disable Accounts	Disable accounts within [Assignment: organization-defined time period] when the accounts:a. Have expired;b. Are no longer associated with a user or individual;c. Are in violation of organizational policy and Have been inactive for [Assignment:	Functional	Equal	Disable Inactive Accounts	IAC-15.3	Automated mechanisms exist to disable inactive accounts after an organization-defined time period.	10		
AC-02(04)	Account Management Automated Audit Actions	Automatically audit account creation, modification, enabling, disabling, and removal actions.	Functional	Equal	Automated Audit Actions	IAC-15.4	Automated mechanisms exist to audit account creation, modification, enabling, disabling and removal actions and notify organization	10		
AC-02(05)	Account Management Inactivity Logout	Require that users log out when [Assignment: organization-defined time period of expected inactivity or description of when to log out].	Functional	Equal	Session Lock	IAC-24	Mechanisms exist to initiate a session lock after an organization-defined time period of inactivity, or upon receiving a request from a user and retain the	10		
AC-02(06)	Account Management Dynamic Privilege Management	Implement [Assignment: organization-defined dynamic privilege management capabilities].	Functional	No Relationship	N/A	N/A	N/A	0		
AC-02(07)	Account Management Privileged User Accounts	a. Establish and administer privileged user accounts in accordance with [Selection (one): a role-based access scheme; an attribute-based access scheme];b. Monitor privileged role or attribute assignments;c. Monitor changes to roles or attributes	Functional	Equal	Role-Based Access Control (RBAC)	IAC-08	Mechanisms exist to enforce Role-Based Access Control (RBAC) for Technology Assets, Applications, Services and Data Assets	10		
AC-02(08)	Account Management Dynamic Account Management	Create, activate, manage, and deactivate [Assignment: organization-defined system accounts] dynamically.	Functional	No Relationship	N/A	N/A	N/A	0		
AC-02(09)	Account Management Restrictions on Use of Shared and Group Accounts	Only permit the use of shared and group accounts that meet [Assignment: organization-defined conditions for establishing shared and group accounts].	Functional	Equal	Restrictions on Shared Groups / Accounts	IAC-15.5	Mechanisms exist to authorize the use of shared/group accounts only under certain organization-defined conditions	10		
AC-02(10)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-02(11)	Account Management Usage Conditions	Enforce [Assignment: organization-defined circumstances and/or usage conditions] for [Assignment: organization-defined system accounts].	Functional	Equal	Usage Conditions	IAC-15.8	Automated mechanisms exist to enforce usage conditions for users and/or roles.	10		
AC-02(12)	Account Management Account Monitoring for Atypical Usage	a. Monitor system accounts for [Assignment: organization-defined atypical usage]; andb. Report atypical usage of system accounts to [Assignment: organization-defined personnel or roles]	Functional	Equal	Anomalous Behavior	MON-16	Mechanisms exist to utilize User & Entity Behavior Analytics (UEBA) and/or User Activity Monitoring (UAM) relative to detected anomalies	10		
AC-02(13)	Account Management Disable Accounts for High-risk Individuals	Disable accounts of individuals within [Assignment: organization-defined time period] of discovery of [Assignment: organization-defined significant risks].	Functional	Intersects With	High-Risk Terminations	HRS-09.2	Mechanisms exist to expedite the process of removing "high risk" individual's access to Technology Assets, Applications, Services and Data Assets	5		
AC-02(13)	Account Management Disable Accounts for High-risk Individuals	Disable accounts of individuals within [Assignment: organization-defined time period] of discovery of [Assignment: organization-defined significant risks].	Functional	Intersects With	Account Disabling for High Risk Individuals	IAC-15.6	Mechanisms exist to disable accounts immediately upon notification for users posing a significant risk to the organization	5		
AC-03	Access Enforcement	Enforce approved authorizations for logical access to information and system resources in accordance with applicable access control policies.	Functional	Intersects With	Access Enforcement	IAC-20	Mechanisms exist to enforce Logical Access Control (LAC) permissions that conform to the principle of "least privilege	5		AC-03
AC-03	Access Enforcement	Enforce approved authorizations for logical access to information and system resources in accordance with applicable access control policies.	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5		AC-03
AC-03	Access Enforcement	Enforce approved authorizations for logical access to information and system resources in accordance with applicable access control policies.	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5		AC-03
AC-03(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-03(02)	Access Enforcement Dual Authorization	Enforce dual authorization for [Assignment: organization-defined privileged commands and/or other organization-defined actions].	Functional	Intersects With	Two-Person Rule	HRS-12.1	Mechanisms exist to enforce a two-person rule for implementing changes to sensitive Technology Assets, Applications and Services (TAPS)	5		
AC-03(02)	Access Enforcement Dual Authorization	Enforce dual authorization for [Assignment: organization-defined privileged commands and/or other organization-defined actions].	Functional	Intersects With	Dual Authorization for Privileged Commands	IAC-20.5	Automated mechanisms exist to enforce dual authorization for privileged commands.	5		
AC-03(03)	Access Enforcement Mandatory Access Control	Enforce [Assignment: organization-defined mandatory access control policy] over the set of covered subjects and objects specified in the policy, and where the policy:a. Is uniformly enforced across the covered subjects and objects within the	Functional	No Relationship	N/A	N/A	N/A	0		

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AC-03(04)	Access Enforcement Discretionary Access Control	Enforce [Assignment: organization-defined discretionary access control policy] over the set of covered subjects and objects specified in the policy, and where the policy specifies that a subject that has been granted access to information can do one	Functional	No Relationship	N/A	N/A	N/A	0		
AC-03(05)	Access Enforcement Security-relevant Information	Prevent access to [Assignment: organization-defined security-relevant information] except during secure, non-operable system states.	Functional	No Relationship	N/A	N/A	N/A	0		
AC-03(06)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-03(07)	Access Enforcement Role-based Access Control	Enforce a role-based access control policy over defined subjects and objects and control access based upon [Assignment: organization-defined roles and users authorized to assume such roles]	Functional	No Relationship	N/A	N/A	N/A	0		
AC-03(08)	Access Enforcement Revocation of Access Authorizations	Enforce the revocation of access authorizations resulting from changes to the security attributes of subjects and objects based on [Assignment: organization-defined rules governing the timing of revocations of access authorizations]	Functional	Equal	Revocation of Access Authorizations	IAC-20.6	Mechanisms exist to revoke logical and physical access authorizations.	10		
AC-03(09)	Access Enforcement Controlled Release	Release information outside of the system only if: a. The receiving [Assignment: organization-defined system or system component] provides [Assignment: organization-defined controls]; and b. [Assignment: organization-defined controls] are	Functional	Equal	Controlled Release	DCH-03.3	Automated mechanisms exist to validate cybersecurity and data protection attributes prior to releasing information to external Technology Assets.	10		
AC-03(10)	Access Enforcement Audited Override of Access Control Mechanisms	Employ an audited override of automated access control mechanisms under [Assignment: organization-defined conditions] by [Assignment: organization-defined roles].	Functional	No Relationship	N/A	N/A	N/A	0		
AC-03(11)	Access Enforcement Restrict Access to Specific Information Types	Restrict access to data repositories containing [Assignment: organization-defined information types].	Functional	Equal	Sensitive / Regulated Data Access Enforcement	CFG-08	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) to restrict sensitive data and regulated	10		
AC-03(12)	Access Enforcement Assert and Enforce Application Access	a. Require applications to assert, as part of the installation process, the access needed to the following system applications and functions: [Assignment: organization-defined system applications and functions]; b. Provide an enforcement	Functional	No Relationship	N/A	N/A	N/A	0		
AC-03(13)	Access Enforcement Attribute-based Access Control	Enforce attribute-based access control policy over defined subjects and objects and control access based upon [Assignment: organization-defined attributes to assume access permissions]	Functional	No Relationship	N/A	N/A	N/A	0		
AC-03(14)	Access Enforcement Individual Access	Provide [Assignment: organization-defined mechanisms] to enable individuals to have access to the following elements of their personally identifiable information: [Assignment: organization-defined elements]	Functional	Equal	Data Subject Empowerment	PRI-06	Mechanisms exist to provide authenticated data subjects the ability to: (1) Access their Personal Data (PD)	10		
AC-03(15)	Access Enforcement Discretionary and Mandatory Access Control	a. Enforce [Assignment: organization-defined mandatory access control policy] over the set of covered subjects and objects specified in the policy; and b. Enforce [Assignment: organization-defined discretionary access control policy] over the set of	Functional	No Relationship	N/A	N/A	N/A	0		
AC-04	Information Flow Enforcement	Enforce approved authorizations for controlling the flow of information within the system and between connected systems based on [Assignment: organization-defined information flow control policies]	Functional	Equal	Data Flow Enforcement - Access Control Lists (ACLs)	NET-04	Mechanisms exist to implement and govern Access Control Lists (ACLs) to provide data flow enforcement that explicitly restrict network traffic to	10		
AC-04(01)	Information Flow Enforcement Object Security and Privacy Attributes	Use [Assignment: organization-defined security and privacy attributes] associated with [Assignment: organization-defined information, source, and destination objects] to enforce [Assignment: organization-defined information flow control	Functional	Equal	Object Security Attributes	NET-04.2	Mechanisms exist to associate security attributes with information, source and destination objects to enforce defined information flow	10		
AC-04(02)	Information Flow Enforcement Processing Domains	Use protected processing domains to enforce [Assignment: organization-defined information flow control policies] as a basis for flow control decisions.	Functional	No Relationship	N/A	N/A	N/A	0		
AC-04(03)	Information Flow Enforcement Dynamic Information Flow Control	Enforce [Assignment: organization-defined information flow control policies].	Functional	No Relationship	N/A	N/A	N/A	0		
AC-04(04)	Information Flow Enforcement Flow Control of Encrypted Information	Prevent encrypted information from bypassing [Assignment: organization-defined information flow control mechanisms] by [Selection (one or more): decrypting the information; blocking the flow of the encrypted information; terminating	Functional	Equal	Content Check for Encrypted Data	NET-04.3	Mechanisms exist to prevent encrypted data from bypassing content-checking mechanisms.	10		
AC-04(05)	Information Flow Enforcement Embedded Data Types	Enforce [Assignment: organization-defined limitations] on embedding data types within other data types.	Functional	Equal	Embedded Data Types	NET-04.4	Mechanisms exist to enforce limitations on embedding data within other data types.	10		
AC-04(06)	Information Flow Enforcement Metadata	Enforce information flow control based on [Assignment: organization-defined metadata].	Functional	Equal	Metadata	NET-04.5	Mechanisms exist to enforce information flow controls based on metadata.	10		
AC-04(07)	Information Flow Enforcement One-way Flow Mechanisms	Enforce one-way information flows through hardware-based flow control mechanisms.	Functional	No Relationship	N/A	N/A	N/A	0		
AC-04(08)	Information Flow Enforcement Security and Privacy Policy Filters	a. Enforce information flow control using [Assignment: organization-defined security or privacy policy filters] as a basis for flow control decisions for [Assignment: organization-defined information flows]; and b. [Selection (one or more): Block; Strip;	Functional	Equal	Policy Decision Point (PDP)	NET-04.7	Automated mechanisms exist to evaluate access requests against established criteria to dynamically and uniformly enforce access rights	10		
AC-04(09)	Information Flow Enforcement Human Reviews	Enforce the use of human reviews for [Assignment: organization-defined information flows] under the following conditions: [Assignment: organization-defined conditions].	Functional	Equal	Human Reviews	NET-04.6	Mechanisms exist to enforce the use of human reviews for Access Control Lists (ACLs) and similar rulesets on a routine basis	10		
AC-04(10)	Information Flow Enforcement Enable and Disable Security or Privacy Policy Filters	Provide the capability for privileged administrators to enable and disable [Assignment: organization-defined security or privacy policy filters] under the following conditions: [Assignment: organization-defined conditions]	Functional	No Relationship	N/A	N/A	N/A	0		
AC-04(11)	Information Flow Enforcement Configuration of Security or Privacy	Provide the capability for privileged administrators to configure [Assignment: organization-defined security or privacy policy filters] to support different security or privacy policies.	Functional	No Relationship	N/A	N/A	N/A	0		
AC-04(12)	Information Flow Enforcement Data Type Identifiers	When transferring information between different security domains, use [Assignment: organization-defined data type identifiers] to validate data essential for information flow decisions.	Functional	Equal	Data Type Identifiers	NET-04.8	Automated mechanisms exist to utilize data type identifiers to validate data essential for information flow decisions when transferring	10		
AC-04(13)	Information Flow Enforcement Decomposition into Policy-relevant Information Flow	When transferring information between different security domains, decompose information into [Assignment: organization-defined policy-relevant subcomponents] for submission to policy enforcement mechanisms.	Functional	Equal	Decomposition Into Policy-Related Subcomponents	NET-04.9	Automated mechanisms exist to decompose information into policy-relevant subcomponents for submission to policy enforcement	10		
AC-04(14)	Information Flow Enforcement Security or Privacy Policy Filter Constraints	When transferring information between different security domains, implement [Assignment: organization-defined security or privacy policy filters] requiring fully enumerated formats that restrict data structure and content.	Functional	No Relationship	N/A	N/A	N/A	0		

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AC-04(15)	Information Flow Enforcement Detection of Unsanctioned	When transferring information between different security domains, examine the information for the presence of [Assignment: organization-defined unsanctioned information] and prohibit the transfer of such information in accordance with	Functional	Equal	Detection of Unsanctioned Information	NET-04.10	Automated mechanisms exist to implement security policy filters requiring fully enumerated formats that restrict data structure and	10		
AC-04(16)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-04(17)	Information Flow Enforcement Domain Authentication	Uniquely identify and authenticate source and destination points by [Selection (one or more): organization; system; application; service; individual] for information transfer.	Functional	Equal	Cross Domain Authentication	NET-04.12	Automated mechanisms exist to uniquely identify and authenticate source and destination points for information transfer.	10		
AC-04(18)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-04(19)	Information Flow Enforcement Validation of Metadata	When transferring information between different security domains, implement [Assignment: organization-defined security or privacy policy filters] on metadata.	Functional	Equal	Metadata Validation	NET-04.13	Automated mechanisms exist to apply cybersecurity and/or data protection filters on metadata.	10		
AC-04(20)	Information Flow Enforcement Approved Solutions	Employ [Assignment: organization-defined solutions in approved configurations] to control the flow of [Assignment: organization-defined information] across security domains.	Functional	Equal	Approved Solutions	NET-04.11	Automated mechanisms exist to examine information for the presence of unsanctioned information and prohibits the transfer of such	10		
AC-04(21)	Information Flow Enforcement Physical or Logical Separation of Information Flows	Separate information flows logically or physically using [Assignment: organization-defined mechanisms and/or techniques] to accomplish [Assignment: organization-defined required separations by types of information]	Functional	Equal	Network Segmentation (macrosegmentation)	NET-06	Mechanisms exist to implement network segmentation within network architectures to isolate Technology Assets, Applications, and/or Services	10		
AC-04(22)	Information Flow Enforcement Access Only	Provide access from a single device to computing platforms, applications, or data residing in multiple different security domains, while preventing information flow between the different security domains.	Functional	No Relationship	N/A	N/A	N/A	0		
AC-04(23)	Information Flow Enforcement Modify Non-releasable Information	When transferring information between different security domains, modify non-releasable information by implementing [Assignment: organization-defined modification action].	Functional	No Relationship	N/A	N/A	N/A	0		
AC-04(24)	Information Flow Enforcement Internal Normalized Format	When transferring information between different security domains, parse incoming data into an internal normalized format and regenerate the data to be consistent with its intended application.	Functional	No Relationship	N/A	N/A	N/A	0		
AC-04(25)	Information Flow Enforcement Data Sanitization	When transferring information between different security domains, sanitize data to minimize [Selection (one or more): delivery of malicious content, command and control of malicious code, malicious code augmentation, and	Functional	Equal	Personal Data (PD) Retention & Disposal	PRJ-05	Mechanisms exist to: (1) Retain Personal Data (PD), including metadata, for an	10		
AC-04(26)	Information Flow Enforcement Audit Filtering Actions	When transferring information between different security domains, record and audit content filtering actions and results for the information being filtered.	Functional	No Relationship	N/A	N/A	N/A	0		
AC-04(27)	Information Flow Enforcement Redundant/Independent Filtering	When transferring information between different security domains, implement content filtering solutions that provide redundant and independent filtering mechanisms for each data flow.	Functional	No Relationship	N/A	N/A	N/A	0		
AC-04(28)	Information Flow Enforcement Linear Filter Pipelines	When transferring information between different security domains, implement a linear content filter pipeline that is enforced with discretionary and mandatory access controls.	Functional	No Relationship	N/A	N/A	N/A	0		
AC-04(29)	Information Flow Enforcement Filter Orchestration Engines	When transferring information between different security domains, employ content filter orchestration engines to ensure that a. Content filtering mechanisms successfully complete execution without errors; and b. Content filtering actions occur	Functional	No Relationship	N/A	N/A	N/A	0		
AC-04(30)	Information Flow Enforcement Filter Mechanisms Using Multiple Processes	When transferring information between different security domains, implement content filtering mechanisms using multiple processes.	Functional	No Relationship	N/A	N/A	N/A	0		
AC-04(31)	Information Flow Enforcement Failed Content Transfer Prevention	When transferring information between different security domains, prevent the transfer of failed content to the receiving domain.	Functional	No Relationship	N/A	N/A	N/A	0		
AC-04(32)	Information Flow Enforcement Process Requirements for Information Transfer	When transferring information between different security domains, the process that transfers information between filter pipelines: a. Does not filter message content; b. Validates filtering metadata; c. Ensures the content associated with the	Functional	No Relationship	N/A	N/A	N/A	0		
AC-05	Separation of Duties	a. Identify and document [Assignment: organization-defined duties of individuals requiring separation]; and b. Define system access authorizations to support separation of duties.	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5		
AC-05	Separation of Duties	a. Identify and document [Assignment: organization-defined duties of individuals requiring separation]; and b. Define system access authorizations to support separation of duties.	Functional	Intersects With	Dual Authorization for Change	CHG-04.3	Mechanisms exist to enforce a two-person rule for implementing changes to critical Technology Assets,	5		
AC-05	Separation of Duties	a. Identify and document [Assignment: organization-defined duties of individuals requiring separation]; and b. Define system access authorizations to support separation of duties.	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5		
AC-05	Separation of Duties	a. Identify and document [Assignment: organization-defined duties of individuals requiring separation]; and b. Define system access authorizations to support separation of duties.	Functional	Intersects With	Separation of Duties (SoD)	HRS-11	Mechanisms exist to implement and maintain Separation of Duties (SoD) to prevent potential inappropriate activity without collusion.	5		
AC-06	Least Privilege	Employ the principle of least privilege, allowing only authorized accesses for users (or processes acting on behalf of users) that are necessary to accomplish assigned organizational tasks.	Functional	Intersects With	Least Privilege	IAC-21	Mechanisms exist to utilize the concept of least privilege, allowing only authorized access to processes necessary to accomplish assigned	5		
AC-06	Least Privilege	Employ the principle of least privilege, allowing only authorized accesses for users (or processes acting on behalf of users) that are necessary to accomplish assigned organizational tasks.	Functional	Intersects With	Access Enforcement	IAC-20	Mechanisms exist to enforce Logical Access Control (LAC) permissions that conform to the principle of "least privilege."	5		
AC-06(01)	Least Privilege Authorize Access to Security Functions	Authorize access for [Assignment: organization-defined individuals or roles] to a. [Assignment: organization-defined security functions (deployed in hardware, software, and firmware)] and b. [Assignment: organization-defined security	Functional	Equal	Authorize Access to Security Functions	IAC-21.1	Mechanisms exist to limit access to security functions to explicitly-authorized privileged users.	10		
AC-06(02)	Least Privilege Non-privileged Access for Nonsecurity Functions	Require that users of system accounts (or roles) with access to [Assignment: organization-defined security functions or security-relevant information] use non-privileged accounts or roles, when accessing nonsecurity functions.	Functional	Equal	Non-Privileged Access for Non-Security Functions	IAC-21.2	Mechanisms exist to prohibit privileged users from using privileged accounts, while performing non-security functions.	10		
AC-06(03)	Least Privilege Network Access to Privileged Commands	Authorize network access to [Assignment: organization-defined privileged commands] only for [Assignment: organization-defined compelling operational needs] and document the rationale for such access in the security plan for the system.	Functional	Equal	Network Access to Privileged Commands	IAC-21.6	Mechanisms exist to authorize remote access to perform privileged commands on critical Technology Assets, Applications, and/or Services	10		
AC-06(04)	Least Privilege Separate Processing Domains	Provide separate processing domains to enable finer-grained allocation of user privileges.	Functional	No Relationship	N/A	N/A	N/A	0		

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AC-06(05)	Least Privilege Privileged Accounts	Restrict privileged accounts on the system to [Assignment: organization-defined personnel or roles].	Functional	Equal	Management Approval For Privileged Accounts	IAC-21.3	Mechanisms exist to restrict the assignment of privileged accounts to management-approved personnel	10		
AC-06(06)	Least Privilege Privileged Access by Non-organizational Users	Prohibit privileged access to the system by non-organizational users.	Functional	Equal	Privileged Access by Non-Organizational Users	IAC-05.2	Mechanisms exist to prohibit privileged access by non-organizational users.	10		
AC-06(07)	Least Privilege Review of User Privileges	a. Review [Assignment: organization-defined frequency] the privileges assigned to [Assignment: organization-defined roles or classes of users] to validate the need for such privileges; and b. Reassign or remove privileges, if necessary, to correctly	Functional	Equal	Periodic Review of Account Privileges	IAC-17	Mechanisms exist to periodically review the privileges assigned to individuals and service accounts to validate the need for such privileges	10		
AC-06(08)	Least Privilege Privilege Levels for Code Execution	Prevent the following software from executing at higher privilege levels than users executing the software: [Assignment: organization-defined software].	Functional	Equal	Privilege Levels for Code Execution	IAC-21.7	Automated mechanisms exist to prevent applications from executing at higher privilege levels than the user's privileges.	10		
AC-06(09)	Least Privilege Log Use of Privileged Functions	Log the execution of privileged functions.	Functional	Equal	Auditing Use of Privileged Functions	IAC-21.4	Mechanisms exist to audit the execution of privileged functions.	10		
AC-06(10)	Least Privilege Prohibit Non-privileged Users from Executing Privileged Functions	Prevent non-privileged users from executing privileged functions.	Functional	Equal	Prohibit Non-Privileged Users from Executing Privileged Functions	IAC-21.5	Mechanisms exist to prevent non-privileged users from executing privileged functions to include	10		
AC-07	Unsuccessful Logon Attempts	a. Enforce a limit of [Assignment: organization-defined number] consecutive invalid logon attempts by a user during a [Assignment: organization-defined time period]; and b. Automatically [Selection (one or more): lock the account or	Functional	Equal	Account Lockout	IAC-22	Mechanisms exist to enforce a limit for consecutive invalid login attempts by a user during an organization-defined time period and automatically locks	10		AC-07
AC-07(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-07(02)	Unsuccessful Logon Attempts Purge or Wipe Mobile Device	Purge or wipe information from [Assignment: organization-defined mobile devices] based on [Assignment: organization-defined purging or wiping requirements and techniques] after [Assignment: organization-defined number] consecutive	Functional	Equal	Remote Purging	MDM-05	Mechanisms exist to remotely purge selected information from mobile devices.	10		
AC-07(03)	Unsuccessful Logon Attempts Biometric Attempt Limiting	Limit the number of unsuccessful biometric logon attempts to [Assignment: organization-defined number].	Functional	No Relationship	N/A	N/A	N/A	0		
AC-07(04)	Unsuccessful Logon Attempts Use of Alternate Authentication Factor	a. Allow the use of [Assignment: organization-defined authentication factors] that are different from the primary authentication factors after the number of organization-defined consecutive invalid logon attempts have been exceeded; and b. Display [Assignment: organization-defined system use	Functional	No Relationship	N/A	N/A	N/A	0		
AC-08	System Use Notification	notification message or banner) to users before granting access to the system that provides privacy and security notices consistent with applicable laws, executive orders, directives	Functional	Equal	System Use Notification (Logon Banner)	SEA-18	Mechanisms exist to utilize system use notification / logon banners that display an approved system use notification message or banner before	10		AC-08
AC-09	Previous Logon Notification	Notify the user, upon successful logon to the system, of the date and time of the last logon.	Functional	Equal	Previous Logon Notification	SEA-19	Mechanisms exist to configure systems that process, store or transmit sensitive and/or regulated data to notify the user upon	10		
AC-09(01)	Previous Logon Notification Unsuccessful Logons	Notify the user, upon successful logon, of the number of unsuccessful logon attempts since the last successful logon.	Functional	No Relationship	N/A	N/A	N/A	0		
AC-09(02)	Previous Logon Notification Successful and Unsuccessful Logons	Notify the user, upon successful logon, of the number of [Selection (one): successful logons; unsuccessful logon attempts; both] during [Assignment: organization-defined time period].	Functional	No Relationship	N/A	N/A	N/A	0		
AC-09(03)	Previous Logon Notification Notification of Account Changes	Notify the user, upon successful logon, of changes to [Assignment: organization-defined security-related characteristics or parameters of the user's account] during [Assignment: organization-defined time period].	Functional	No Relationship	N/A	N/A	N/A	0		
AC-09(04)	Previous Logon Notification Additional Logon Information	Notify the user, upon successful logon, of the following additional information: [Assignment: organization-defined additional information].	Functional	No Relationship	N/A	N/A	N/A	0		
AC-10	Concurrent Session Control	Limit the number of concurrent sessions for each [Assignment: organization-defined account and/or account type] to [Assignment: organization-defined number].	Functional	Equal	Concurrent Session Control	IAC-23	Mechanisms exist to limit the number of concurrent sessions for each system account.	10		
AC-11	Device Lock	a. Prevent further access to the system by [Selection (one or more): initiating a device lock after [Assignment: organization-defined time period] of inactivity; requiring the user to initiate a device lock before leaving the system unattended]; and b. Retain	Functional	Intersects With	Session Lock	IAC-24	Mechanisms exist to initiate a session lock after an organization-defined time period of inactivity, or upon receiving a request from a user and retain the	5		
AC-11(01)	Device Lock Pattern-hiding Displays	Conceal, via the device lock, information previously visible on the display with a publicly viewable image.	Functional	Equal	Pattern-Hiding Displays	IAC-24.1	Mechanisms exist to implement pattern-hiding displays to conceal information previously visible on the display during the session lock.	10		
AC-12	Session Termination	Automatically terminate a user session after [Assignment: organization-defined conditions or trigger events requiring session disconnect].	Functional	Equal	Session Termination	IAC-25	Automated mechanisms exist to log out users, both locally on the network and for remote sessions, at the end of the session or after an organization-	10		
AC-12(01)	Session Termination User-initiated Logouts	Provide a logout capability for user-initiated communications sessions whenever authentication is used to gain access to [Assignment: organization-defined information resources].	Functional	Equal	User-Initiated Logouts / Message Displays	IAC-25.1	Mechanisms exist to provide a logout capability and display an explicit logout message to users indicating the reliable termination of the session.	10		
AC-12(02)	Session Termination Termination Message	Display an explicit logout message to users indicating the termination of authenticated communications sessions.	Functional	No Relationship	N/A	N/A	N/A	0		
AC-12(03)	Session Termination Timeout Warning Message	Display an explicit message to users indicating that the session will end in [Assignment: organization-defined time until end of session].	Functional	No Relationship	N/A	N/A	N/A	0		
AC-13		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-14	Permitted Actions Without Identification or Authentication	a. Identify [Assignment: organization-defined user actions] that can be performed on the system without identification or authentication consistent with organizational mission and business functions; and b. Document and provide supporting	Functional	Equal	Permitted Actions Without Identification or Authorization	IAC-26	Mechanisms exist to identify and document the supporting rationale for specific user actions that can be performed on a system without	10		AC-14
AC-14(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-15		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-16	Security and Privacy Attributes	a. Provide the means to associate [Assignment: organization-defined types of security and privacy attributes] with [Assignment: organization-defined security and privacy attribute	Functional	Equal	Cybersecurity & Data Protection Attributes	DCH-05	Mechanisms exist to bind cybersecurity and data protection attributes to information as it is stored, transmitted and processed.	10		
AC-16(01)	Security and Privacy Attributes Dynamic Attribute Association	Dynamically associate security and privacy attributes with [Assignment: organization-defined subjects and objects] in accordance with the following security and privacy policies as information is created and combined: [Assignment: organization-	Functional	Equal	Dynamic Attribute Association	DCH-05.1	Mechanisms exist to dynamically associate cybersecurity and data protection attributes with individuals and objects as information is created	10		

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
AC-16(02)	Security and Privacy Attributes Attribute Value Changes by Authorized Individuals	Provide authorized individuals (or processes acting on behalf of individuals) the capability to define or change the value of associated security and privacy attributes.	Functional	Equal	Attribute Value Changes By Authorized Individuals	DCH-05.2	Mechanisms exist to provide authorized individuals (or processes acting on behalf of individuals) the capability to define or change the	10		
AC-16(03)	Security and Privacy Attributes Maintenance of Attribute Associations	Maintain the association and integrity of [Assignment: organization-defined security and privacy attributes] to [Assignment: organization-defined subjects and objects].	Functional	Equal	Maintenance of Attribute Associations By System	DCH-05.3	Mechanisms exist to maintain the association and integrity of cybersecurity and data protection attributes to individuals and objects	10		
AC-16(04)	Security and Privacy Attributes Association of Attributes by Authorized Individuals	Provide the capability to associate [Assignment: organization-defined security and privacy attributes] with [Assignment: organization-defined subjects and objects] by authorized individuals (or processes acting on behalf of individuals)	Functional	Equal	Association of Attributes By Authorized Individuals	DCH-05.4	Mechanisms exist to provide the capability to associate cybersecurity and data protection attributes with individuals and objects by authorized	10		
AC-16(05)	Security and Privacy Attributes Attribute Displays on Objects to Be Output	Display security and privacy attributes in human-readable form on each object that the system transmits to output devices to identify [Assignment: organization-defined special dissemination, handling, or distribution instructions] using	Functional	Equal	Attribute Displays for Output Devices	DCH-05.5	Mechanisms exist to display cybersecurity and data protection attributes in human-readable form on each object that the system transmits	10		
AC-16(06)	Security and Privacy Attributes Maintenance of Attribute Association	Require personnel to associate and maintain the association of [Assignment: organization-defined security and privacy attributes] with [Assignment: organization-defined subjects and objects] in accordance with [Assignment: organization-defined	Functional	Equal	Data Subject Attribute Associations	DCH-05.6	Mechanisms exist to require personnel to associate and maintain the association of cybersecurity and data protection attributes with	10		
AC-16(07)	Security and Privacy Attributes Consistent Attribute Interpretation	Provide a consistent interpretation of security and privacy attributes transmitted between distributed system components.	Functional	Equal	Consistent Attribute Interpretation	DCH-05.7	Mechanisms exist to provide a consistent, organizationally agreed upon interpretation of cybersecurity and data protection attributes	10		
AC-16(08)	Security and Privacy Attributes Association Techniques and Technologies	Implement [Assignment: organization-defined techniques and technologies] in associating security and privacy attributes to information.	Functional	Equal	Identity Association Techniques & Technologies	DCH-05.8	Mechanisms exist to associate cybersecurity and data protection attributes to information.	10		
AC-16(09)	Security and Privacy Attributes Attribute Reassignment — Regrading	Change security and privacy attributes associated with information only via regrading mechanisms validated using [Assignment: organization-defined techniques or procedures].	Functional	Equal	Attribute Reassignment	DCH-05.9	Mechanisms exist to reclassify data as required, due to changing business/technical requirements.	10		
AC-16(10)	Security and Privacy Attributes Attribute Configuration by Authorized Individuals	Provide authorized individuals the capability to define or change the type and value of security and privacy attributes available for association with subjects and objects.	Functional	Equal	Attribute Configuration By Authorized Individuals	DCH-05.10	Mechanisms exist to provide authorized individuals the capability to define or change the type and value of cybersecurity and data protection	10		
AC-17	Remote Access	a. Establish and document usage restrictions, configuration/connection requirements, and implementation guidance for each type of remote access allowed; andb. Authorize each type of remote access to the system prior to	Functional	Intersects With	Remote Access	NET-14	Mechanisms exist to define, control and review organization-approved, secure remote access methods.	5		AC-17
AC-17(01)	Remote Access Monitoring and Control	Employ automated mechanisms to monitor and control remote access methods.	Functional	Equal	Automated Monitoring & Control	NET-14.1	Automated mechanisms exist to monitor and control remote access sessions.	10		
AC-17(02)	Remote Access Protection of Confidentiality and Integrity Using	Implement cryptographic mechanisms to protect the confidentiality and integrity of remote access sessions.	Functional	Equal	Protection of Confidentiality / Integrity Using Encryption	NET-14.2	Cryptographic mechanisms exist to protect the confidentiality and integrity of remote access sessions (e.g., VPN)	10		
AC-17(03)	Remote Access Managed Access Control Points	Route remote accesses through authorized and managed network access control points.	Functional	Equal	Managed Access Control Points	NET-14.3	Mechanisms exist to route all remote accesses through managed network access control points (e.g., VPN concentrator)	10		
AC-17(04)	Remote Access Privileged Commands and Access	a. Authorize the execution of privileged commands and access to security-relevant information via remote access only in a format that provides assessable evidence and for the following needs: [Assignment: organization-defined needs]; andb.	Functional	Equal	Remote Privileged Commands & Sensitive Data Access	NET-14.4	Mechanisms exist to restrict the execution of privileged commands and access to security-relevant information via remote access only for	10		
AC-17(05)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-17(06)	Remote Access Protection of Mechanism Information	Protect information about remote access mechanisms from unauthorized use and disclosure.	Functional	Intersects With	Remote Access	NET-14	Mechanisms exist to define, control and review organization-approved, secure remote access methods.	5		
AC-17(07)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-17(08)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-17(09)	Remote Access Disconnect or Disable Access	Provide the capability to disconnect or disable remote access to the system within [Assignment: organization-defined time period].	Functional	Equal	Expeditious Disconnect / Disable Capability	NET-14.8	Mechanisms exist to provide the capability to expeditiously disconnect or disable a user's remote access session.	10		AC-17(09)
AC-17(10)	Remote Access Authenticate Remote Commands	Implement [Assignment: organization-defined mechanisms] to authenticate [Assignment: organization-defined remote commands].	Functional	No Relationship	N/A	N/A	N/A	0		
AC-18	Wireless Access	a. Establish configuration requirements, connection requirements, and implementation guidance for each type of wireless access; andb. Authorize each type of wireless access to the system prior to allowing such connections.	Functional	Intersects With	Wireless Networking	NET-15	Mechanisms exist to control authorized wireless usage and monitor for unauthorized wireless access.	5		AC-18
AC-18	Wireless Access	a. Establish configuration requirements, connection requirements, and implementation guidance for each type of wireless access; andb. Authorize each type of wireless access to the system prior to allowing such connections.	Functional	Intersects With	Wireless Access Authentication & Encryption	CRY-07	Mechanisms exist to protect the confidentiality and integrity of wireless networking technologies by	5		AC-18
AC-18(01)	Wireless Access Authentication and Encryption	Protect wireless access to the system using authentication of [Selection (one or more): users; devices] and encryption.	Functional	Equal	Authentication & Encryption	NET-15.1	Mechanisms exist to secure Wi-Fi (e.g., IEEE 802.11) and prevent unauthorized access by:	10		
AC-18(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-18(03)	Wireless Access Disable Wireless Networking	Disable, when not intended for use, wireless networking capabilities embedded within system components prior to issuance and deployment.	Functional	Equal	Disable Wireless Networking	NET-15.2	Mechanisms exist to disable unnecessary wireless networking capabilities that are internally embedded within system components.	10		
AC-18(04)	Wireless Access Restrict Configurations by Users	Identify and explicitly authorize users allowed to independently configure wireless networking capabilities.	Functional	Equal	Restrict Configuration By Users	NET-15.3	Mechanisms exist to identify and explicitly authorize users who are allowed to independently configure wireless networking capabilities.	10		
AC-18(05)	Wireless Access Antennas and Transmission Power Levels	Select radio antennas and calibrate transmission power levels to reduce the probability that signals from wireless access points can be received outside of organization-controlled boundaries.	Functional	Equal	Wireless Boundaries	NET-15.4	Mechanisms exist to confine wireless communications to organization-controlled boundaries.	10		
AC-19	Access Control for Mobile Devices	a. Establish configuration requirements, connection requirements, and implementation guidance for organization-controlled mobile devices, to include when such devices are outside of controlled areas; andb. Authorize the connection of	Functional	Equal	Access Control For Mobile Devices	MDM-02	Mechanisms exist to enforce access control requirements for the connection of mobile devices to	10		AC-19
AC-19(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-19(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-19(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AC-19(04)	Access Control for Mobile Devices Restrictions for Classified Information	a. Prohibit the use of unclassified mobile devices in facilities containing systems processing, storing, or transmitting classified information unless specifically permitted by the authorizing official; andb. Enforce the following restrictions on	Functional	No Relationship	N/A	N/A	N/A	0		

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AC-19(05)	Access Control for Mobile Devices Full Device or Container-based Encryption	Employ [Selection: full-device encryption; container-based encryption] to protect the confidentiality and integrity of information on [Assignment: organization-defined mobile devices].	Functional	Equal	Full Device & Container-Based Encryption	MDM-03	Cryptographic mechanisms exist to protect the confidentiality and integrity of information on mobile devices through full-device or	10		
AC-20	Use of External Systems	a. [Selection (one or more): Establish [Assignment: organization-defined terms and conditions]; Identify [Assignment: organization-defined controls asserted to be implemented on external systems], consistent with the trust relationships	Functional	Equal	Use of External Technology Assets, Applications and/or Services (TAAS)	DCH-13	Mechanisms exist to govern how external parties, including Technology Assets, Applications and/or Services	10		AC-20
AC-20(01)	Use of External Systems Limits on Authorized Use	Permit authorized individuals to use an external system to access the system or to process, store, or transmit organization-controlled information only after: a. Verification of the implementation of controls on the external system as specified	Functional	Equal	Limits of Authorized Use	DCH-13.1	Mechanisms exist to prohibit external parties, including Technology Assets, Applications and/or Services (TAAS),	10		
AC-20(02)	Use of External Systems Portable Storage Devices — Restricted Use	Restrict the use of organization-controlled portable storage devices by authorized individuals on external systems using [Assignment: organization-defined restrictions].	Functional	Intersects With	Portable Storage Devices	DCH-13.2	Mechanisms exist to restrict or prohibit the use of portable storage devices by users on external systems.	5		
AC-20(03)	Use of External Systems Non-Organizationally Owned Systems —	Restrict the use of non-organizationally owned systems or system components to process, store, or transmit organizational information using [Assignment: organization-defined restrictions].	Functional	Equal	Non-Organizationally Owned Technology Assets, Applications and/or Services (TAAS)	DCH-13.4	Mechanisms exist to restrict the use of non-organizationally owned Technology Assets, Applications	10		
AC-20(04)	Use of External Systems Network Accessible Storage Devices — Prohibited	Prohibit the use of [Assignment: organization-defined network accessible storage devices] in external systems.	Functional	No Relationship	N/A	N/A	N/A	0		
AC-20(05)	Use of External Systems Portable Storage Devices — Prohibited Use	Prohibit the use of organization-controlled portable storage devices by authorized individuals on external systems.	Functional	Equal	Portable Storage Devices	DCH-13.2	Mechanisms exist to restrict or prohibit the use of portable storage devices by users on external systems.	10		
AC-21	Information Sharing	a. Enable authorized users to determine whether access authorizations assigned to a sharing partner match the information's access and use restrictions for [Assignment: organization-defined information sharing circumstances where	Functional	Intersects With	Information Sharing With Third Parties	PRI-07	Personal Data (PD) to third-parties only for the purposes identified in the data privacy notice and with the	5		
AC-21	Information Sharing	a. Enable authorized users to determine whether access authorizations assigned to a sharing partner match the information's access and use restrictions for [Assignment: organization-defined information sharing circumstances where	Functional	Intersects With	Information Sharing	DCH-14	Mechanisms exist to utilize a process to assist users in making information sharing decisions to ensure data is appropriately protected.	5		
AC-21(01)	Information Sharing Automated Decision Support	Employ [Assignment: organization-defined automated mechanisms] to enforce information-sharing decisions by authorized users based on access authorizations of sharing partners and access restrictions on information to be shared	Functional	No Relationship	N/A	N/A	N/A	0		
AC-21(02)	Information Sharing Information Search and Retrieval	Implement information search and retrieval services that enforce [Assignment: organization-defined information sharing restrictions].	Functional	Equal	Information Search & Retrieval	DCH-14.1	Mechanisms exist to ensure Technology Assets, Applications and/or Services (TAAS) implement	10		
AC-22	Publicly Accessible Content	a. Designate individuals authorized to make information publicly accessible; b. Train authorized individuals to ensure that publicly accessible information does not contain nonpublic information; c. Review the proposed content of information prior	Functional	Equal	Publicly Accessible Content	DCH-15	Mechanisms exist to control publicly-accessible content.	10		AC-22
AC-23	Data Mining Protection	Employ [Assignment: organization-defined data mining prevention and detection techniques] for [Assignment: organization-defined data storage objects] to detect and protect against unauthorized data mining	Functional	Intersects With	Data Mining Protection	DCH-16	Mechanisms exist to protect data storage objects against unauthorized data mining and data harvesting techniques	5		
AC-23	Data Mining Protection	Employ [Assignment: organization-defined data mining prevention and detection techniques] for [Assignment: organization-defined data storage objects] to detect and protect against unauthorized data mining	Functional	Intersects With	Usage Restrictions of Personal Data (PD)	PRI-05.4	Mechanisms exist to restrict collecting, receiving, processing, storing, transmitting, sharing and/or	5		
AC-24	Access Control Decisions	[Selection (one or more): Establish procedures; Implement mechanisms] to ensure [Assignment: organization-defined access control decisions] are applied to each access request prior to access enforcement	Functional	Intersects With	Management Approval For New or Changed Accounts	IAC-28.1	Mechanisms exist to ensure management approvals are required for new accounts or changes in permissions to existing accounts	5		
AC-24(01)	Access Control Decisions Transmit Access Authorization Information	Transmit [Assignment: organization-defined access authorization information] using [Assignment: organization-defined controls] to [Assignment: organization-defined systems] that enforce access control decisions	Functional	No Relationship	N/A	N/A	N/A	0		
AC-24(02)	Access Control Decisions No User or Process Identity	Enforce access control decisions based on [Assignment: organization-defined security or privacy attributes] that do not include the identity of the user or process acting on behalf of the user	Functional	No Relationship	N/A	N/A	N/A	0		
AC-25	Reference Monitor	Implement a reference monitor for [Assignment: organization-defined access control policies] that is tamperproof, always invoked, and small enough to be subject to analysis and testing, the completeness of which can be assured	Functional	Equal	Reference Monitor	IAC-27	Mechanisms exist to implement a reference monitor that is tamperproof, always-invoked, small enough to be subject to analysis / testing and the	10		
AT-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] awareness and training policy that a. Addresses	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRIP), including policies,	5		
AT-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] awareness and training policy that a. Addresses	Functional	Subset Of	Security, Compliance & Resilience-Minded Workforce	SAT-01	Mechanisms exist to facilitate the implementation of security workforce development and awareness controls.	10		
AT-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] awareness and training policy that a. Addresses	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5		AT-01
AT-02	Literacy Training and Awareness	a. Provide security and privacy literacy training to system users (including managers, senior executives, and contractors); 1. As part of initial training for new users and [Assignment: organization-defined frequency] thereafter; and 2. When	Functional	Equal	Security, Compliance & Resilience Awareness Training	SAT-02	Mechanisms exist to provide all employees and contractors appropriate security, compliance and	10		AT-02
AT-02(01)	Literacy Training and Awareness Practical Exercises	Provide practical exercises in literacy training that simulate events and incidents.	Functional	Intersects With	Simulated Cyber Attack Scenario Training	SAT-02.1	Mechanisms exist to include simulated actual cyber-attacks through practical exercises that are aligned with current threat scenarios	5		
AT-02(02)	Literacy Training and Awareness Insider Threat	Provide literacy training on recognizing and reporting potential indicators of insider threat.	Functional	Equal	Insider Threat Awareness	THR-05	Mechanisms exist to utilize security awareness training on recognizing and reporting potential indicators of insider threat	10		AT-02(02)
AT-02(03)	Literacy Training and Awareness Social Engineering and Mining	Provide literacy training on recognizing and reporting potential and actual instances of social engineering and social mining.	Functional	Equal	Social Engineering & Mining	SAT-02.2	Mechanisms exist to include awareness training on recognizing and reporting potential and actual instances of social engineering and	10		
AT-02(04)	Literacy Training and Awareness Suspicious Communications and	Provide literacy training on recognizing suspicious communications and anomalous behavior in organizational systems using [Assignment: organization-defined indicators of malicious code]	Functional	Intersects With	Suspicious Communications & Anomalous System Behavior	SAT-03.2	Mechanisms exist to provide training to personnel on organization-defined indicators of malware to recognize suspicious communications and	5		
AT-02(05)	Literacy Training and Awareness Advanced Persistent Threat	Provide literacy training on the advanced persistent threat.	Functional	Intersects With	Suspicious Communications & Anomalous System Behavior	SAT-03.2	Mechanisms exist to provide training to personnel on organization-defined indicators of malware to recognize suspicious communications and	5		

Secure Controls Framework (SCF)

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

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AU-06	Audit Record Review, Analysis, and Reporting	a. Review and analyze system audit records [Assignment: organization-defined frequency] for indications of [Assignment: organization-defined inappropriate or unusual activity] and the potential impact of the inappropriate or unusual activity; b.	Functional	Intersects With	Audit Level Adjustments	MON-02.6	Mechanisms exist to adjust the level of audit review, analysis and reporting based on evolving threat information from law enforcement industry.	5		AU-06
AU-06(01)	Audit Record Review, Analysis, and Reporting Automated Process Integration	Integrate audit record review, analysis, and reporting processes using [Assignment: organization-defined automated mechanisms].	Functional	Intersects With	Sensitive Event Log Information	MON-03.1	Mechanisms exist to protect sensitive and/or regulated data contained in log files.	5		
AU-06(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AU-06(03)	Audit Record Review, Analysis, and Reporting Correlate Audit Record	Analyze and correlate audit records across different repositories to gain organization-wide situational awareness.	Functional	Intersects With	Correlate Monitoring Information	MON-02.1	Automated mechanisms exist to correlate both technical and non-technical information from across the enterprise by a Security Incident Event Manager.	5		
AU-06(04)	Audit Record Review, Analysis, and Reporting Central Review and Analysis	Provide and implement the capability to centrally review and analyze audit records from multiple components within the system.	Functional	Equal	Central Review & Analysis	MON-02.2	Automated mechanisms exist to centrally collect, review and analyze audit records from multiple sources.	10		
AU-06(05)	Audit Record Review, Analysis, and Reporting Integrated Analysis of Audit	Integrate analysis of audit records with analysis of [Selection (one or more): vulnerability scanning information; performance data; system monitoring information; [Assignment: organization-defined data/information collected from other sources]] to	Functional	Equal	Integration of Scanning & Other Monitoring Information	MON-02.3	Automated mechanisms exist to integrate the analysis of audit records with analysis of vulnerability scanners, network performance system	10		
AU-06(06)	Audit Record Review, Analysis, and Reporting Correlation with Physical	Correlate information from audit records with information obtained from monitoring physical access to further enhance the ability to identify suspicious, inappropriate, unusual, or malevolent activity.	Functional	Equal	Correlation with Physical Monitoring	MON-02.4	Automated mechanisms exist to correlate information from audit records with information obtained from monitoring physical access to	10		
AU-06(07)	Audit Record Review, Analysis, and Reporting Permitted Actions	Specify the permitted actions for each [Selection (one or more): system process; role; user] associated with the review, analysis, and reporting of audit record information.	Functional	Equal	Permitted Actions	MON-02.5	Mechanisms exist to specify the permitted actions for both users and Technology Assets, Applications and/or Services (TAAS) associated	10		
AU-06(08)	Audit Record Review, Analysis, and Reporting Full Text Analysis of Privileged	Perform a full text analysis of logged privileged commands in a physically distinct component or subsystem of the system, or other system that is dedicated to that analysis.	Functional	Equal	Privileged Functions Logging	MON-03.3	Mechanisms exist to log and review the actions of users and/or services with elevated privileges.	10		
AU-06(09)	Audit Record Review, Analysis, and Reporting Correlation with Information from	Correlate information from nontechnical sources with audit record information to enhance organization-wide situational awareness.	Functional	Intersects With	Correlate Monitoring Information	MON-02.1	Automated mechanisms exist to correlate both technical and non-technical information from across the enterprise by a Security Incident Event Manager.	5		
AU-06(10)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AU-07	Audit Record Reduction and Report Generation	Provide and implement an audit record reduction and report generation capability that a. Supports on-demand audit record review, analysis, and reporting requirements and after-the-fact investigations of incidents; and b. Does not alter the original	Functional	Intersects With	Monitoring Reporting	MON-06	Mechanisms exist to provide an event log report generation capability to aid in detecting and assessing anomalous activities.	5		
AU-07(01)	Audit Record Reduction and Report Generation Automatic Processing	Provide and implement the capability to process, sort, and search audit records for events of interest based on the following content: [Assignment: organization-defined fields within audit records]	Functional	Intersects With	Monitoring Reporting	MON-06	Mechanisms exist to provide an event log report generation capability to aid in detecting and assessing anomalous activities.	5		
AU-07(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AU-08	Time Stamps	a. Use internal system clocks to generate time stamps for audit records; and b. Record time stamps for audit records that meet [Assignment: organization-defined granularity of time measurement] and that use Coordinated Universal Time. have a	Functional	Intersects With	Clock Synchronization	SEA-20	Mechanisms exist to utilize time-synchronization technology to synchronize all critical system clocks.	5		AU-08
AU-08	Time Stamps	a. Use internal system clocks to generate time stamps for audit records; and b. Record time stamps for audit records that meet [Assignment: organization-defined granularity of time measurement] and that use Coordinated Universal Time. have a	Functional	Intersects With	Time Stamps	MON-07	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) to use an	5		AU-08
AU-08(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AU-08(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AU-09	Protection of Audit Information	a. Protect audit information and audit logging tools from unauthorized access, modification, and deletion; and b. Alert [Assignment: organization-defined personnel or roles] upon detection of unauthorized access, modification, or deletion of	Functional	Equal	Protection of Event Logs	MON-08	Mechanisms exist to protect event logs and audit tools from unauthorized access, modification and deletion.	10		AU-09
AU-09(01)	Protection of Audit Information Hardware Write-once Media	Write audit trails to hardware-enforced, write-once media.	Functional	No Relationship	N/A	N/A	N/A	0		
AU-09(02)	Protection of Audit Information Store on Separate Physical Systems or	Store audit records [Assignment: organization-defined frequency] in a repository that is part of a physically different system or system component than the system or component being audited.	Functional	Intersects With	Event Log Backup on Separate Physical Systems / Components	MON-08.1	Mechanisms exist to back up event logs onto a physically different system or system component than the Security Incident Event Manager.	5		
AU-09(03)	Protection of Audit Information Cryptographic Protection	Implement cryptographic mechanisms to protect the integrity of audit information and audit tools.	Functional	Equal	Cryptographic Protection of Event Log Information	MON-08.3	Cryptographic mechanisms exist to protect the integrity of event logs and audit tools.	10		
AU-09(04)	Protection of Audit Information Access by Subset of Privileged Users	Authorize access to management of audit logging functionality to only [Assignment: organization-defined subset of privileged users or roles].	Functional	Equal	Access by Subset of Privileged Users	MON-08.2	Mechanisms exist to restrict access to the management of event logs to privileged users with a specific business need.	10		
AU-09(05)	Protection of Audit Information Dual Authorization	Enforce dual authorization for [Selection (one or more): movement; deletion] of [Assignment: organization-defined audit information].	Functional	Equal	Dual Authorization for Event Log Movement	MON-08.4	Automated mechanisms exist to enforce dual authorization for the movement or deletion of event logs.	10		
AU-09(06)	Protection of Audit Information Read-only Access	Authorize read-only access to audit information to [Assignment: organization-defined subset of privileged users or roles].	Functional	No Relationship	N/A	N/A	N/A	0		
AU-09(07)	Protection of Audit Information Store on Component with Different Operating	Store audit information on a component running a different operating system than the system or component being audited.	Functional	No Relationship	N/A	N/A	N/A	0		
AU-10	Non-repudiation	Provide irrefutable evidence that an individual (or process acting on behalf of an individual) has performed [Assignment: organization-defined actions to be covered by non-repudiation].	Functional	Equal	Non-Repudiation	MON-09	Mechanisms exist to utilize a non-repudiation capability to protect against an individual falsely denying having performed a particular action.	10		
AU-10(01)	Non-repudiation Association of Identities	a. Bind the identity of the information producer with the information to [Assignment: organization-defined strength of binding]; and b. Provide the means for authorized individuals to determine the identity of the producer of the information.	Functional	Intersects With	Identity Binding	MON-09.1	Mechanisms exist to bind the identity of the information producer to the information generated.	5		
AU-10(02)	Non-repudiation Validate Binding of Information Producer Identity	a. Validate the binding of the information producer identity to the information at [Assignment: organization-defined frequency]; and b. Perform [Assignment: organization-defined actions] in the event of a validation error.	Functional	Intersects With	Identity Binding	MON-09.1	Mechanisms exist to bind the identity of the information producer to the information generated.	5		
AU-10(03)	Non-repudiation Chain of Custody	Maintain reviewer or releaser credentials within the established chain of custody for information reviewed or released.	Functional	Intersects With	Chain of Custody & Forensics	IRO-08	Mechanisms exist to perform digital forensics and maintain the integrity of the chain of custody, in accordance with applicable laws, regulations and	5		

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AU-10(04)	Non-repudiation Validate Binding of Information Reviewer Identity	a. Validate the binding of the information reviewer identity to the information at the transfer or release points prior to release or transfer between [Assignment: organization-defined security domains]; and b. Perform [Assignment: organization-defined	Functional	No Relationship	N/A	N/A	N/A	0		
AU-10(05)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AU-11	Audit Record Retention	Retain audit records for [Assignment: organization-defined time period consistent with records retention policy] to provide support for after-the-fact investigations of incidents and to meet regulatory and organizational information retention	Functional	Equal	Event Log Retention	MON-10	Mechanisms exist to retain event logs for a time period consistent with records retention requirements to provide support for after-the-fact	10		AU-11
AU-11(01)	Audit Record Retention Long-term Retrieval Capability	Employ [Assignment: organization-defined measures] to ensure that long-term audit records generated by the system can be retrieved.	Functional	No Relationship	N/A	N/A	N/A	0		
AU-12	Audit Record Generation	a. Provide audit record generation capability for the event types the system is capable of auditing as defined in AU-2a on [Assignment: organization-defined system components]; b. Allow [Assignment: organization-defined personnel or roles] to Compile audit records from [Assignment: organization-defined system components] into a system-wide (logical or physical) audit trail that is time-correlated to within [Assignment: organization-defined level of tolerance for the relationship	Functional	Intersects With	Monitoring Reporting	MON-06	Mechanisms exist to provide an event log report generation capability to aid in detecting and assessing anomalous activities	5		AU-12
AU-12(01)	Audit Record Generation System-wide and Time-correlated Audit Trail	Compile audit records from [Assignment: organization-defined system components] into a system-wide (logical or physical) audit trail that is time-correlated to within [Assignment: organization-defined level of tolerance for the relationship	Functional	Equal	System-Wide / Time-Correlated Audit Trail	MON-02.7	Automated mechanisms exist to compile audit records into an organization-wide audit trail that is time-correlated	10		
AU-12(02)	Audit Record Generation Standardized Formats	Produce a system-wide (logical or physical) audit trail composed of audit records in a standardized format.	Functional	No Relationship	N/A	N/A	N/A	0		
AU-12(03)	Audit Record Generation Changes by Authorized Individuals	Provide and implement the capability for [Assignment: organization-defined individuals or roles] to change the logging to be performed on [Assignment: organization-defined system components] based on [Assignment: organization-defined	Functional	Equal	Changes by Authorized Individuals	MON-02.8	Mechanisms exist to provide privileged users or roles the capability to change the auditing to be	10		
AU-12(04)	Audit Record Generation Query Parameter Audits of Personally Identifiable	Provide and implement the capability for auditing the parameters of user query events for data sets containing personally identifiable information.	Functional	Equal	Query Parameter Audits of Personal Data (PD)	MON-06.1	Mechanisms exist to provide and implement the capability for auditing the parameters of user query events in data sets containing Personal Data	10		
AU-13	Monitoring for Information Disclosure	a. Monitor [Assignment: organization-defined open-source information and/or information sites] [Assignment: organization-defined frequency] for evidence of unauthorized disclosure of organizational information; and b. If an information disclosure is	Functional	Equal	Monitoring For Information Disclosure	MON-11	Mechanisms exist to monitor for evidence of unauthorized exfiltration or disclosure of non-public information	10		
AU-13(01)	Monitoring for Information Disclosure Use of Automated Tools	Monitor open-source information and information sites using [Assignment: organization-defined automated mechanisms].	Functional	No Relationship	N/A	N/A	N/A	0		
AU-13(02)	Monitoring for Information Disclosure Review of Monitored Sites	Review the list of open-source information sites being monitored [Assignment: organization-defined frequency].	Functional	No Relationship	N/A	N/A	N/A	0		
AU-13(03)	Monitoring for Information Disclosure Unauthorized Replication of	Employ discovery techniques, processes, and tools to determine if external entities are replicating organizational information in an unauthorized manner.	Functional	No Relationship	N/A	N/A	N/A	0		
AU-14	Session Audit	a. Provide and implement the capability for [Assignment: organization-defined users or roles] to [Selection (one or more): record; view; hear; log] the content of a user session under [Assignment: organization-defined circumstances]; and b.	Functional	Equal	Session Audit	MON-12	Mechanisms exist to provide session audit capabilities that can: (1) Capture and log all content related to a user session; and	10		
AU-14(01)	Session Audit System Start-up	Initiate session audits automatically at system start-up.	Functional	No Relationship	N/A	N/A	N/A	0		
AU-14(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AU-14(03)	Session Audit Remote Viewing and Listening	Provide and implement the capability for authorized users to remotely view and hear content related to an established user session in real time.	Functional	Equal	Real-Time Session Monitoring	MON-01.17	Mechanisms exist to enable authorized personnel the ability to remotely view and hear content related to an established user session	10		
AU-15		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
AU-16	Cross-organizational Audit Logging	Employ [Assignment: organization-defined methods] for coordinating [Assignment: organization-defined audit information] among external organizations when audit information is transmitted across organizational boundaries.	Functional	Intersects With	Cross-Organizational Monitoring	MON-14	Mechanisms exist to coordinate sanitized event logs among external organizations to identify anomalous events when event logs are shared	5		
AU-16(01)	Cross-organizational Audit Logging Identity Preservation	Preserve the identity of individuals in cross-organizational audit trails.	Functional	Intersects With	Cross-Organizational Monitoring	MON-14	Mechanisms exist to coordinate sanitized event logs among external organizations to identify anomalous events when event logs are shared	5		
AU-16(02)	Cross-organizational Audit Logging Sharing of Audit Information	Provide cross-organizational audit information to [Assignment: organization-defined organizations] based on [Assignment: organization-defined cross-organizational sharing agreements].	Functional	Equal	Sharing of Event Logs	MON-14.1	Mechanisms exist to share event logs with third-party organizations based on specific cross-organizational sharing agreements	10		
AU-16(03)	Cross-organizational Audit Logging Dissociability	Implement [Assignment: organization-defined measures] to disassociate individuals from audit information transmitted across organizational boundaries.	Functional	No Relationship	N/A	N/A	N/A	0		
CA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level assessment, authorization, and monitoring policy	Functional	Subset Of	Information Assurance (IA) Operations	IAO-01	Mechanisms exist to facilitate the implementation of security, compliance and resilience assessment and authorization	10		CA-01
CA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level assessment, authorization, and monitoring policy	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRIP), including policies, standards and procedures necessary for secure, compliant and resilient	5		CA-01
CA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]; 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level assessment, authorization, and monitoring policy	Functional	Intersects With	Resilience Program Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5		CA-01
CA-02	Control Assessments	a. Select the appropriate assessor or assessment team for the type of assessment to be conducted; b. Develop a control assessment plan that describes the scope of the assessment including: 1. Controls and control enhancements under	Functional	Intersects With	Functional Review Of Security, Compliance & Resilience Controls	CPL-03.2	Mechanisms exist to regularly review Technology Assets, Applications and/or Services (TAAS) for adherence	5		CA-02
CA-02	Control Assessments	a. Select the appropriate assessor or assessment team for the type of assessment to be conducted; b. Develop a control assessment plan that describes the scope of the assessment including: 1. Controls and control enhancements under	Functional	Intersects With	Technical Verification	IAO-06	Mechanisms exist to perform Information Assurance Program (IAP) activities to evaluate the design, implementation and effectiveness of	5		CA-02
CA-02	Control Assessments	a. Select the appropriate assessor or assessment team for the type of assessment to be conducted; b. Develop a control assessment plan that describes the scope of the assessment including: 1. Controls and control enhancements under	Functional	Intersects With	Security, Compliance & Resilience in Project Management	PRM-04	Mechanisms exist to assess security, compliance and resilience controls as part of Technology Assets, Applications and/or Services (TAAS)	5		CA-02
CA-02	Control Assessments	a. Select the appropriate assessor or assessment team for the type of assessment to be conducted; b. Develop a control assessment plan that describes the scope of the assessment including: 1. Controls and control enhancements under	Functional	Intersects With	Assessments	IAO-02	Mechanisms exist to formally assess the security, compliance and resilience controls in Technology Assets, Applications and/or Services	5		CA-02

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CA-02	Control Assessments	a. Select the appropriate assessor or assessment team for the type of assessment to be conducted;b. Develop a control assessment plan that describes the scope of the assessment including:1. Controls and control enhancements under	Functional	Intersects With	Control Conformity Monitoring	CPL-03	Mechanisms exist to validate that Technology Assets, Applications, Services and/or Data (TAASD) conform	5		CA-02
CA-02(01)	Control Assessments Independent Assessors	Employ independent assessors or assessment teams to conduct control assessments.	Functional	Equal	Assessor Independence	IAO-02.1	Mechanisms exist to ensure assessors or assessment teams have the appropriate independence to conduct security, compliance and/or resilience	10		
CA-02(02)	Control Assessments Specialized Assessments	Include as part of control assessments, [Assignment: organization-defined frequency], [Selection (one): announced; unannounced], [Selection (one or more): in-depth monitoring; security instrumentation; automated security test cases]	Functional	Intersects With	Specialized Assessments	IAO-02.2	Mechanisms exist to conduct specialized assessments for: (1) Statutory, regulatory and contractual compliance obligations	5		
CA-02(03)	Control Assessments Leveraging Results from External Organizations	Leverage the results of control assessments performed by [Assignment: organization-defined external organization(s)] on [Assignment: organization-defined system] when the assessment meets [Assignment: organization-defined	Functional	Equal	Third-Party Assessment Reciprocity	IAO-02.3	Mechanisms exist to accept and respond to the results of external assessments that are performed by impartial external organizations	10		
CA-03	Information Exchange	a. Approve and manage the exchange of information between the system and other systems using [Selection (one or more): interconnection security agreements; information exchange security agreements; memoranda of understanding or	Functional	Intersects With	Interconnection Security Agreements (ISAs)	NET-05	Mechanisms exist to authorize connections from systems to other systems using Interconnection Security Agreements (ISAs), or similar	5		CA-03
CA-03(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CA-03(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CA-03(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CA-03(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CA-03(05)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CA-03(06)	Information Exchange Transfer Authorizations	Verify that individuals or systems transferring data between interconnecting systems have the requisite authorizations (i.e., write permissions or privileges) prior to accepting such data.	Functional	Equal	Transfer Authorizations	DCH-14.2	Mechanisms exist to verify that individuals or Technology Assets, Applications and/or Services (TAAS)	10		
CA-03(07)	Information Exchange Transitive Information Exchanges	a. Identify transitive (downstream) information exchanges with other systems through the systems identified in CA-3a; andb. Take measures to ensure that transitive (downstream) information exchanges cease when the controls on identified	Functional	No Relationship	N/A	N/A	N/A	0		
CA-04		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CA-05	Plan of Action and Milestones	a. Develop a plan of action and milestones for the system to document the planned remediation actions of the organization to correct weaknesses or deficiencies noted during the assessment of the controls and to reduce or eliminate known	Functional	Intersects With	Capabilities Deficiency Tracking	IAO-05	Mechanisms exist to govern identified deficiencies (e.g., Plan of Action and Milestones (POA&M) or similar methodology) that formally	5		CA-05
CA-05(01)	Plan of Action and Milestones Automation Support for Accuracy and	Ensure the accuracy, currency, and availability of the plan of action and milestones for the system using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Deficiency Tracking Automation	IAO-05.1	Automated mechanisms exist to help ensure tracked deficiencies are: (1) Accurate; (2) Up-to-date; and	10		
CA-06	Authorization	a. Assign a senior official as the authorizing official for the system;b. Assign a senior official as the authorizing official for common controls available for inheritance by organizational systems;c. Ensure that the authorizing official for the system	Functional	Equal	Security Authorization	IAO-07	Mechanisms exist to ensure Technology Assets, Applications and/or Services (TAAS) are officially authorized access for the system	10		CA-06
CA-06(01)	Authorization Joint Authorization — Intra-organization	Employ a joint authorization process for the system that includes multiple authorizing officials from the same organization conducting the authorization.	Functional	No Relationship	N/A	N/A	N/A	0		
CA-06(02)	Authorization Joint Authorization — Inter-organization	Employ a joint authorization process for the system that includes multiple authorizing officials with at least one authorizing official from an organization external to the organization conducting the authorization.	Functional	No Relationship	N/A	N/A	N/A	0		
CA-07	Continuous Monitoring	Develop a system-level continuous monitoring strategy and implement continuous monitoring in accordance with the organization-level continuous monitoring strategy that includes:a. Establishing the following system-level metrics to be	Functional	Intersects With	Security, Compliance & Resilience Controls Oversight	CPL-02	Mechanisms exist to provide a security, compliance and resilience controls oversight function that reports to the organization's executive	5		CA-07
CA-07(01)	Continuous Monitoring Independent Assessment	Employ independent assessors or assessment teams to monitor the controls in the system on an ongoing basis.	Functional	Intersects With	Independent Assessors	CPL-03.1	Mechanisms exist to utilize independent assessors to evaluate security, compliance and resilience at	5		
CA-07(01)	Continuous Monitoring Types of Assessments	Employ independent assessors or assessment teams to monitor the controls in the system on an ongoing basis.	Functional	Intersects With	Security, Compliance & Resilience Controls Oversight	CPL-02	Mechanisms exist to provide a security, compliance and resilience controls oversight function that reports to the organization's executive	5		
CA-07(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CA-07(03)	Continuous Monitoring Trend Analyses	Employ trend analyses to determine if control implementations, the frequency of continuous monitoring activities, and the types of activities used in the continuous monitoring process need to be modified based on empirical data.	Functional	Equal	Trend Analysis Reporting	MON-06.2	Mechanisms exist to employ trend analyses to determine if security control implementations, the frequency of continuous monitoring	10		
CA-07(04)	Continuous Monitoring Risk Monitoring	Ensure risk monitoring is an integral part of the continuous monitoring strategy that includes the following:a. Effectiveness monitoring;b. Compliance monitoring; andc. Change monitoring	Functional	Equal	Risk Monitoring	RSK-11	Mechanisms exist to ensure risk monitoring as an integral part of the continuous monitoring strategy that includes monitoring the effectiveness	10		CA-07(04)
CA-07(05)	Continuous Monitoring Consistency Analysis	Employ the following actions to validate that policies are established and implemented controls are operating in a consistent manner: [Assignment: organization-defined actions].	Functional	No Relationship	N/A	N/A	N/A	0		
CA-07(06)	Continuous Monitoring Automation Support for Monitoring	Ensure the accuracy, currency, and availability of monitoring results for the system using [Assignment: organization-defined automated mechanisms].	Functional	No Relationship	N/A	N/A	N/A	0		
CA-08	Penetration Testing	Conduct penetration testing [Assignment: organization-defined frequency] on [Assignment: organization-defined systems or system components].	Functional	Equal	Penetration Testing	VPM-07	Mechanisms exist to conduct penetration testing on Technology Assets, Applications and/or Services	10		
CA-08(01)	Penetration Testing Independent Penetration Testing Agent or Team	Employ an independent penetration testing agent or team to perform penetration testing on the system or system components.	Functional	Equal	Independent Penetration Agent or Team	VPM-07.1	Mechanisms exist to utilize an independent assessor or penetration team to perform penetration testing.	10		
CA-08(02)	Penetration Testing Red Team Exercises	Employ the following red-team exercises to simulate attempts by adversaries to compromise organizational systems in accordance with applicable rules of engagement: [Assignment: organization-defined red team exercises]	Functional	Equal	Red Team Exercises	VPM-10	Mechanisms exist to utilize "red team" exercises to simulate attempts by adversaries to compromise	10		
CA-08(03)	Penetration Testing Facility Penetration Testing	Employ a penetration testing process that includes [Assignment: organization-defined frequency] [Selection (one or more): announced; unannounced] attempts to bypass or circumvent controls associated with physical access points to	Functional	No Relationship	N/A	N/A	N/A	0		
CA-09	Internal System Connections	a. Authorize internal connections of [Assignment: organization-defined system components or classes of components] to the system;b. Document, for each internal connection, the interface characteristics, security and privacy requirements, and the	Functional	Equal	Internal System Connections	NET-05.2	Mechanisms exist to control internal system connections through authorizing internal connections of systems and documenting for each	10		CA-09
CA-09(01)	Internal System Connections Compliance Checks	Perform security and privacy compliance checks on constituent system components prior to the establishment of the internal connection.	Functional	Equal	Endpoint Security Validation	NET-14.7	Automated mechanisms exist to validate the security posture of the endpoint devices (e.g., software	10		

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FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
CM-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level configuration management policy that a	Functional	Subset Of	Configuration Management Program	CFG-01	Mechanisms exist to facilitate the implementation of configuration management controls.	10		CM-01
CM-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level configuration management policy that a	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies,	5		CM-01
CM-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level configuration management policy that a	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5		CM-01
CM-02	Baseline Configuration	a. Develop, document, and maintain under configuration control, a current baseline configuration of the system; andb. Review and update the baseline configuration of the system: 1. [Assignment: organization-defined frequency]: 2. When required	Functional	Intersects With	Reviews & Updates	CFG-02.1	Mechanisms exist to review and update baseline configurations: (1) At least annually; (2) When required due to so: or	5		CM-02
CM-02	Baseline Configuration	a. Develop, document, and maintain under configuration control, a current baseline configuration of the system; andb. Review and update the baseline configuration of the system: 1. [Assignment: organization-defined frequency]: 2. When required	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for Technology Assets, Applications and/or Services (TAAS)	5		CM-02
CM-02(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CM-02(02)	Baseline Configuration Automation Support for Accuracy and Currency	Maintain the currency, completeness, accuracy, and availability of the baseline configuration of the system using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Automated Central Management & Verification	CFG-02.2	Automated mechanisms exist to govern and report on baseline configurations of Technology Assets, Applications and/or Services (TAAS)	10		
CM-02(03)	Baseline Configuration Retention of Previous Configurations	Retain [Assignment: organization-defined number] of previous versions of baseline configurations of the system to support rollback.	Functional	Equal	Retention Of Previous Configurations	CFG-02.3	Mechanisms exist to retain previous versions of baseline configuration to support roll back.	10		
CM-02(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CM-02(05)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CM-02(06)	Baseline Configuration Development and Test Environments	Maintain a baseline configuration for system development and test environments that is managed separately from the operational baseline configuration.	Functional	Equal	Development & Test Environment Configurations	CFG-02.4	Mechanisms exist to manage baseline configurations for development and test environments separately from operational baseline configurations to	10		
CM-02(07)	Baseline Configuration Configure Systems and Components for High-risk Areas	a. Issue [Assignment: organization-defined systems or system components] with [Assignment: organization-defined configurations] to [Assignment: organization-defined locations that the organization deems to be of significant risk]; andb. Apply the	Functional	Equal	Configure Technology Assets, Applications and/or Services (TAAS) for High-Risk Areas	CFG-02.5	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) utilized in high-risk areas.	10		
CM-03	Configuration Change Control	a. Determine and document the types of changes to the system that are configuration-controlled;b. Review proposed configuration-controlled changes to the system and approve or disapprove such changes with explicit consideration for security	Functional	Subset Of	Change Management Program	CHG-01	Mechanisms exist to facilitate the implementation of a change management program.	10		
CM-03	Configuration Change Control	a. Determine and document the types of changes to the system that are configuration-controlled;b. Review proposed configuration-controlled changes to the system and approve or disapprove such changes with explicit consideration for security	Functional	Intersects With	Configuration Change Control	CHG-02	Mechanisms exist to govern the technical configuration change control processes.	5		
CM-03(01)	Configuration Change Control Automated Documentation, Notification and	Use [Assignment: organization-defined automated mechanisms] to: a. Document proposed changes to the system;b. Notify [Assignment: organization-defined approval authorities of proposed changes to the system and request	Functional	Equal	Prohibition Of Changes	CHG-02.1	Mechanisms exist to prohibit unauthorized changes, unless organization-approved change requests are received.	10		
CM-03(02)	Configuration Change Control Testing, Validation, and Documentation of	Test, validate, and document changes to the system before finalizing the implementation of the changes.	Functional	Intersects With	Control Functionality Verification	CHG-06	Mechanisms exist to verify the functionality of security, compliance and resilience controls following implemented changes to assure	5		
CM-03(02)	Configuration Change Control Testing, Validation, and Documentation of	Test, validate, and document changes to the system before finalizing the implementation of the changes.	Functional	Intersects With	Test, Validate & Document Changes	CHG-02.2	Mechanisms exist to appropriately test and document proposed changes in a non-production environment before changes are implemented in a	5		
CM-03(03)	Configuration Change Control Automated Change Implementation	Implement changes to the current system baseline and deploy the updated baseline across the installed base using [Assignment: organization-defined automated mechanisms].	Functional	No Relationship	N/A	N/A	N/A	0		
CM-03(04)	Configuration Change Control Security and Privacy Representatives	Require [Assignment: organization-defined security and privacy representatives] to be members of the [Assignment: organization-defined configuration change control element].	Functional	Equal	Security, Compliance & Resilience Representative for Asset Lifecycle	CHG-02.3	Mechanisms exist to include a cybersecurity and/or data protection representative in the configuration	10		
CM-03(05)	Configuration Change Control Automated Security Response	Implement the following security responses automatically if baseline configurations are changed in an unauthorized manner: [Assignment: organization-defined security responses].	Functional	Equal	Automated Security Response	CHG-02.4	Automated mechanisms exist to implement remediation actions upon the detection of unauthorized baseline configurations change(s)	10		
CM-03(06)	Configuration Change Control Cryptography Management	Ensure that cryptographic mechanisms used to provide the following controls are under configuration management: [Assignment: organization-defined controls].	Functional	Equal	Cryptographic Management	CHG-02.5	Mechanisms exist to govern assets involved in providing cryptographic protections according to the organization's configuration	10		
CM-03(07)	Configuration Change Control Review System Changes	Review changes to the system [Assignment: organization-defined frequency] or when [Assignment: organization-defined circumstances] to determine whether unauthorized changes have occurred.	Functional	Intersects With	Test, Validate & Document Changes	CHG-02.2	Mechanisms exist to appropriately test and document proposed changes in a non-production environment before changes are implemented in a	5		
CM-03(08)	Configuration Change Control Prevent or Restrict Configuration Changes	Prevent or restrict changes to the configuration of the system under the following circumstances: [Assignment: organization-defined circumstances].	Functional	Equal	Configuration Enforcement	CFG-06	Automated mechanisms exist to monitor, enforce and report on configurations for endpoint devices.	10		
CM-03(08)	Configuration Change Control Prevent or Restrict Configuration Changes	Prevent or restrict changes to the configuration of the system under the following circumstances: [Assignment: organization-defined circumstances].	Functional	Intersects With	Integrity Assurance & Enforcement (IAE)	CFG-06.1	Automated mechanisms exist to identify unauthorized deviations from an approved baseline and implement automated resilience actions to	5		
CM-04	Impact Analyses	Analyze changes to the system to determine potential security and privacy impacts prior to change implementation.	Functional	Equal	Security Impact Analysis for Changes	CHG-03	Mechanisms exist to analyze proposed changes for potential security impacts, prior to the implementation of the change.	10		CM-04
CM-04(01)	Impact Analyses Separate Test Environments	Analyze changes to the system in a separate test environment before implementation in an operational environment, looking for security and privacy impacts due to flaws, weaknesses, incompatibility, or intentional malice	Functional	Equal	Separation of Development, Testing and Operational Environments	TDA-08	Mechanisms exist to manage separate development, testing and operational environments to reduce the risks of	10		
CM-04(02)	Impact Analyses Verification of Controls	After system changes, verify that the impacted controls are implemented correctly, operating as intended, and producing the desired outcome with regard to meeting the security and privacy requirements for the system	Functional	Equal	Technical Verification	IAO-06	Mechanisms exist to perform Information Assurance Program (IAP) activities to evaluate the design, implementation and effectiveness of	10		
CM-05	Access Restrictions for Change	Define, document, approve, and enforce physical and logical access restrictions associated with changes to the system.	Functional	Intersects With	Governing Access Restriction for Change	END-03.2	Mechanisms exist to define, document, approve and enforce access restrictions associated with	5		CM-05
CM-05	Access Restrictions for Change	Define, document, approve, and enforce physical and logical access restrictions associated with changes to the system.	Functional	Intersects With	Access Restriction For Change	CHG-04	Mechanisms exist to enforce configuration restrictions in an effort to restrict the ability of users to conduct unauthorized changes.	5		CM-05

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CM-05(01)	Access Restrictions for Change Automated Access Enforcement and Audit Records	a. Enforce access restrictions using [Assignment: organization-defined automated mechanisms]; andb. Automatically generate audit records of the enforcement actions.	Functional	Equal	Automated Access Enforcement / Auditing	CHG-04.1	Mechanisms exist to perform after-the-fact reviews of configuration change logs to discover any unauthorized changes.	10		
CM-05(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CM-05(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CM-05(04)	Access Restrictions for Change Dual Authorization	Enforce dual authorization for implementing changes to [Assignment: organization-defined system components and system-level information].	Functional	Equal	Dual Authorization for Change	CHG-04.3	Mechanisms exist to enforce a two-person rule for implementing changes to critical Technology Assets, Applications and/or Services (TAAS).	10		
CM-05(05)	Access Restrictions for Change Privilege Limitation for Production and	a. Limit privileges to change system components and system-related information within a production or operational environment; andb. Review and reevaluate privileges [Assignment: organization-defined frequency].	Functional	Equal	Permissions To Implement Changes	CHG-04.4	Mechanisms exist to limit operational privileges for implementing changes.	10		
CM-05(06)	Access Restrictions for Change Limit Library Privileges	Limit privileges to change software resident within software libraries.	Functional	Equal	Library Privileges	CHG-04.5	Mechanisms exist to restrict software library privileges to those individuals with a pertinent business need for access.	10		
CM-05(07)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CM-06	Configuration Settings	a. Establish and document configuration settings for components employed within the system that reflect the most restrictive mode consistent with operational requirements using [Assignment: organization-defined common secure	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for	5		CM-06
CM-06	Configuration Settings	a. Establish and document configuration settings for components employed within the system that reflect the most restrictive mode consistent with operational requirements using [Assignment: organization-defined common secure	Functional	Intersects With	Approved Configuration Deviations	CFG-02.7	Mechanisms exist to document, assess risk and approve or deny deviations to standardized configurations.	5		CM-06
CM-06(01)	Configuration Settings Automated Management, Application and	Manage, apply, and verify configuration settings for [Assignment: organization-defined system components] using [Assignment: organization-defined automated mechanisms].	Functional	Intersects With	Automated Central Management & Verification	CFG-02.2	Automated mechanisms exist to govern and report on baseline configurations of Technology Assets, Applications and/or Services (TAAS).	5		
CM-06(02)	Configuration Settings Respond to Unauthorized Changes	Take the following actions in response to unauthorized changes to [Assignment: organization-defined configuration settings]: [Assignment: organization-defined actions].	Functional	Equal	Respond To Unauthorized Changes	CFG-02.8	Mechanisms exist to respond to unauthorized changes to configuration settings as security incidents.	10		
CM-06(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CM-06(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CM-07	Least Functionality	a. Configure the system to provide only [Assignment: organization-defined mission essential capabilities]; andb. Prohibit or restrict the use of the following functions, ports, protocols, software, and/or services: [Assignment: organization-	Functional	Equal	Least Functionality	CFG-03	Mechanisms exist to configure systems to provide only essential capabilities by specifically prohibiting or restricting the use of ports,	10		CM-07
CM-07(01)	Least Functionality Periodic Review	a. Review the system [Assignment: organization-defined frequency] to identify unnecessary and/or nonsecure functions, ports, protocols, software, and services; andb. Disable or remove [Assignment: organization-defined functions, ports	Functional	Equal	Periodic Review	CFG-03.1	Mechanisms exist to periodically review system configurations to identify and disable unnecessary and/or non-secure functions, ports,	10		
CM-07(02)	Least Functionality Prevent Program Execution	Prevent program execution in accordance with [Selection (one or more): [Assignment: organization-defined policies, rules of behavior, and/or access agreements regarding software program usage and restrictions]: rules authorizing the terms and	Functional	Intersects With	Prevent Program Execution	SEA-06	Automated mechanisms exist to prevent the execution of unauthorized software programs.	5		
CM-07(02)	Least Functionality Prevent Program Execution	Prevent program execution in accordance with [Selection (one or more): [Assignment: organization-defined policies, rules of behavior, and/or access agreements regarding software program usage and restrictions]: rules authorizing the terms and	Functional	Intersects With	Prevent Unauthorized Software Execution	CFG-03.2	Mechanisms exist to configure systems to prevent the execution of unauthorized software programs.	5		
CM-07(03)	Least Functionality Registration Compliance	Ensure compliance with [Assignment: organization-defined registration requirements for functions, ports, protocols, and services].	Functional	No Relationship	N/A	N/A	N/A	0		
CM-07(04)	Least Functionality Unauthorized Software — Deny-by-exception	a. Identify [Assignment: organization-defined software programs not authorized to execute on the system];b. Employ an allow-all, deny-by-exception policy to prohibit the execution of	Functional	Equal	Explicitly Allow / Deny Applications	CFG-03.3	Mechanisms exist to explicitly allow (allowlist / whitelist) and/or block (denylist / blacklist) applications that	10		
CM-07(05)	Least Functionality Authorized Software — Allow-by-exception	a. Identify [Assignment: organization-defined software programs authorized to execute on the system];b. Employ a deny-all, permit-by-exception policy to allow the execution of authorized	Functional	Equal	Explicitly Allow / Deny Applications	CFG-03.3	Mechanisms exist to explicitly allow (allowlist / whitelist) and/or block (denylist / blacklist) applications that	10		
CM-07(06)	Least Functionality Confined Environments with Limited Privileges	Require that the following user-installed software execute in a confined physical or virtual machine environment with limited privileges: [Assignment: organization-defined user-installed software].	Functional	Intersects With	Configure Technology Assets, Applications and/or Services (TAAS) for High-Risk Areas	CFG-02.5	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) utilized in high-	5		
CM-07(07)	Least Functionality Code Execution in Protected Environments	Allow execution of binary or machine-executable code only in confined physical or virtual machine environments and with the explicit approval of [Assignment: organization-defined personnel or roles] when such code is a. Obtained from sources	Functional	Intersects With	Configure Technology Assets, Applications and/or Services (TAAS) for High-Risk Areas	CFG-02.5	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) utilized in high-	5		
CM-07(08)	Least Functionality Binary or Machine Executable Code	a. Prohibit the use of binary or machine-executable code from sources with limited or no warranty or without the provision of source code; andb. Allow exceptions only for compelling mission or operational requirements and with the approval of	Functional	Equal	Binary or Machine-Executable Code	END-06.7	Mechanisms exist to prohibit the use of binary or machine-executable code from sources with limited or no warranty and without access to	10		
CM-07(09)	Least Functionality Prohibiting The Use of Unauthorized Hardware	a. Identify [Assignment: organization-defined hardware components authorized for system use];b. Prohibit the use or connection of unauthorized hardware components;c. Review and update the list of authorized hardware components	Functional	Intersects With	Configure Technology Assets, Applications and/or Services (TAAS) for High-Risk Areas	CFG-02.5	Mechanisms exist to configure Technology Assets, Applications and/or Services (TAAS) utilized in high-	5		
CM-08	System Component Inventory	a. Develop and document an inventory of system components that:1. Accurately reflects the system;2. Includes all components within the system;3. Does not include duplicate accounting of components or components assigned to any	Functional	Intersects With	Asset Inventories	AST-02	Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data	5		CM-08
CM-08	System Component Inventory	a. Develop and document an inventory of system components that:1. Accurately reflects the system;2. Includes all components within the system;3. Does not include duplicate accounting of components or components assigned to any	Functional	Intersects With	Component Duplication Avoidance	AST-02.3	Mechanisms exist to establish and maintain an authoritative source and repository to provide a trusted source and accountability for approved and	5		CM-08
CM-08(01)	System Component Inventory Updates During Installation and Removal	Update the inventory of system components as part of component installations, removals, and system updates.	Functional	Equal	Updates During Installations / Removals	AST-02.1	Mechanisms exist to update asset inventories as part of component installations, removals and asset upgrades.	10		
CM-08(02)	System Component Inventory Automated Maintenance	Maintain the currency, completeness, accuracy, and availability of the inventory of system components using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Configuration Management Database (CMDB)	AST-02.9	Mechanisms exist to implement and manage a Configuration Management Database (CMDB), or similar	10		
CM-08(03)	System Component Inventory Automated Unauthorized	a. Detect the presence of unauthorized hardware, software, and firmware components within the system using [Assignment: organization-defined automated mechanisms] [Assignment: organization-defined frequency]; andb. Take the following	Functional	Intersects With	Automated Unauthorized Component Detection	AST-02.2	Automated mechanisms exist to detect and alert upon the detection of unauthorized hardware, software and	5		
CM-08(03)	System Component Inventory Automated Unauthorized	a. Detect the presence of unauthorized hardware, software, and firmware components within the system using [Assignment: organization-defined automated mechanisms] [Assignment: organization-defined frequency]; andb. Take the following	Functional	Intersects With	Software Installation Alerts	END-03.1	Mechanisms exist to generate an alert when new software is detected.	5		

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

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CM-08(03)	System Component Inventory Automated Unauthorized Component Detection	a. Detect the presence of unauthorized hardware, software, and firmware components within the system using [Assignment: organization-defined automated mechanisms] [Assignment: organization-defined frequency]; andb. Take the following Include in the system component inventory information, a means for identifying by [Selection (one or more): name; position; role], individuals responsible and accountable for administering those components	Functional	Intersects With	Unauthorized Installation Alerts	CFG-05.1	Mechanisms exist to configure systems to generate an alert when the unauthorized installation of software is detected	5		
CM-08(04)	System Component Inventory Accountability Information		Functional	Equal	Accountability Information	AST-03.1	Mechanisms exist to include capturing the name, position and/or role of individuals responsible/accountable for administering assets as part of the	10		
CM-08(05)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CM-08(06)	System Component Inventory Assessed Configurations and Approved Deviations	Include assessed component configurations and any approved deviations to current deployed configurations in the system component inventory.	Functional	Equal	Approved Baseline Deviations	AST-02.4	Mechanisms exist to document and govern instances of approved deviations from established baseline configurations	10		
CM-08(07)	System Component Inventory Centralized Repository	Provide a centralized repository for the inventory of system components.	Functional	Intersects With	Configuration Management Database (CMDB)	AST-02.9	Mechanisms exist to implement and manage a Configuration Management Database (CMDB), or similar technology, to monitor and govern	5		
CM-08(08)	System Component Inventory Automated Location Tracking	Support the tracking of system components by geographic location using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Automated Location Tracking	AST-02.10	Mechanisms exist to track the geographic location of system components.	10		
CM-08(09)	System Component Inventory Assignment of Components to Systems	a. Assign system components to a system; andb. Receive an acknowledgement from [Assignment: organization-defined personnel or roles] of this assignment.	Functional	Equal	Component Assignment	AST-02.11	Mechanisms exist to bind components to a specific system.	10		
CM-09	Configuration Management Plan	Develop, document, and implement a configuration management plan for the system that:a. Addresses roles, responsibilities, and configuration management processes and procedures;b. Establishes a process for identifying	Functional	Subset Of	Configuration Management Program	CFG-01	Mechanisms exist to facilitate the implementation of configuration management controls.	10		
CM-09	Configuration Management Plan	Develop, document, and implement a configuration management plan for the system that:a. Addresses roles, responsibilities, and configuration management processes and procedures;b. Establishes a process for identifying	Functional	Intersects With	Stakeholder Notification of Changes	CHG-05	Mechanisms exist to ensure stakeholders are made aware of and understand the impact of proposed changes	5		
CM-09(01)	Configuration Management Plan Assignment of Responsibility	Assign responsibility for developing the configuration management process to organizational personnel that are not directly involved in system development.	Functional	Equal	Assignment of Responsibility	CFG-01.1	Mechanisms exist to implement a segregation of duties for configuration management that prevents	10		
CM-10	Software Usage Restrictions	a. Use software and associated documentation in accordance with contract agreements and copyright laws;b. Track the use of software and associated documentation protected by quantity licenses to control copying and distribution; andc. Control and	Functional	Equal	Software Usage Restrictions	CFG-04	Mechanisms exist to enforce software usage restrictions to comply with applicable contract agreements and copyright laws	10		CM-10
CM-10(01)	Software Usage Restrictions Open-source Software	Establish the following restrictions on the use of open-source software: [Assignment: organization-defined restrictions].	Functional	Equal	Open Source Software	CFG-04.1	Mechanisms exist to establish parameters for the secure use of open source software.	10		
CM-11	User-installed Software	a. Establish [Assignment: organization-defined policies] governing the installation of software by users;b. Enforce software installation policies through the following methods: [Assignment: organization-defined methods]; andc. Monitor	Functional	Intersects With	Prohibit Installation Without Privileged Status	END-03	Automated mechanisms exist to prohibit software installations without explicitly assigned privileged status.	5		
CM-11	User-installed Software	a. Establish [Assignment: organization-defined policies] governing the installation of software by users;b. Enforce software installation policies through the following methods: [Assignment: organization-defined methods]; andc. Monitor	Functional	Intersects With	User-Installed Software	CFG-05	Mechanisms exist to restrict the ability of non-privileged users to install unauthorized software.	5		CM-11
CM-11(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CM-11(02)	User-installed Software Software Installation with Privileged Status	Allow user installation of software only with explicit privileged status.	Functional	Intersects With	User-Installed Software	CFG-05	Mechanisms exist to restrict the ability of non-privileged users to install unauthorized software.	5		
CM-11(02)	User-installed Software Software Installation with Privileged Status	Allow user installation of software only with explicit privileged status.	Functional	Intersects With	Restrict Roles Permitted To Install Software	CFG-05.2	Mechanisms exist to configure systems to prevent the installation of software, unless the action is performed by a privileged user or	5		
CM-11(02)	User-installed Software Software Installation with Privileged Status	Allow user installation of software only with explicit privileged status.	Functional	Intersects With	Prohibit Installation Without Privileged Status	END-03	Automated mechanisms exist to prohibit software installations without explicitly assigned privileged status.	5		
CM-11(03)	User-installed Software Automated Enforcement and Monitoring	Enforce and monitor compliance with software installation policies using [Assignment: organization-defined automated mechanisms].	Functional	Intersects With	Configuration Enforcement	CFG-06	Automated mechanisms exist to monitor, enforce and report on configurations for endpoint devices.	5		
CM-11(03)	User-installed Software Automated Enforcement and Monitoring	Enforce and monitor compliance with software installation policies using [Assignment: organization-defined automated mechanisms].	Functional	Intersects With	Integrity Assurance & Enforcement (IAE)	CFG-06.1	Automated mechanisms exist to identify unauthorized deviations from an approved baseline and implement automated resiliency actions to	5		
CM-11(03)	User-installed Software Automated Enforcement and Monitoring	Enforce and monitor compliance with software installation policies using [Assignment: organization-defined automated mechanisms].	Functional	Intersects With	Software Installation Alerts	END-03.1	Mechanisms exist to generate an alert when new software is detected.	5		
CM-11(03)	User-installed Software Automated Enforcement and Monitoring	Enforce and monitor compliance with software installation policies using [Assignment: organization-defined automated mechanisms].	Functional	Intersects With	Unauthorized Installation Alerts	CFG-05.1	Mechanisms exist to configure systems to generate an alert when the unauthorized installation of software is detected	5		
CM-12	Information Location	a. Identify and document the location of [Assignment: organization-defined information] and the specific system components on which the information is processed and stored;b. Identify and document the users who have access to	Functional	Equal	Information Location	DCH-24	Mechanisms exist to identify and document the location of information and the specific system components on which the information resides	10		
CM-12(01)	Information Location Automated Tools to Support Information Location	Use automated tools to identify [Assignment: organization-defined information by information type] on [Assignment: organization-defined system components] to ensure controls are in place to protect organizational information and individual	Functional	Equal	Automated Tools to Support Information Location	DCH-24.1	Automated mechanisms exist to identify by data classification type to ensure adequate security, compliance and resiliency controls are in place to	10		
CM-13	Data Action Mapping	Develop and document a map of system data actions.	Functional	Equal	Data Action Mapping	AST-02.8	Mechanisms exist to create and maintain a map of Technology Assets, Applications and/or Services (TAAS)	10		
CM-14	Signed Components	Prevent the installation of [Assignment: organization-defined software and firmware components] without verification that the component has been digitally signed using a certificate that is recognized and approved by the organization	Functional	Intersects With	Signed Components	CHG-04.2	Mechanisms exist to prevent the installation of software and firmware components without verification that the component has been digitally	5		
CP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] contingency planning policy that:a. Addresses	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCR), including policies,	5		CP-01
CP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] contingency planning policy that:a. Addresses	Functional	Subset Of	Business Continuity Management System (BCMS)	BCD-01	Mechanisms exist to facilitate the implementation of contingency planning controls to help ensure	10		CP-01

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Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

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Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
CP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] contingency planning policy that a. Addresses	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5		CP-01
CP-02	Contingency Plan	a. Develop a contingency plan for the system that:1. Identifies essential mission and business functions and associated contingency requirements;2. Provides recovery objectives, restoration priorities, and metrics;3. Addresses contingency	Functional	Subset Of	Business Continuity Management System (BCMS)	BCD-01	Mechanisms exist to facilitate the implementation of contingency planning controls to help ensure	10		CP-02
CP-02	Contingency Plan	a. Develop a contingency plan for the system that:1. Identifies essential mission and business functions and associated contingency requirements;2. Provides recovery objectives, restoration priorities, and metrics;3. Addresses contingency	Functional	Intersects With	Ongoing Contingency Planning	BCD-06	Mechanisms exist to update contingency plans due to changes affecting:	5		CP-02
CP-02(01)	Contingency Plan Coordinate with Related Plans	Coordinate contingency plan development with organizational elements responsible for related plans.	Functional	Equal	Coordinate with Related Plans	BCD-01.1	Mechanisms exist to coordinate contingency plan development with internal and external elements responsible for related plans.	10		
CP-02(02)	Contingency Plan Capacity Planning	Conduct capacity planning so that necessary capacity for information processing, telecommunications, and environmental support exists during contingency operations.	Functional	Equal	Capacity Planning	CAP-03	Mechanisms exist to conduct capacity planning so that necessary capacity for information processing, telecommunications and	10		
CP-02(03)	Contingency Plan Resume Mission and Business Functions	Plan for the resumption of [Selection (one): all; essential] mission and business functions within [Assignment: organization-defined time period] of contingency plan activation.	Functional	Intersects With	Resume All Missions & Business Functions	BCD-02.1	Mechanisms exist to resume all missions and business functions within Recovery Time Objectives (RTO) of the contingency plan's	5		
CP-02(03)	Contingency Plan Resume Mission and Business Functions	Plan for the resumption of [Selection (one): all; essential] mission and business functions within [Assignment: organization-defined time period] of contingency plan activation.	Functional	Intersects With	Resume Essential Missions & Business Functions	BCD-02.3	Mechanisms exist to resume essential missions and business functions within an organization-defined time period of contingency plan activation.	5		
CP-02(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CP-02(05)	Contingency Plan Continue Mission and Business Functions	Plan for the continuance of [Selection (one): all; essential] mission and business functions with minimal or no loss of operational continuity and sustains that continuity until full system restoration at primary processing and/or storage sites.	Functional	Equal	Continue Essential Mission & Business Functions	BCD-02.2	Mechanisms exist to continue essential missions and business functions with little or no loss of operational continuity and sustain	10		
CP-02(06)	Contingency Plan Alternate Processing and Storage Sites	Plan for the transfer of [Selection (one): all; essential] mission and business functions to alternate processing and/or storage sites with minimal or no loss of operational continuity and sustain that continuity through system restoration to primary.	Functional	Equal	Transfer to Alternate Processing / Storage Site	BCD-01.3	Mechanisms exist to redeploy personnel to other roles during a disruptive event or in the execution of	10		
CP-02(07)	Contingency Plan Coordinate with External Service Providers	Coordinate the contingency plan with the contingency plans of external service providers to ensure that contingency requirements can be satisfied.	Functional	Equal	Coordinate With External Service Providers	BCD-01.2	Mechanisms exist to coordinate internal contingency plans with the contingency plans of external service providers to ensure that contingency	10		
CP-02(08)	Contingency Plan Identify Critical Assets	Identify critical system assets supporting [Selection (one): all; essential] mission and business functions.	Functional	Equal	Identify Critical Assets	BCD-02	Mechanisms exist to identify and document the critical Technology Assets, Applications, Services and/or	10		
CP-03	Contingency Training	a. Provide contingency training to system users consistent with assigned roles and responsibilities;1. Within [Assignment: organization-defined time period] of assuming a contingency role or responsibility;2. When required by system changes.	Functional	Equal	Contingency Training	BCD-03	Mechanisms exist to adequately train contingency personnel and applicable stakeholders in their contingency roles and responsibilities.	10		CP-03
CP-03(01)	Contingency Training Simulated Events	Incorporate simulated events into contingency training to facilitate effective response by personnel in crisis situations.	Functional	Equal	Simulated Events	BCD-03.1	Mechanisms exist to incorporate simulated events into contingency training to facilitate effective response by personnel in crisis situations.	10		
CP-03(02)	Contingency Training Mechanisms Used in Training Environments	Employ mechanisms used in operations to provide a more thorough and realistic contingency training environment.	Functional	Equal	Automated Training Environments	BCD-03.2	Automated mechanisms exist to provide a more thorough and realistic contingency training environment.	10		
CP-04	Contingency Plan Testing	a. Test the contingency plan for the system [Assignment: organization-defined frequency] using the following tests to determine the effectiveness of the plan and the readiness to execute the plan: [Assignment: organization-defined tests] b.	Functional	Intersects With	Contingency Plan Root Cause Analysis (RCA) & Lessons Learned	BCD-05	Mechanisms exist to conduct a Root Cause Analysis (RCA) and "lessons learned" activity every time the contingency plan is activated.	5		CP-04
CP-04	Contingency Plan Testing	a. Test the contingency plan for the system [Assignment: organization-defined frequency] using the following tests to determine the effectiveness of the plan and the readiness to execute the plan: [Assignment: organization-defined tests] b.	Functional	Intersects With	Contingency Plan Testing & Exercises	BCD-04	Mechanisms exist to conduct tests and/or exercises to evaluate the contingency plan's effectiveness and the organization's readiness to	5		CP-04
CP-04(01)	Contingency Plan Testing Coordinate with Related Plans	Coordinate contingency plan testing with organizational elements responsible for related plans.	Functional	Equal	Coordinated Testing with Related Plans	BCD-04.1	Mechanisms exist to coordinate contingency plan testing with internal and external elements responsible for related plans.	10		
CP-04(02)	Contingency Plan Testing Alternate Processing Site	Test the contingency plan at the alternate processing site:a. To familiarize contingency personnel with the facility and available resources; andb. To evaluate the capabilities of the alternate processing site to support contingency operations.	Functional	Equal	Alternate Storage & Processing Sites	BCD-04.2	Mechanisms exist to test contingency plans at alternate storage & processing sites to both familiarize contingency personnel with the facility	10		
CP-04(03)	Contingency Plan Testing Automated Testing	Test the contingency plan using [Assignment: organization-defined automated mechanisms].	Functional	No Relationship	N/A	N/A	N/A	0		
CP-04(04)	Contingency Plan Testing Full Recovery and Reconstitution	Include a full recovery and reconstitution of the system to a known state as part of contingency plan testing.	Functional	No Relationship	N/A	N/A	N/A	0		
CP-04(05)	Contingency Plan Testing Self-challenge	Employ [Assignment: organization-defined mechanisms] to [Assignment: organization-defined system or system component] to disrupt and adversely affect the system or system component.	Functional	No Relationship	N/A	N/A	N/A	0		
CP-05		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CP-06	Alternate Storage Site	a. Establish an alternate storage site, including necessary agreements to permit the storage and retrieval of system backup information; andb. Ensure that the alternate storage site provides controls equivalent to that of the primary site.	Functional	Equal	Alternate Storage Site	BCD-08	Mechanisms exist to establish an alternate storage site that includes both the assets and necessary	10		
CP-06(01)	Alternate Storage Site Separation from Primary Site	Identify an alternate storage site that is sufficiently separated from the primary storage site to reduce susceptibility to the same threats.	Functional	Equal	Separation from Primary Storage Site	BCD-08.1	Mechanisms exist to separate the alternate storage site from the primary storage site to reduce susceptibility to similar threats.	10		
CP-06(02)	Alternate Storage Site Recovery Time and Recovery Point Objectives	Configure the alternate storage site to facilitate recovery operations in accordance with recovery time and recovery point objectives.	Functional	Intersects With	Recovery Time / Point Objectives (RTO / RPO)	BCD-01.4	Mechanisms exist to facilitate recovery operations in accordance with Recovery Time Objectives (RTOs) and Recovery Point Objectives (RPOs).	5		
CP-06(03)	Alternate Storage Site Accessibility	Identify potential accessibility problems to the alternate storage site in the event of an area-wide disruption or disaster and outline explicit mitigation actions.	Functional	Equal	Primary Storage Site Accessibility	BCD-08.2	Mechanisms exist to identify and mitigate potential accessibility problems to the alternate storage sites in the event of an area-wide	10		
CP-07	Alternate Processing Site	a. Establish an alternate processing site, including necessary agreements to permit the transfer and resumption of [Assignment: organization-defined system operations] for essential mission and business functions within [Assignment:	Functional	Equal	Alternate Processing Site	BCD-09	Mechanisms exist to establish an alternate processing site that provides security measures equivalent to that of the primary site.	10		

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

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CP-07(01)	Alternate Processing Site Separation from Primary Site	Identify an alternate processing site that is sufficiently separated from the primary processing site to reduce susceptibility to the same threats.	Functional	Equal	Separation from Primary Processing Site	BCD-09.1	Mechanisms exist to separate the alternate processing site from the primary processing site to reduce susceptibility to similar threats.	10		
CP-07(02)	Alternate Processing Site Accessibility	Identify potential accessibility problems to alternate processing sites in the event of an area-wide disruption or disaster and outlines explicit mitigation actions.	Functional	Equal	Alternate Processing Site Accessibility	BCD-09.2	Mechanisms exist to identify and mitigate potential accessibility problems to the alternate processing sites and possible mitigation actions.	10		
CP-07(03)	Alternate Processing Site Priority of Service	Develop alternate processing site agreements that contain priority-of-service provisions in accordance with availability requirements (including recovery time objectives).	Functional	Equal	Alternate Site Priority of Service	BCD-09.3	Mechanisms exist to address priority-of-service provisions in alternate processing and storage sites that support availability requirements.	10		
CP-07(04)	Alternate Processing Site Preparation for Use	Prepare the alternate processing site so that the site can serve as the operational site supporting essential mission and business functions.	Functional	Equal	Preparation for Use	BCD-09.4	Mechanisms exist to prepare the alternate processing alternate to support essential missions and business functions so that the	10		
CP-07(05)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CP-07(06)	Alternate Processing Site Inability to Return to Primary Site	Plan and prepare for circumstances that preclude returning to the primary processing site.	Functional	Equal	Inability to Return to Primary Site	BCD-09.5	Mechanisms exist to plan and prepare for both natural and manmade circumstances that preclude returning to the primary site.	10		
CP-08	Telecommunications Services	Establish alternate telecommunications services, including necessary agreements to permit the resumption of [Assignment: organization-defined system operations] for essential mission and business functions within [Assignment: organization-defined time period].	Functional	Intersects With	Telecommunications Services Availability	BCD-10	Mechanisms exist to reduce the likelihood of a single point of failure with primary telecommunications services.	5		
CP-08(01)	Telecommunications Services Priority of Service Provisions	a. Develop primary and alternate telecommunications service agreements that contain priority-of-service provisions in accordance with availability requirements (including recovery time objectives); and b. Request Telecommunications Service	Functional	Equal	Telecommunications Priority of Service Provisions	BCD-10.1	Mechanisms exist to formalize primary and alternate telecommunications service agreements contain priority-of-service provisions that support	10		
CP-08(02)	Telecommunications Services Single Points of Failure	Obtain alternate telecommunications services to reduce the likelihood of sharing a single point of failure with primary telecommunications services.	Functional	Intersects With	Telecommunications Services Availability	BCD-10	Mechanisms exist to reduce the likelihood of a single point of failure with primary telecommunications services.	5		
CP-08(03)	Telecommunications Services Separation of Primary and Alternate Providers	Obtain alternate telecommunications services from providers that are separated from primary service providers to reduce susceptibility to the same threats.	Functional	Equal	Separation of Primary / Alternate Providers	BCD-10.2	Mechanisms exist to obtain alternate telecommunications services from providers that are separated from primary service providers to reduce	10		
CP-08(04)	Telecommunications Services Provider Contingency Plan	a. Require primary and alternate telecommunications service providers to have contingency plans; b. Review provider contingency plans to ensure that the plans meet organizational contingency requirements; and c. Obtain evidence of	Functional	Equal	Provider Contingency Plan	BCD-10.3	Mechanisms exist to contractually require external service providers to have contingency plans that meet organizational contingency	10		
CP-08(05)	Telecommunications Services Alternate Telecommunication Service Testing	Test alternate telecommunication services [Assignment: organization-defined frequency].	Functional	No Relationship	N/A	N/A	N/A	0		
CP-09	System Backup	a. Conduct backups of user-level information contained in [Assignment: organization-defined system components] [Assignment: organization-defined frequency consistent with recovery time and recovery point objectives]; b. Conduct	Functional	Intersects With	Data Backups	BCD-11	Mechanisms exist to create recurring backups of data, software and/or system images, as well as verify the	5		CP-09
CP-09(01)	System Backup Testing for Reliability and Integrity	Test backup information [Assignment: organization-defined frequency] to verify media reliability and information integrity.	Functional	Equal	Testing for Reliability & Integrity	BCD-11.1	Mechanisms exist to routinely test backups that verify the reliability of the backup process, as well as the integrity and availability of the data.	10		
CP-09(02)	System Backup Test Restoration Using Sampling	Use a sample of backup information in the restoration of selected system functions as part of contingency plan testing.	Functional	Equal	Test Restoration Using Sampling	BCD-11.5	Mechanisms exist to utilize sampling of available backups to test recovery capabilities as part of business continuity plan testing.	10		
CP-09(03)	System Backup Separate Storage for Critical Information	Store backup copies of [Assignment: organization-defined critical system software and other security-related information] in a separate facility or in a fire rated container that is not collocated with the operational system.	Functional	Equal	Separate Storage for Critical Information	BCD-11.2	Mechanisms exist to store backup copies of critical software and other security-related information in a separate facility or in a fire-rated	10		
CP-09(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CP-09(05)	System Backup Transfer to Alternate Storage Site	Transfer system backup information to the alternate storage site [Assignment: organization-defined time period and transfer rate consistent with the recovery time and recovery point objectives].	Functional	Equal	Transfer to Alternate Storage Site	BCD-11.6	Mechanisms exist to transfer backup data to the alternate storage site at a rate that is capable of meeting both Recovery Time Objectives (RTOs) and	10		
CP-09(06)	System Backup Redundant Secondary System	Conduct system backup by maintaining a redundant secondary system that is not collocated with the primary system and that can be activated without loss of information or disruption to operations.	Functional	Equal	Redundant Secondary System	BCD-11.7	Mechanisms exist to maintain a failover capability, which is not collocated with the primary	10		
CP-09(07)	System Backup Dual Authorization for Deletion or Destruction	Enforce dual authorization for the deletion or destruction of [Assignment: organization-defined backup information].	Functional	Equal	Dual Authorization For Backup Media Destruction	BCD-11.8	Mechanisms exist to implement and enforce dual authorization for the deletion or destruction of sensitive backup media and data.	10		
CP-09(08)	System Backup Cryptographic Protection	Implement cryptographic mechanisms to prevent unauthorized disclosure and modification of [Assignment: organization-defined backup information].	Functional	Equal	Cryptographic Protection	BCD-11.4	Cryptographic mechanisms exist to prevent the unauthorized disclosure and/or modification of backup information.	10		
CP-10	System Recovery and Reconstitution	Provide for the recovery and reconstitution of the system to a known state within [Assignment: organization-defined time period consistent with recovery time and recovery point objectives] after a disruption, compromise, or failure.	Functional	Intersects With	Technology Assets, Applications and/or Services (TAAS) Recovery &	BCD-12	Mechanisms exist to ensure the secure recovery and reconstitution of Technology Assets, Applications	5		CP-10
CP-10	System Recovery and Reconstitution	Provide for the recovery and reconstitution of the system to a known state within [Assignment: organization-defined time period consistent with recovery time and recovery point objectives] after a disruption, compromise, or failure.	Functional	Intersects With	Business Continuity Management System (BCMS)	BCD-01	Mechanisms exist to facilitate the implementation of contingency planning controls to help ensure	5		CP-10
CP-10	System Recovery and Reconstitution	Provide for the recovery and reconstitution of the system to a known state within [Assignment: organization-defined time period consistent with recovery time and recovery point objectives] after a disruption, compromise, or failure.	Functional	Intersects With	Recovery Time / Point Objectives (RTO / RPO)	BCD-01.4	Mechanisms exist to facilitate recovery operations in accordance with Recovery Time Objectives (RTOs) and Recovery Point Objectives (RPOs).	5		CP-10
CP-10(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CP-10(02)	System Recovery and Reconstitution Transaction Recovery	Implement transaction recovery for systems that are transaction-based.	Functional	Equal	Transaction Recovery	BCD-12.1	Mechanisms exist to utilize specialized backup mechanisms that will allow transaction recovery for transaction-based systems.	10		
CP-10(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CP-10(04)	System Recovery and Reconstitution Restore Within Time Period	Provide the capability to restore system components within [Assignment: organization-defined restoration time periods] from configuration-controlled and integrity-protected information representing a known operational state for the	Functional	Equal	Restore Within Time Period	BCD-12.4	Mechanisms exist to restore Technology Assets, Applications, Services and/or Data (TAASD) within organization-defined restoration time	10		
CP-10(05)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
CP-10(06)	System Recovery and Reconstitution Component Protection	Protect system components used for recovery and reconstitution.	Functional	Equal	Backup & Restoration Hardware Protection	BCD-13	Mechanisms exist to protect backup and restoration hardware and software.	10		

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CP-11	Alternate Communications Protocols	Provide the capability to employ [Assignment: organization-defined alternate communications protocols] in support of maintaining continuity of operations.	Functional	Intersects With	Telecommunications Services Availability	BCD-10	Mechanisms exist to reduce the likelihood of a single point of failure with primary telecommunications services.	5		
CP-12	Safe Mode	When [Assignment: organization-defined conditions] are detected, enter a safe mode of operation with [Assignment: organization-defined restrictions of safe mode of operation].	Functional	Intersects With	Fail Secure	SEA-07.2	Mechanisms exist to enable systems to fail to an organization-defined known-state for types of failures, preserving system state information in the process.	5		CP-12
CP-13	Alternative Security Mechanisms	Employ [Assignment: organization-defined alternative or supplemental security mechanisms] for satisfying [Assignment: organization-defined security functions] when the primary means of implementing the security function is unavailable or	Functional	Equal	Alternative Security Measures	BCD-07	Mechanisms exist to implement alternative or compensating controls to satisfy security functions when the primary means of implementing the	10		
IA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] identification and authentication policy that a	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards, and procedures, related to	5		IA-01
IA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] identification and authentication policy that a	Functional	Subset Of	Identity & Access Management (IAM)	IAC-01	Mechanisms exist to facilitate the implementation of identification and access management controls.	10		IA-01
IA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] identification and authentication policy that a	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5		IA-01
IA-02	Identification and Authentication (organizational Users)	Uniquely identify and authenticate organizational users and associate that unique identification with processes acting on behalf of those users.	Functional	Equal	Identification & Authentication for Organizational Users	IAC-02	Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) organizational users and processes acting on behalf of	10		IA-02
IA-02(01)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Multi-Factor Authentication (MFA)	IAC-06	Automated mechanisms exist to enforce Multi-Factor Authentication (MFA) for:	5		IA-02(01)
IA-02(01)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Local Access to Privileged Accounts	IAC-06.3	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate local access for privileged accounts.	5		IA-02(01)
IA-02(01)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Information Assurance Enabled Products	TDA-02.2	Mechanisms exist to limit the use of commercially-provided Information Assurance (IA) and IA-enabled IT products to those products that have	5		IA-02(01)
IA-02(01)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Out-of-Band Multi-Factor Authentication	IAC-06.4	Mechanisms exist to implement Multi-Factor Authentication (MFA) for access to privileged and non-privileged accounts such that one of	5		IA-02(01)
IA-02(01)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Network Access to Privileged Accounts	IAC-06.1	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate network access for privileged accounts.	5		IA-02(01)
IA-02(01)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Network Access to Non-Privileged Accounts	IAC-06.2	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate network access for non-privileged accounts.	5		IA-02(01)
IA-02(01)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to privileged accounts.	Functional	Intersects With	Hardware Token-Based Authentication	IAC-10.7	Automated mechanisms exist to ensure organization-defined token quality requirements are satisfied for hardware token-based authentication.	5		IA-02(01)
IA-02(02)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Information Assurance Enabled Products	TDA-02.2	Mechanisms exist to limit the use of commercially-provided Information Assurance (IA) and IA-enabled IT products to those products that have	5		IA-02(02)
IA-02(02)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Network Access to Non-Privileged Accounts	IAC-06.2	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate network access for non-privileged accounts.	5		IA-02(02)
IA-02(02)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Out-of-Band Multi-Factor Authentication	IAC-06.4	Mechanisms exist to implement Multi-Factor Authentication (MFA) for access to privileged and non-privileged accounts such that one of	5		IA-02(02)
IA-02(02)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Hardware Token-Based Authentication	IAC-10.7	Automated mechanisms exist to ensure organization-defined token quality requirements are satisfied for hardware token-based authentication.	5		IA-02(02)
IA-02(02)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Network Access to Privileged Accounts	IAC-06.1	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate network access for privileged accounts.	5		IA-02(02)
IA-02(02)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Multi-Factor Authentication (MFA)	IAC-06	Automated mechanisms exist to enforce Multi-Factor Authentication (MFA) for:	5		IA-02(02)
IA-02(02)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for access to non-privileged accounts.	Functional	Intersects With	Local Access to Privileged Accounts	IAC-06.3	Mechanisms exist to utilize Multi-Factor Authentication (MFA) to authenticate local access for privileged accounts.	5		IA-02(02)
IA-02(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
IA-02(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
IA-02(05)	Identification and Authentication (organizational Users)	When shared accounts or authenticators are employed, require users to be individually authenticated before granting access to the shared accounts or resources.	Functional	Equal	Group Authentication	IAC-02.1	Mechanisms exist to require individuals to be authenticated with an individual authenticator when a group authenticator is utilized.	10		
IA-02(06)	Identification and Authentication (organizational Users)	Implement multi-factor authentication for [Selection (one or more): local; network; remote] access to [Selection (one or more): privileged accounts; non-privileged accounts] such that a. One of the factors is provided by a device separate from	Functional	Intersects With	Out-of-Band Multi-Factor Authentication	IAC-06.4	Mechanisms exist to implement Multi-Factor Authentication (MFA) for access to privileged and non-privileged accounts such that one of	5		
IA-02(07)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
IA-02(08)	Identification and Authentication (organizational Users)	Implement replay-resistant authentication mechanisms for access to [Selection (one or more): privileged accounts; non-privileged accounts].	Functional	Equal	Replay-Resistant Authentication	IAC-02.2	Automated mechanisms exist to employ replay-resistant authentication.	10		IA-02(08)
IA-02(09)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
IA-02(10)	Identification and Authentication (organizational Users)	Provide a single sign-on capability for [Assignment: organization-defined system accounts and services].	Functional	Equal	Single Sign-On (SSO) Transparent Authentication	IAC-13.1	Mechanisms exist to provide a transparent authentication (e.g., Single Sign-On (SSO)) capability to the participating Trusted Computing	10		
IA-02(11)	Single Sign-on	Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn

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Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

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FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
IA-02(12)	Identification and Authentication (organizational Users) Acceptance of PIV Credentials	Accept and electronically verify Personal Identity Verification-compliant credentials.	Functional	Intersects With	Acceptance of PIV Credentials	IAC-02.3	Mechanisms exist to accept and electronically verify organizational Personal Identity Verification (PIV) credentials.	5		IA-02(12)
IA-02(13)	Identification and Authentication (organizational Users) Out-of-band	Implement the following out-of-band authentication mechanisms under [Assignment: organization-defined conditions]; [Assignment: organization-defined out-of-band authentication].	Functional	Equal	Out-of-Band Authentication (OOBA)	IAC-02.4	Mechanisms exist to implement Out-of-Band Authentication (OOBA) under specific conditions.	10		
IA-03	Device Identification and Authentication	Uniquely identify and authenticate [Assignment: organization-defined devices and/or types of devices] before establishing a [Selection (one or more): local; remote; network] connection.	Functional	Intersects With	Identification & Authentication for Devices	IAC-04	Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) devices before establishing a connection using	5		IA-03
IA-03(01)	Device Identification and Authentication Cryptographic Bidirectional	Authenticate [Assignment: organization-defined devices and/or types of devices] before establishing [Selection (one or more): local; remote; network] connection using bidirectional authentication that is cryptographically based.	Functional	Intersects With	Identification & Authentication for Devices	IAC-04	Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) devices before establishing a connection using	5		
IA-03(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
IA-03(03)	Device Identification and Authentication Dynamic Address Allocation	a. Where addresses are allocated dynamically, standardize dynamic address allocation lease information and the lease duration assigned to devices in accordance with [Assignment: organization-defined lease information and lease duration].	Functional	Intersects With	Network Access Control (NAC)	AST-02.5	Automated mechanisms exist to employ Network Access Control (NAC), or a similar technology, which is capable of detecting unauthorized	5		
IA-03(04)	Device Identification and Authentication Device Attestation	Handle device identification and authentication based on attestation by [Assignment: organization-defined configuration management process].	Functional	Intersects With	Identification & Authentication for Devices	IAC-04	Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) devices before establishing a connection using	5		
IA-03(04)	Device Identification and Authentication Device Attestation	Handle device identification and authentication based on attestation by [Assignment: organization-defined configuration management process].	Functional	Intersects With	Device Attestation	IAC-04.1	Mechanisms exist to ensure device identification and authentication is accurate by centrally-managing the joining of systems to the domain as	5		
IA-04	Identifier Management	Manage system identifiers by:a. Receiving authorization from [Assignment: organization-defined personnel or roles] to assign an individual, group, role, service, or device identifier;b. Selecting an identifier that identifies an individual, group, role	Functional	Intersects With	Authenticate, Authorize and Audit (AAA)	IAC-01.2	Mechanisms exist to strictly govern the use of Authenticate, Authorize and Audit (AAA) solutions, both on-premises and those hosted by an	5		IA-04
IA-04	Identifier Management	Manage system identifiers by:a. Receiving authorization from [Assignment: organization-defined personnel or roles] to assign an individual, group, role, service, or device identifier;b. Selecting an identifier that identifies an individual, group, role	Functional	Intersects With	Identifier Management (User Names)	IAC-09	Mechanisms exist to govern naming standards for usernames and Technology Assets, Applications and/or Services (TAAs)	5		IA-04
IA-04(01)	Identifier Management Prohibit Account Identifiers as Public Identifiers	Prohibit the use of system account identifiers that are the same as public identifiers for individual accounts.	Functional	No Relationship	N/A	N/A	N/A	0		
IA-04(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
IA-04(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
IA-04(04)	Identifier Management Identify User Status	Manage individual identifiers by uniquely identifying each individual as [Assignment: organization-defined characteristic identifying individual status].	Functional	Intersects With	Authenticate, Authorize and Audit (AAA)	IAC-01.2	Mechanisms exist to strictly govern the use of Authenticate, Authorize and Audit (AAA) solutions, both on-premises and those hosted by an	5		
IA-04(04)	Identifier Management Identify User Status	Manage individual identifiers by uniquely identifying each individual as [Assignment: organization-defined characteristic identifying individual status].	Functional	Intersects With	User Identity (ID) Management	IAC-09.1	Mechanisms exist to ensure proper user identification management for non-consumer users and administrative	5		
IA-04(04)	Identifier Management Identify User Status	Manage individual identifiers by uniquely identifying each individual as [Assignment: organization-defined characteristic identifying individual status].	Functional	Intersects With	Identity User Status	IAC-09.2	Mechanisms exist to identify contractors and other third-party users through unique username characteristics	5		
IA-04(05)	Identifier Management Dynamic Management	Manage individual identifiers dynamically in accordance with [Assignment: organization-defined dynamic identifier policy].	Functional	Intersects With	Dynamic Management	IAC-09.3	Mechanisms exist to dynamically manage usernames and system identifiers.	5		
IA-04(06)	Identifier Management Cross-organization Management	Coordinate with the following external organizations for cross-organization management of identifiers: [Assignment: organization-defined external organizations].	Functional	Equal	Cross-Organization Management	IAC-09.4	Mechanisms exist to coordinate username identifiers with external organizations for cross-organization management of identifiers	10		
IA-04(07)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
IA-04(08)	Identifier Management Pairwise Pseudonymous Identifiers	Generate pairwise pseudonymous identifiers.	Functional	Equal	Pairwise Pseudonymous Identifiers (PPID)	IAC-09.6	Mechanisms exist to generate pairwise pseudonymous identifiers with no identifying information about a data subject to discourage activity	10		
IA-04(09)	Identifier Management Attribute Maintenance and Protection	Maintain the attributes for each uniquely identified individual, device, or service in [Assignment: organization-defined protected central storage].	Functional	No Relationship	N/A	N/A	N/A	0		
IA-05	Authenticator Management	Manage system authenticators by:a. Verifying, as part of the initial authenticator distribution, the identity of the individual, group, role, service, or device receiving the authenticator;b. Establishing initial authenticator content for any authenticators	Functional	Intersects With	Authenticator Management	IAC-10	Mechanisms exist to: (1) Securely manage authenticators for users and devices; and (2) Enforce the strength of	5		IA-05
IA-05	Authenticator Management	Manage system authenticators by:a. Verifying, as part of the initial authenticator distribution, the identity of the individual, group, role, service, or device receiving the authenticator;b. Establishing initial authenticator content for any authenticators	Functional	Intersects With	Default Authenticators	IAC-10.8	Mechanisms exist to ensure default authenticators are changed as part of account creation or system installation	5		IA-05
IA-05(01)	Authenticator Management Password-based Authentication	For password-based authentication:a. Maintain a list of commonly-used, expected, or compromised passwords and update the list [Assignment: organization-defined frequency] and when organizational passwords are suspected to have been	Functional	Intersects With	Automated Support For Password Strength	IAC-10.4	Automated mechanisms exist to determine if password authenticators are sufficiently strong enough to satisfy organization-defined password	5		IA-05(01)
IA-05(01)	Authenticator Management Password-based Authentication	For password-based authentication:a. Maintain a list of commonly-used, expected, or compromised passwords and update the list [Assignment: organization-defined frequency] and when organizational passwords are suspected to have been	Functional	Intersects With	Password-Based Authentication	IAC-10.1	Mechanisms exist to enforce complexity, length and lifespan considerations to ensure strong criteria for password-based	5		IA-05(01)
IA-05(01)	Authenticator Management Password-based Authentication	For password-based authentication:a. Maintain a list of commonly-used, expected, or compromised passwords and update the list [Assignment: organization-defined frequency] and when organizational passwords are suspected to have been	Functional	Intersects With	Authenticator Management	IAC-10	Mechanisms exist to: (1) Securely manage authenticators for users and devices; and (2) Enforce the strength of	5		IA-05(01)
IA-05(02)	Authenticator Management Public Key-based Authentication	a. For public key-based authentication:1. Enforce authorized access to the corresponding private key; and2. Map the authenticated identity to the account of the individual or group; andb. When public key infrastructure (PKI) is used:1. Validate	Functional	Equal	PKI-Based Authentication	IAC-10.2	Automated mechanisms exist to validate certificates by constructing and verifying a certification path to an accepted trust anchor including	10		
IA-05(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
IA-05(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
IA-05(05)	Authenticator Management Change Authenticators Prior to Delivery	Require developers and installers of system components to provide unique authenticators or change default authenticators prior to delivery and installation.	Functional	Intersects With	Default Authenticators	IAC-10.8	Mechanisms exist to ensure default authenticators are changed as part of account creation or system installation	5		
IA-05(06)	Authenticator Management Protection of Authenticators	Protect authenticators commensurate with the security category of the information to which use of the authenticator permits access.	Functional	Intersects With	User Responsibilities for Account Management	IAC-18	Mechanisms exist to compel users to follow accepted practices in the use of authentication mechanisms (e.g., passwords, passphrases, physical or	5		

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

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IA-05(06)	Authenticator Management Protection of Authenticators	Protect authenticators commensurate with the security category of the information to which use of the authenticator permits access.	Functional	Intersects With	Protection of Authenticators	IAC-10.5	Mechanisms exist to protect authenticators commensurate with the sensitivity of the information to which use of the authenticator permits access.	5		
IA-05(07)	Authenticator Management No Embedded Unencrypted Static Authenticators	Ensure that unencrypted static authenticators are not embedded in applications or other forms of static storage.	Functional	Equal	No Embedded Unencrypted Static Authenticators	IAC-10.6	Mechanisms exist to ensure that unencrypted, static authenticators are not embedded in applications, scripts or stored on function keys.	10		
IA-05(08)	Authenticator Management Multiple System Accounts	Implement [Assignment: organization-defined security controls] to manage the risk of compromise due to individuals having accounts on multiple systems.	Functional	Intersects With	Multiple System Accounts	IAC-10.9	Mechanisms exist to implement security safeguards to manage the risk of compromise due to individuals having accounts on multiple systems.	5		
IA-05(08)	Authenticator Management Multiple System Accounts	Implement [Assignment: organization-defined security controls] to manage the risk of compromise due to individuals having accounts on multiple systems.	Functional	Intersects With	Privileged Account Identifiers	IAC-09.5	Mechanisms exist to uniquely manage privileged accounts to identify the account as a privileged user or service.	5		
IA-05(09)	Authenticator Management Federated Credential Management	Use the following external organizations to federate credentials: [Assignment: organization-defined external organizations].	Functional	Equal	Federated Credential Management	IAC-13.2	Mechanisms exist to federate credentials to allow cross-organization authentication of individuals and devices.	10		
IA-05(10)	Authenticator Management Dynamic Credential Binding	Bind identities and authenticators dynamically using the following rules: [Assignment: organization-defined binding rules].	Functional	Intersects With	Dynamic Management	IAC-09.3	Mechanisms exist to dynamically manage usernames and system identifiers.	5		
IA-05(11)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
IA-05(12)	Authenticator Management Biometric Authentication	For biometric-based authentication, employ mechanisms that satisfy the following biometric quality requirements [Assignment: organization-defined biometric quality requirements].	Functional	Equal	Biometric Authentication	IAC-10.12	Mechanisms exist to ensure biometric-based authentication satisfies organization-defined biometric quality requirements for false positives and false negatives.	10		
IA-05(13)	Authenticator Management Expiration of Cached Authenticators	Prohibit the use of cached authenticators after [Assignment: organization-defined time period].	Functional	Equal	Expiration of Cached Authenticators	IAC-10.10	Automated mechanisms exist to prohibit the use of cached authenticators after organization-defined time period.	10		
IA-05(14)	Authenticator Management Managing Content of PKI Trust Stores	For PKI-based authentication, employ an organization-wide methodology for managing the content of PKI trust stores installed across all platforms, including networks, operating systems, browsers, and applications.	Functional	No Relationship	N/A	N/A	N/A	0		
IA-05(15)	Authenticator Management GSA-approved Products and Services	Use only General Services Administration-approved products and services for identity, credential, and access management.	Functional	No Relationship	N/A	N/A	N/A	0		
IA-05(16)	Authenticator Management In-person or Trusted External Party	Require that the issuance of [Assignment: organization-defined types of and/or specific authenticators] be conducted [Selection (one): in person; by a trusted external party] before [Assignment: organization-defined registration authority].	Functional	No Relationship	N/A	N/A	N/A	0		
IA-05(17)	Authenticator Management Presentation Attack Detection for	Employ presentation attack detection mechanisms for biometric-based authentication.	Functional	No Relationship	N/A	N/A	N/A	0		
IA-05(18)	Authenticator Management Password Managers	a. Employ [Assignment: organization-defined password managers] to generate and manage passwords; and b. Protect the passwords using [Assignment: organization-defined controls].	Functional	Equal	Password Managers	IAC-10.11	Mechanisms exist to protect and store passwords via a password manager tool.	10		
IA-06	Authentication Feedback	Observe feedback of authentication information during the authentication process to protect the information from possible exploitation and use by unauthorized individuals.	Functional	Equal	Authentication Feedback	IAC-11	Mechanisms exist to observe the feedback of authentication information during the authentication process to protect the information.	10		IA-06
IA-07	Cryptographic Module Authentication	Implement mechanisms for authentication to a cryptographic module that meet the requirements of applicable laws, executive orders, directives, policies, regulations, standards, and guidelines for such authentication.	Functional	Intersects With	Cryptographic Module Authentication	IAC-12	Mechanisms exist to ensure cryptographic modules adhere to applicable statutory, regulatory and contractual requirements for security.	5		IA-07
IA-07	Cryptographic Module Authentication	Implement mechanisms for authentication to a cryptographic module that meet the requirements of applicable laws, executive orders, directives, policies, regulations, standards, and guidelines for such authentication.	Functional	Intersects With	Automated Authentication Through Cryptographic Modules	CRY-02	Automated mechanisms exist to enable systems to authenticate to a cryptographic module.	5		IA-07
IA-08	Identification and Authentication (non-organizational Users)	Uniquely identify and authenticate non-organizational users or processes acting on behalf of non-organizational users.	Functional	Equal	Identification & Authentication for Non-Organizational Users	IAC-03	Mechanisms exist to uniquely identify and centrally Authenticate, Authorize and Audit (AAA) third-party users and processes that provide services to the organization.	10		IA-08
IA-08(01)	Identification and Authentication (non-organizational Users) Acceptance of PIV	Accept and electronically verify Personal Identity Verification-compliant credentials from other federal agencies.	Functional	Equal	Acceptance of PIV Credentials from Other Organizations	IAC-03.1	Mechanisms exist to accept and electronically verify Personal Identity Verification (PIV) credentials from third parties.	10		IA-08(01)
IA-08(02)	Identification and Authentication (non-organizational Users) Acceptance of	a. Accept only external authenticators that are NIST-compliant; and b. Document and maintain a list of accepted external authenticators.	Functional	Equal	Acceptance of Third-Party Credentials	IAC-03.2	Automated mechanisms exist to accept Federal Identity, Credential and Access Management (FICAM)-approved third-party credentials.	10		IA-08(02)
IA-08(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
IA-08(04)	Identification and Authentication (non-organizational Users) Use of Defined Profiles	Conform to the following profiles for identity management [Assignment: organization-defined identity management profiles].	Functional	Equal	Use of FICAM-Issued Profiles	IAC-03.3	Mechanisms exist to conform systems to Federal Identity, Credential and Access Management (FICAM)-issued profiles.	10		IA-08(04)
IA-08(05)	Identification and Authentication (non-organizational Users) Acceptance of PIV	Accept and verify federated or PKI credentials that meet [Assignment: organization-defined policy].	Functional	Equal	Acceptance of PIV Credentials	IAC-02.3	Mechanisms exist to accept and electronically verify organizational Personal Identity Verification (PIV) credentials.	10		
IA-08(06)	Identification and Authentication (non-organizational Users) Disassociability	Implement the following measures to disassociate user attributes or identifier assertion relationships among individuals, credential service providers, and relying parties: [Assignment: organization-defined measures].	Functional	Equal	Disassociability	IAC-03.4	Mechanisms exist to disassociate user attributes or credential assertion relationships among individuals, credential service providers and relying parties.	10		
IA-09	Service Identification and Authentication	Uniquely identify and authenticate [Assignment: organization-defined system services and applications] before establishing communications with devices, users, or other services or applications.	Functional	Equal	Identification & Authentication for Third-Party Technology Assets, Applications and Services	IAC-05	Mechanisms exist to identify and authenticate third-party Technology Assets, Applications and/or Services Assets.	10		
IA-09(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
IA-09(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
IA-10	Adaptive Authentication	Require individuals accessing the system to employ [Assignment: organization-defined supplemental authentication techniques or mechanisms] under specific [Assignment: organization-defined circumstances or situations].	Functional	Equal	Adaptive Identification & Authentication	IAC-13	Mechanisms exist to allow individuals to utilize alternative methods of authentication under specific circumstances or situations.	10		
IA-11	Re-authentication	Require users to re-authenticate when [Assignment: organization-defined circumstances or situations requiring re-authentication].	Functional	Equal	Re-Authentication	IAC-14	Mechanisms exist to force users and devices to re-authenticate according to organization-defined circumstances that necessitate re-authentication.	10		IA-11
IA-12	Identity Proofing	a. Identity proof users that require accounts for logical access to systems based on appropriate identity assurance level requirements as specified in applicable standards and guidelines; b. Resolve user identities to a unique individual.	Functional	Equal	Identity Proofing (Identity Verification)	IAC-28	Mechanisms exist to verify the identity of a user before issuing authenticators or modifying access permissions.	10		

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IA-12(01)	Identity Proofing Supervisor Authorization	Require that the registration process to receive an account for logical access includes supervisor or sponsor authorization.	Functional	Intersects With	Management Approval For New or Changed Accounts	IAC-28.1	Mechanisms exist to ensure management approvals are required for new accounts or changes in permissions to existing accounts	5		
IA-12(02)	Identity Proofing Identity Evidence	Require evidence of individual identification be presented to the registration authority.	Functional	Equal	Identity Evidence	IAC-28.2	Mechanisms exist to require evidence of individual identification to be presented to the registration authority	10		
IA-12(03)	Identity Proofing Identity Evidence Validation and Verification	Require that the presented identity evidence be validated and verified through [Assignment: organizational defined methods of validation and verification].	Functional	Equal	Identity Evidence Validation & Verification	IAC-28.3	Mechanisms exist to require that the presented identity evidence be validated and verified through organizational defined methods of	10		
IA-12(04)	Identity Proofing In-person Validation and Verification	Require that the validation and verification of identity evidence be conducted in person before a designated registration authority.	Functional	Intersects With	User Provisioning & De-Provisioning	IAC-07	Mechanisms exist to utilize a formal user registration and de-registration process that governs the assignment of access rights	5		
IA-12(04)	Identity Proofing In-person Validation and Verification	Require that the validation and verification of identity evidence be conducted in person before a designated registration authority.	Functional	Intersects With	In-Person or Trusted Third-Party Registration	IAC-10.3	Mechanisms exist to conduct in-person or trusted third-party identify verification before user accounts for third parties are created	5		
IA-12(04)	Identity Proofing In-person Validation and Verification	Require that the validation and verification of identity evidence be conducted in person before a designated registration authority.	Functional	Intersects With	In-Person Validation & Verification	IAC-28.4	Mechanisms exist to require that the validation and verification of identity evidence be conducted in person before a designated registration	5		
IA-12(05)	Identity Proofing Address Confirmation	Require that a [Selection (one): registration code; notice of proofing] be delivered through an out-of-band channel to verify the users address (physical or digital) of record.	Functional	Equal	Address Confirmation	IAC-28.5	Mechanisms exist to require that a notice of proofing be delivered through an out-of-band channel to verify the user's address (physical or	10		
IA-12(06)	Identity Proofing Accept Externally-proofed Identities	Accept externally-proofed identities at [Assignment: organization-defined identity assurance level].	Functional	No Relationship	N/A	N/A	N/A	0		
IA-13	Identity Providers and Authorization Servers	Employ identity providers and authorization servers to manage user, device, and non-person entity (NPE) identities, attributes, and access rights supporting authentication and authorization decisions in accordance with [Assignment: organization-defined	Functional	No Relationship	N/A	N/A	N/A	0		
IA-13(01)	Protection of Cryptographic Keys	Cryptographic keys that protect access tokens are generated, managed, and protected from disclosure and misuse.	Functional	No Relationship	N/A	N/A	N/A	0		
IA-13(02)	Verification of Identity and Access Tokens	The source and integrity of identity assertions and access tokens are verified before granting access to system and information resources.	Functional	No Relationship	N/A	N/A	N/A	0		
IA-13(03)	Token Management	In accordance with [Assignment: organization-defined identification and authentication policy], assertions and access tokens are:a. generated;b. issued; c. refreshed;d. revoked;e. time-restricted; andf. audience-restricted.	Functional	No Relationship	N/A	N/A	N/A	0		
IR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level]incident response policy that:a. Addresses	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5		IR-01
IR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level]incident response policy that:a. Addresses	Functional	Subset Of	Incident Response Operations	IRO-01	Mechanisms exist to implement and govern processes and documentation to facilitate an organization-wide response capability for cybersecurity	10		IR-01
IR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level]incident response policy that:a. Addresses	Functional	Intersects With	IRP Update	IRO-04.2	Mechanisms exist to regularly review and modify incident response practices to incorporate lessons learned, business process changes	5		IR-01
IR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level]incident response policy that:a. Addresses	Functional	Intersects With	Root Cause Analysis (RCA) & Lessons Learned	IRO-13	Mechanisms exist to incorporate lessons learned from analyzing and resolving cybersecurity and data protection incidents to reduce the	5		IR-01
IR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level]incident response policy that:a. Addresses	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRP), including policies,	5		IR-01
IR-02	Incident Response Training	a. Provide incident response training to system users consistent with assigned roles and responsibilities:1. Within [Assignment: organization-defined time period] of assuming an incident response role or responsibility or acquiring system access;2	Functional	Intersects With	Incident Response Training	IRO-05	Mechanisms exist to train personnel in their incident response roles and responsibilities.	5		IR-02
IR-02(01)	Incident Response Training Simulated Events	Incorporate simulated events into incident response training to facilitate the required response by personnel in crisis situations.	Functional	Equal	Simulated Incidents	IRO-05.1	Mechanisms exist to incorporate simulated events into incident response training to facilitate effective response by personnel in crisis	10		
IR-02(02)	Incident Response Training Automated Training Environments	Provide an incident response training environment using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Automated Incident Response Training Environments	IRO-05.2	Automated mechanisms exist to provide a more thorough and realistic incident response training environment	10		
IR-02(03)	Incident Response Training Breach	Provide incident response training on how to identify and respond to a breach, including the organization's process for reporting a breach.	Functional	Intersects With	Incident Response Training	IRO-05	Mechanisms exist to train personnel in their incident response roles and responsibilities.	5		
IR-03	Incident Response Testing	Test the effectiveness of the incident response capability for the system [Assignment: organization-defined frequency] using the following tests: [Assignment: organization-defined tests].	Functional	Intersects With	Incident Response Testing	IRO-06	Mechanisms exist to formally test incident response capabilities through realistic exercises to determine the operational effectiveness of those	5		
IR-03(01)	Incident Response Testing Automated Testing	Test the incident response capability using [Assignment: organization-defined automated mechanisms].	Functional	No Relationship	N/A	N/A	N/A	0		
IR-03(02)	Incident Response Testing Coordination with Related Plans	Coordinate incident response testing with organizational elements responsible for related plans.	Functional	Equal	Coordination with Related Plans	IRO-06.1	Mechanisms exist to coordinate incident response testing with organizational elements responsible for related plans	10		
IR-03(03)	Incident Response Testing Continuous Improvement	Use qualitative and quantitative data from testing to:a. Determine the effectiveness of incident response processes;b. Continuously improve incident response processes; andc. Provide incident response measures and metrics that are	Functional	Equal	Continuous Incident Response Improvements	IRO-04.3	Mechanisms exist to use qualitative and quantitative data from incident response testing to: (1) Determine the effectiveness of	10		
IR-04	Incident Handling	a. Implement an incident handling capability for incidents that is consistent with the incident response plan and includes preparation, detection and analysis, containment, eradication, and recovery.b. Coordinate incident handling activities with	Functional	Equal	Incident Handling	IRO-02	Mechanisms exist to cover: (1) Preparation; (2) Automated event detection or manual incident report intake	10		IR-04
IR-04(01)	Incident Handling Automated Incident Handling Processes	Support the incident handling process using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Automated Incident Handling Processes	IRO-02.1	Automated mechanisms exist to support the incident handling process.	10		

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FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
IR-04(02)	Incident Handling Dynamic Reconfiguration	Include the following types of dynamic reconfiguration for [Assignment: organization-defined system components] as part of the incident response capability: [Assignment: organization-defined types of dynamic reconfiguration]	Functional	Equal	Dynamic Reconfiguration	IRO-02.3	Automated mechanisms exist to dynamically reconfigure system components as part of the incident response capability.	10		
IR-04(03)	Incident Handling Continuity of Operations	Identify [Assignment: organization-defined classes of incidents] and take the following actions in response to those incidents to ensure continuation of organizational mission and business functions: [Assignment: organization-defined actions to take in response to incidents]	Functional	Intersects With	Business Continuity Management System (BCMS)	BCD-01	Mechanisms exist to facilitate the implementation of contingency planning controls to help ensure the continuation of organizational missions and business functions.	5		
IR-04(03)	Incident Handling Continuity of Operations	Identify [Assignment: organization-defined classes of incidents] and take the following actions in response to those incidents to ensure continuation of organizational mission and business functions: [Assignment: organization-defined actions to take in response to incidents]	Functional	Intersects With	Incident Classification & Prioritization	IRO-02.4	Mechanisms exist to identify classes of incidents and actions to take to ensure the continuation of organizational missions and business functions.	5		
IR-04(04)	Incident Handling Information Correlation	Correlate incident information and individual incident responses to achieve an organization-wide perspective on incident awareness and response.	Functional	Intersects With	Centralized Collection of Security Event Logs	MON-02	Mechanisms exist to utilize a Security Incident Event Manager (SIEM), or similar automated tool, to support the centralized collection of security event logs.	5		
IR-04(04)	Incident Handling Information Correlation	Correlate incident information and individual incident responses to achieve an organization-wide perspective on incident awareness and response.	Functional	Intersects With	Correlate Monitoring Information	MON-02.1	Automated mechanisms exist to correlate both technical and non-technical information from across the enterprise by a Security Incident Event Manager (SIEM).	5		
IR-04(05)	Incident Handling Automatic Disabling of System	Implement a configurable capability to automatically disable the system if [Assignment: organization-defined security violations] are detected.	Functional	Intersects With	Automated Response to Suspicious Events	MON-01.11	Automated mechanisms exist to implement pre-determined corrective actions in response to detected security violations.	5		
IR-04(05)	Incident Handling Automatic Disabling of System	Implement a configurable capability to automatically disable the system if [Assignment: organization-defined security violations] are detected.	Functional	Intersects With	Automatic Disabling of Technology Assets, Applications and/or Services (TAAS)	IRO-02.6	Mechanisms exist to automatically disable Technology Assets, Applications and/or Services (TAAS), in response to detected security violations.	5		
IR-04(06)	Incident Handling Insider Threats	Implement an incident handling capability for incidents involving insider threats.	Functional	Intersects With	Insider Threat Response Capability	IRO-02.2	Mechanisms exist to implement and govern an insider threat program.	5		
IR-04(07)	Incident Handling Insider Threats — Intra organization Coordination	Coordinate an incident handling capability for insider threats that includes the following organizational entities [Assignment: organization-defined entities].	Functional	Intersects With	Insider Threat Response Capability	IRO-02.2	Mechanisms exist to implement and govern an insider threat program.	5		
IR-04(08)	Incident Handling Correlation with External Organizations	Coordinate with [Assignment: organization-defined external organizations] to correlate and share [Assignment: organization-defined incident information] to achieve a cross-organization perspective on incident awareness and more effective incident response.	Functional	Equal	Correlation with External Organizations	IRO-02.5	Mechanisms exist to coordinate with approved third-parties to achieve a cross-organization perspective on incident awareness and more effective incident response.	10		
IR-04(09)	Incident Handling Dynamic Response Capability	Employ [Assignment: organization-defined dynamic response capabilities] to respond to incidents.	Functional	No Relationship	N/A	N/A	N/A	0		
IR-04(10)	Incident Handling Supply Chain Coordination	Coordinate incident handling activities involving supply chain events with other organizations involved in the supply chain.	Functional	Intersects With	Third-Party Incident Response & Recovery Capabilities	TPM-11	Mechanisms exist to ensure response/recovery planning and testing are conducted with critical suppliers/providers.	5		
IR-04(10)	Incident Handling Supply Chain Coordination	Coordinate incident handling activities involving supply chain events with other organizations involved in the supply chain.	Functional	Intersects With	Supply Chain Coordination	IRO-10.4	Mechanisms exist to provide cybersecurity and data protection incident information to the provider of the supply chain.	5		
IR-04(11)	Incident Handling Integrated Incident Response Team	Establish and maintain an integrated incident response team that can be deployed to any location identified by the organization in [Assignment: organization-defined time period].	Functional	Equal	Integrated Security Incident Response Team (ISIRT)	IRO-07	Mechanisms exist to establish an integrated team of cybersecurity, IT and business function representatives that are capable of addressing incidents.	10		
IR-04(12)	Incident Handling Malicious Code and Forensic Analysis	Analyze malicious code and/or other residual artifacts remaining in the system after the incident.	Functional	Intersects With	Root Cause Analysis (RCA) & Lessons Learned	IRO-13	Mechanisms exist to incorporate lessons learned from analyzing and resolving cybersecurity and data protection incidents to reduce the risk of future incidents.	5		
IR-04(12)	Incident Handling Malicious Code and Forensic Analysis	Analyze malicious code and/or other residual artifacts remaining in the system after the incident.	Functional	Intersects With	Chain of Custody & Forensics	IRO-08	Mechanisms exist to perform digital forensics and maintain the integrity of the chain of custody, in accordance with applicable laws, regulations and policies.	5		
IR-04(13)	Incident Handling Behavior Analysis	Analyze anomalous or suspected adversarial behavior in or related to [Assignment: organization-defined environments or resources].	Functional	Intersects With	Honeypots	SEA-11	Mechanisms exist to utilize honeypots that are specifically designed to be the target of malicious attacks for the purpose of detecting, deflecting and analyzing threats.	5		
IR-04(13)	Incident Handling Behavior Analysis	Analyze anomalous or suspected adversarial behavior in or related to [Assignment: organization-defined environments or resources].	Functional	Intersects With	Anomalous Behavior	MON-16	Mechanisms exist to utilize User & Entity Behavior Analytics (UEBA) and/or User Activity Monitoring (UAM) to detect anomalous behavior.	5		
IR-04(13)	Incident Handling Behavior Analysis	Analyze anomalous or suspected adversarial behavior in or related to [Assignment: organization-defined environments or resources].	Functional	Intersects With	Honeyclients	SEA-12	Mechanisms exist to utilize honeyclients that proactively seek to identify malicious websites and/or web-based malicious code.	5		
IR-04(14)	Incident Handling Security Operations Center	Establish and maintain a security operations center.	Functional	Equal	Security Operations Center (SOC)	OPS-04	Mechanisms exist to establish and maintain a Security Operations Center (SOC) that facilitates a 24x7 response capability.	10		
IR-04(15)	Incident Handling Public Relations and Reputation Repair	a. Manage public relations associated with an incident; and b. Employ measures to repair the reputation of the organization.	Functional	Equal	Public Relations & Reputation Repair	IRO-16	Mechanisms exist to proactively manage public relations associated with incidents and employ appropriate measures to respond further.	10		
IR-05	Incident Monitoring	Track and document incidents.	Functional	Equal	Situational Awareness For Incidents	IRO-09	Mechanisms exist to document, monitor and report the status of cybersecurity and data protection incidents to internal stakeholders all	10		IR-05
IR-05(01)	Incident Monitoring Automated Tracking, Data Collection, and Analysis	Track incidents and collect and analyze incident information using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Automated Tracking, Data Collection & Analysis	IRO-09.1	Automated mechanisms exist to assist in the tracking, collection and analysis of information from actual and potential cybersecurity and data protection incidents.	10		
IR-06	Incident Reporting	a. Require personnel to report suspected incidents to the organizational incident response capability within [Assignment: organization-defined time period]; and b. Report incident information to [Assignment: organization-defined authorities]	Functional	Intersects With	Incident Stakeholder Reporting	IRO-10	Mechanisms exist to timely-report incidents to applicable: (1) Internal stakeholders; (2) Affected clients & third-parties; and (3) Applicable regulatory and law enforcement agencies.	5		IR-06
IR-06	Incident Reporting	a. Require personnel to report suspected incidents to the organizational incident response capability within [Assignment: organization-defined time period]; and b. Report incident information to [Assignment: organization-defined authorities]	Functional	Intersects With	Regulatory & Law Enforcement Contacts	IRO-14	Mechanisms exist to maintain incident response contacts with applicable regulatory and law enforcement agencies.	5		IR-06
IR-06	Incident Reporting	a. Require personnel to report suspected incidents to the organizational incident response capability within [Assignment: organization-defined time period]; and b. Report incident information to [Assignment: organization-defined authorities]	Functional	Intersects With	Contacts With Authorities	GOV-06	Mechanisms exist to identify and document appropriate contacts with relevant law enforcement and regulatory bodies.	5		IR-06
IR-06(01)	Incident Reporting Automated Reporting	Report incidents using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Automated Reporting	IRO-10.1	Automated mechanisms exist to assist in the reporting of cybersecurity and data protection incidents.	10		

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

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IR-06(02)	Incident Reporting Vulnerabilities Related to Incidents	Report system vulnerabilities associated with reported incidents to [Assignment: organization-defined personnel or roles].	Functional	Intersects With	Root Cause Analysis (RCA) & Lessons Learned	IRO-13	Mechanisms exist to incorporate lessons learned from analyzing and resolving cybersecurity and data protection incidents to reduce the	5		
IR-06(02)	Incident Reporting Vulnerabilities Related to Incidents	Report system vulnerabilities associated with reported incidents to [Assignment: organization-defined personnel or roles].	Functional	Intersects With	Vulnerabilities Related To Incidents	IRO-10.3	Mechanisms exist to report system vulnerabilities associated with reported cybersecurity and data protection incidents to organization	5		
IR-06(03)	Incident Reporting Supply Chain Coordination	Provide incident information to the provider of the product or service and other organizations involved in the supply chain or supply chain governance for systems or system components related to the incident	Functional	Intersects With	Supply Chain Coordination	IRO-10.4	Mechanisms exist to provide cybersecurity and data protection incident information to the provider of the Technology Assets, Assets, and/or	5		
IR-07	Incident Response Assistance	Provide an incident response support resource, integral to the organizational incident response capability, that offers advice and assistance to users of the system for the handling and reporting of incidents.	Functional	Equal	Incident Reporting Assistance	IRO-11	Mechanisms exist to provide incident response advice and assistance to users of Technology Assets, Assets, and/or	10		IR-07
IR-07(01)	Incident Response Assistance Automation Support for Availability of	Increase the availability of incident response information and support using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Automation Support of Availability of Information / Support	IRO-11.1	Automated mechanisms exist to increase the availability of incident response-related information and support	10		
IR-07(02)	Incident Response Assistance Coordination With External Providers	a. Establish a direct, cooperative relationship between its incident response capability and external providers of system protection capability; andb. Identify organizational incident response team members to the external provider	Functional	Equal	Coordination With External Providers	IRO-11.2	Mechanisms exist to establish a direct, cooperative relationship between the organization's incident response capability and external	10		
IR-08	Incident Response Plan	a. Develop an incident response plan that:1. Provides the organization with a roadmap for implementing its incident response capability;2. Describes the structure and organization of the incident response capability;3. Provides a high-level include the following in the Incident Response Plan for breaches involving personally identifiable information:a. A process to determine if notice to individuals or other organizations, including oversight organizations, is needed;b. An assessment Respond to information spills by:a. Assigning [Assignment: organization-defined personnel or roles] with responsibility for responding to information spills;b. Identifying the specific information involved in the system contamination;c. Alerting Respond to information spills by:a. Assigning [Assignment: organization-defined personnel or roles] with responsibility for responding to information spills;b. Identifying the specific information involved in the system contamination;c. Alerting	Functional	Equal	Incident Response Plan (IRP)	IRO-04	Mechanisms exist to maintain and make available a current and viable Incident Response Plan (IRP) to all stakeholders	10		IR-08
IR-08(01)	Incident Response Plan Breaches	Include the following in the Incident Response Plan for breaches involving personally identifiable information:a. A process to determine if notice to individuals or other organizations, including oversight organizations, is needed;b. An assessment Respond to information spills by:a. Assigning [Assignment: organization-defined personnel or roles] with responsibility for responding to information spills;b. Identifying the specific information involved in the system contamination;c. Alerting	Functional	Equal	Data Breach	IRO-04.1	Mechanisms exist to address data breaches, or other incidents involving the unauthorized disclosure of sensitive or regulated data according	10		
IR-09	Information Spillage Response	Respond to information spills by:a. Assigning [Assignment: organization-defined personnel or roles] with responsibility for responding to information spills;b. Identifying the specific information involved in the system contamination;c. Alerting	Functional	Intersects With	Sensitive / Regulated Data Spill Response	IRO-12	Mechanisms exist to respond to sensitive and/or regulated data spills.	5		
IR-09	Information Spillage Response	Respond to information spills by:a. Assigning [Assignment: organization-defined personnel or roles] with responsibility for responding to information spills;b. Identifying the specific information involved in the system contamination;c. Alerting	Functional	Intersects With	Sensitive / Regulated Data Spill Responsible Personnel	IRO-12.1	Mechanisms exist to formally assign personnel or roles with responsibility for responding to sensitive and/or	5		
IR-09(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
IR-09(02)	Information Spillage Response Training	Provide information spillage response training [Assignment: organization-defined frequency].	Functional	Equal	Sensitive / Regulated Data Spill Training	IRO-12.2	Mechanisms exist to ensure incident response training material provides coverage for sensitive and/or	10		
IR-09(03)	Information Spillage Response Post-spill Operations	Implement the following procedures to ensure that organizational personnel impacted by information spills can continue to carry out assigned tasks while contaminated systems are undergoing corrective actions: [Assignment: organization-defined corrective actions]	Functional	Equal	Post-Sensitive / Regulated Data Spill Operations	IRO-12.3	Mechanisms exist to ensure that organizational personnel impacted by sensitive and/or regulated data spills	10		
IR-09(04)	Information Spillage Response Exposure to Unauthorized Personnel	Employ the following controls for personnel exposed to information not within assigned access authorizations: [Assignment: organization-defined controls].	Functional	Equal	Sensitive / Regulated Data Exposure to Unauthorized Personnel	IRO-12.4	Mechanisms exist to address security safeguards for personnel exposed to sensitive and/or regulated data that is	10		
IR-10		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
MA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] maintenance policy that:a. Addresses purpose	Functional	Subset Of	Maintenance Operations	MNT-01	Mechanisms exist to develop, disseminate, review & update procedures to facilitate the implementation of maintenance	10		MA-01
MA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] maintenance policy that:a. Addresses purpose	Functional	Intersects With	Remote Maintenance Notifications	MNT-05.2	Mechanisms exist to require maintenance personnel to notify affected stakeholders when remote, non-local maintenance is planned	5		MA-01
MA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] maintenance policy that:a. Addresses purpose	Functional	Intersects With	Auditing Remote Maintenance	MNT-05.1	Mechanisms exist to audit remote, non-local maintenance and diagnostic sessions, as well as review the maintenance action performed during	5		MA-01
MA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] maintenance policy that:a. Addresses purpose	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCR), including policies, standards, and procedures necessary	5		MA-01
MA-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] maintenance policy that:a. Addresses purpose	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5		MA-01
MA-02	Controlled Maintenance	a. Schedule, document, and review records of maintenance, repair, and replacement on system components in accordance with manufacturer or vendor specifications and/or organizational requirements;b. Approve and monitor all	Functional	Equal	Controlled Maintenance	MNT-02	Mechanisms exist to conduct controlled maintenance activities throughout the lifecycle of the Technology Assets, Assets, and/or	10		MA-02
MA-02(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
MA-02(02)	Controlled Maintenance Automated Maintenance Activities	a. Schedule, conduct, and document maintenance, repair, and replacement actions for the system using [Assignment: organization-defined automated mechanisms]; andb. Produce up-to-date, accurate, and complete records of all maintenance	Functional	Equal	Automated Maintenance Activities	MNT-02.1	Automated mechanisms exist to schedule, conduct and document maintenance and repairs.	10		
MA-03	Maintenance Tools	a. Approve, control, and monitor the use of system maintenance tools; andb. Review previously approved system maintenance tools [Assignment: organization-defined frequency].	Functional	Intersects With	Maintenance Tools	MNT-04	Mechanisms exist to control and monitor the use of system maintenance tools.	5		
MA-03(01)	Maintenance Tools Inspect Tools	Inspect the maintenance tools used by maintenance personnel for improper or unauthorized modifications.	Functional	Equal	Inspect Tools	MNT-04.1	Mechanisms exist to inspect maintenance tools carried into a facility by maintenance personnel for improper or unauthorized	10		
MA-03(02)	Maintenance Tools Inspect Media	Check media containing diagnostic and test programs for malicious code before the media are used in the system.	Functional	Equal	Inspect Media	MNT-04.2	Mechanisms exist to check media containing diagnostic and test programs for malicious code before the media are used	10		
MA-03(03)	Maintenance Tools Prevent Unauthorized Removal	Prevent the removal of maintenance equipment containing organizational information by:a. Verifying that there is no organizational information contained on the equipment;b. Sanitizing or destroying the equipment;c. Retaining the	Functional	Equal	Prevent Unauthorized Removal	MNT-04.3	Mechanisms exist to prevent or control the removal of equipment undergoing maintenance that contains	10		
MA-03(04)	Maintenance Tools Restricted Tool Use	Restrict the use of maintenance tools to authorized personnel only.	Functional	Equal	Restrict Tool Usage	MNT-04.4	Automated mechanisms exist to restrict the use of maintenance tools to authorized maintenance personnel and/or roles	10		
MA-03(05)	Maintenance Tools Execution With Privilege	Monitor the use of maintenance tools that execute with increased privilege.	Functional	Intersects With	Maintenance Tools	MNT-04	Mechanisms exist to control and monitor the use of system maintenance tools.	5		

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MA-03(06)	Maintenance Tools Software Updates and Patches	Inspect maintenance tools to ensure the latest software updates and patches are installed.	Functional	Intersects With	Maintenance Tools	MNT-04	Mechanisms exist to control and monitor the use of system maintenance tools.	5		
MA-04	Nonlocal Maintenance	a. Approve and monitor nonlocal maintenance and diagnostic activities;b. Allow the use of nonlocal maintenance and diagnostic tools only as consistent with organizational policy and documented in the security plan for the system;c. Employ	Functional	Intersects With	Remote Maintenance	MNT-05	Mechanisms exist to authorize, monitor and control remote, non-local maintenance and diagnostic activities.	5		MA-04
MA-04	Nonlocal Maintenance	a. Approve and monitor nonlocal maintenance and diagnostic activities;b. Allow the use of nonlocal maintenance and diagnostic tools only as consistent with organizational policy and documented in the security plan for the system;c. Employ	Functional	Intersects With	Remote Maintenance Notifications	MNT-05.2	Mechanisms exist to require maintenance personnel to notify affected stakeholders when remote, non-local maintenance is planned.	5		MA-04
MA-04	Nonlocal Maintenance	a. Approve and monitor nonlocal maintenance and diagnostic activities;b. Allow the use of nonlocal maintenance and diagnostic tools only as consistent with organizational policy and documented in the security plan for the system;c. Employ	Functional	Intersects With	Auditing Remote Maintenance	MNT-05.1	Mechanisms exist to audit remote, non-local maintenance and diagnostic sessions, as well as review the maintenance action performed during	5		MA-04
MA-04(01)	Nonlocal Maintenance Logging and Review	a. Log [Assignment: organization-defined audit events] for nonlocal maintenance and diagnostic sessions; andb. Review the audit records of the maintenance and diagnostic sessions to detect anomalous behavior	Functional	Intersects With	Auditing Remote Maintenance	MNT-05.1	Mechanisms exist to audit remote, non-local maintenance and diagnostic sessions, as well as review the maintenance action performed during	5		
MA-04(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
MA-04(03)	Nonlocal Maintenance Comparable Security & Sanitization	a. Require that nonlocal maintenance and diagnostic services be performed from a system that implements a security capability comparable to the capability implemented on the system being serviced; orb. Remove the component to be	Functional	Equal	Remote Maintenance Comparable Security & Sanitization	MNT-05.6	Mechanisms exist to require Technology Assets, Applications and/or Services (TAAS) performing	10		
MA-04(04)	Nonlocal Maintenance Authentication and Separation of Maintenance Sessions	Protect nonlocal maintenance sessions by:a. Employing [Assignment: organization-defined authenticators that are replay resistant]; andb. Separating the maintenance sessions from other network sessions with the system by either:1.	Functional	Equal	Separation of Maintenance Sessions	MNT-05.7	Mechanisms exist to protect maintenance sessions through replay-resistant sessions that are physically or logically separated.	10		
MA-04(05)	Nonlocal Maintenance Approvals and Notifications	a. Require the approval of each nonlocal maintenance session by [Assignment: organization-defined personnel or roles]; andb. Notify the following personnel or roles of the date and time of planned nonlocal maintenance: [Assignment: organization-	Functional	Equal	Remote Maintenance Pre-Approval	MNT-05.5	Mechanisms exist to require maintenance personnel to obtain pre-approval and scheduling for remote, non-local maintenance sessions.	10		
MA-04(06)	Nonlocal Maintenance Cryptographic Protection	Implement the following cryptographic mechanisms to protect the integrity and confidentiality of nonlocal maintenance and diagnostic communications: [Assignment: organization-defined cryptographic mechanisms]	Functional	Equal	Remote Maintenance Cryptographic Protection	MNT-05.3	Cryptographic mechanisms exist to protect the integrity and confidentiality of remote, non-local maintenance and diagnostic	10		
MA-04(07)	Nonlocal Maintenance Disconnect Verification	Verify session and network connection termination after the completion of nonlocal maintenance and diagnostic sessions.	Functional	Equal	Remote Maintenance Disconnect Verification	MNT-05.4	Mechanisms exist to provide remote disconnect verification to ensure remote, non-local maintenance and diagnostic sessions are properly	10		
MA-05	Maintenance Personnel	a. Establish a process for maintenance personnel authorization and maintain a list of authorized maintenance organizations or personnel;b. Verify that non-escorted personnel performing maintenance on the system possess the required access	Functional	Equal	Authorized Maintenance Personnel	MNT-06	Mechanisms exist to maintain a current list of authorized maintenance organizations or personnel.	10		MA-05
MA-05(01)	Maintenance Personnel Individuals Without Appropriate Access	a. Implement procedures for the use of maintenance personnel that lack appropriate security clearances or are not U.S. citizens, that include the following requirements:1. Maintenance personnel who do not have needed access authorizations	Functional	Intersects With	Maintenance Personnel Without Appropriate Access	MNT-06.1	Mechanisms exist to ensure the risks associated with maintenance personnel who do not have appropriate access authorizations	5		
MA-05(02)	Maintenance Personnel Security Clearances for Classified Systems Maintenance	Verify that personnel performing maintenance and diagnostic activities on a system processing, storing, or transmitting classified information possess security clearances and formal access approvals for at least the highest classification level and	Functional	Intersects With	Maintenance Personnel Without Appropriate Access	MNT-06.1	Mechanisms exist to ensure the risks associated with maintenance personnel who do not have appropriate access authorizations	5		
MA-05(03)	Maintenance Personnel Citizenship Requirements for Classified Systems	Verify that personnel performing maintenance and diagnostic activities on a system processing, storing, or transmitting classified information are U.S. citizens.	Functional	Intersects With	Maintenance Personnel Without Appropriate Access	MNT-06.1	Mechanisms exist to ensure the risks associated with maintenance personnel who do not have appropriate access authorizations	5		
MA-05(04)	Maintenance Personnel Foreign Nationals	Ensure that:a. Foreign nationals with appropriate security clearances are used to conduct maintenance and diagnostic activities on classified systems only when the systems are jointly owned and operated by the United States and foreign	Functional	Intersects With	Maintenance Personnel Without Appropriate Access	MNT-06.1	Mechanisms exist to ensure the risks associated with maintenance personnel who do not have appropriate access authorizations	5		
MA-05(05)	Maintenance Personnel Non-system Maintenance	Ensure that non-escorted personnel performing maintenance activities not directly associated with the system but in the physical proximity of the system, have required access authorizations	Functional	Equal	Non-System Related Maintenance	MNT-06.2	Mechanisms exist to ensure that non-escorted personnel performing non-IT maintenance activities in the physical	10		
MA-06	Timely Maintenance	Obtain maintenance support and/or spare parts for [Assignment: organization-defined system components] within [Assignment: organization-defined time period] of failure.	Functional	Equal	Timely Maintenance	MNT-03	Mechanisms exist to obtain maintenance support and/or spare parts for Technology Assets,	10		
MA-06(01)	Timely Maintenance Preventive Maintenance	Perform preventive maintenance on [Assignment: organization-defined system components] at [Assignment: organization-defined time intervals].	Functional	Equal	Preventative Maintenance	MNT-03.1	Mechanisms exist to perform preventive maintenance on critical Technology Assets, Applications	10		
MA-06(02)	Timely Maintenance Predictive Maintenance	Perform predictive maintenance on [Assignment: organization-defined system components] at [Assignment: organization-defined time intervals].	Functional	Equal	Predictive Maintenance	MNT-03.2	Mechanisms exist to perform predictive maintenance on critical Technology Assets, Applications	10		
MA-06(03)	Timely Maintenance Automated Support for Predictive Maintenance	Transfer predictive maintenance data to a maintenance management system using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Automated Support For Predictive Maintenance	MNT-03.3	Automated mechanisms exist to transfer predictive maintenance data to a computerized maintenance management system	10		
MA-07	Field Maintenance	Restrict or prohibit field maintenance on [Assignment: organization-defined systems or system components] to [Assignment: organization-defined trusted maintenance facilities]	Functional	Equal	Field Maintenance	MNT-08	Mechanisms exist to securely conduct field maintenance on geographically deployed assets.	10		MA-07
MP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] media protection policy that:a. Addresses	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures, aligned	5		MP-01
MP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] media protection policy that:a. Addresses	Functional	Subset Of	Data Protection	DCH-01	Mechanisms exist to facilitate the implementation of data protection controls.	10		MP-01
MP-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] media protection policy that:a. Addresses	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5		MP-01
MP-02	Media Access	Restrict access to [Assignment: organization-defined types of digital and/or non-digital media] to [Assignment: organization-defined personnel or roles].	Functional	Intersects With	Media Access	DCH-03	Mechanisms exist to control and restrict access to digital and non-digital media to authorized individuals.	5		MP-02
MP-02	Media Access	Restrict access to [Assignment: organization-defined types of digital and/or non-digital media] to [Assignment: organization-defined personnel or roles].	Functional	Intersects With	Endpoint Device Management (EDM)	END-01	Mechanisms exist to facilitate the implementation of Endpoint Device Management (EDM) controls.	5		MP-02
MP-02(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
MP-02(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

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FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
MP-03	Media Marking	a. Mark system media indicating the distribution limitations, handling caveats, and applicable security markings (if any) of the information; andb. Exempt [Assignment: organization-defined types of system media] from marking if the media	Functional	Intersects With	Media Marking	DCH-04	Mechanisms exist to mark media in accordance with data protection requirements so that personnel are alerted to distribution limitations	5		
MP-03	Media Marking	a. Mark system media indicating the distribution limitations, handling caveats, and applicable security markings (if any) of the information; andb. Exempt [Assignment: organization-defined types of system media] from marking if the media	Functional	Intersects With	Automated Marking	DCH-04.1	Automated mechanisms exist to mark physical media and digital files to indicate the distribution limitations, handling requirements and applicable	5		
MP-04	Media Storage	a. Physically control and securely store [Assignment: organization-defined types of digital and/or non-digital media] within [Assignment: organization-defined controlled areas]; andb. Protect system media types defined in MP-04a until the	Functional	Equal	Media Storage	DCH-06	Mechanisms exist to: (1) Physically control and securely store digital and non-digital media within controlled areas using	10		
MP-04(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
MP-04(02)	Media Storage Automated Restricted Access	Restrict access to media storage areas and log access attempts and access granted using [Assignment: organization-defined automated mechanisms].	Functional	No Relationship	N/A	N/A	N/A	0		
MP-05	Media Transport	a. Protect and control [Assignment: organization-defined types of system media] during transport outside of controlled areas using [Assignment: organization-defined controls];b. Maintain accountability for system media during transport outside of	Functional	Equal	Media Transportation	DCH-07	Mechanisms exist to protect and control digital and non-digital media during transport outside of controlled areas using appropriate security	10		
MP-05(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
MP-05(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
MP-05(03)	Media Transport Custodians	Employ an identified custodian during transport of system media outside of controlled areas.	Functional	Equal	Custodians	DCH-07.1	Mechanisms exist to identify custodians throughout the transport of digital or non-digital media.	10		
MP-05(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
MP-06	Media Sanitization	a. Sanitize [Assignment: organization-defined system media] prior to disposal, release out of organizational control, or release for reuse using [Assignment: organization-defined sanitization techniques and procedures]; andb. Employ	Functional	Intersects With	Physical Media Disposal	DCH-08	Mechanisms exist to securely dispose of media when it is no longer required, using formal procedures.	5		MP-06
MP-06	Media Sanitization	a. Sanitize [Assignment: organization-defined system media] prior to disposal, release out of organizational control, or release for reuse using [Assignment: organization-defined sanitization techniques and procedures]; andb. Employ	Functional	Intersects With	System Media Sanitization	DCH-09	Mechanisms exist to sanitize system media with the strength and integrity commensurate with the classification or sensitivity of the information prior	5		MP-06
MP-06	Media Sanitization	a. Sanitize [Assignment: organization-defined system media] prior to disposal, release out of organizational control, or release for reuse using [Assignment: organization-defined sanitization techniques and procedures]; andb. Employ	Functional	Intersects With	Sanitization of Personal Data (PD)	DCH-09.3	Mechanisms exist to facilitate the sanitization of Personal Data (PD).	5		MP-06
MP-06(01)	Media Sanitization Review, Approve, Track, Document, and Verify	Review, approve, track, document, and verify media sanitization and disposal actions.	Functional	Equal	System Media Sanitization Documentation	DCH-09.1	Mechanisms exist to supervise, track, document and verify system media sanitization and disposal actions.	10		
MP-06(02)	Media Sanitization Equipment Testing	Test sanitization equipment and procedures [Assignment: organization-defined frequency] to ensure that the intended sanitization is being achieved.	Functional	Equal	Equipment Testing	DCH-09.2	Mechanisms exist to test sanitization equipment and procedures to verify that the intended result is achieved.	10		
MP-06(03)	Media Sanitization Nondestructive Techniques	Apply nondestructive sanitization techniques to portable storage devices prior to connecting such devices to the system under the following circumstances: [Assignment: organization-defined circumstances requiring sanitization of portable storage devices prior to first use]	Functional	Intersects With	First Time Use Sanitization	DCH-09.4	Mechanisms exist to apply nondestructive sanitization techniques to portable storage devices prior to first use	5		
MP-06(03)	Media Sanitization Nondestructive Techniques	Apply nondestructive sanitization techniques to portable storage devices prior to connecting such devices to the system under the following circumstances: [Assignment: organization-defined circumstances requiring sanitization of portable storage devices prior to first use]	Functional	Intersects With	System Media Sanitization	DCH-09	Mechanisms exist to sanitize system media with the strength and integrity commensurate with the classification or sensitivity of the information prior	5		
MP-06(03)	Media Sanitization Nondestructive Techniques	Apply nondestructive sanitization techniques to portable storage devices prior to connecting such devices to the system under the following circumstances: [Assignment: organization-defined circumstances requiring sanitization of portable storage devices prior to first use]	Functional	Intersects With	Sanitization of Personal Data (PD)	DCH-09.3	Mechanisms exist to facilitate the sanitization of Personal Data (PD).	5		
MP-06(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
MP-06(05)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
MP-06(06)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
MP-06(07)	Media Sanitization Dual Authorization	Enforce dual authorization for the sanitization of [Assignment: organization-defined system media].	Functional	Equal	Dual Authorization for Sensitive Data Destruction	DCH-09.5	Mechanisms exist to enforce dual authorization for the destruction, disposal or sanitization of digital media that contains sensitive and/or	10		
MP-06(08)	Media Sanitization Remote Purging or Wiping of Information	Provide the capability to purge or wipe information from [Assignment: organization-defined systems or system components] [Selection (one): remotely; under the following conditions: [Assignment: organization-defined conditions]]	Functional	Equal	Remote Purging	MDM-05	Mechanisms exist to remotely purge selected information from mobile devices.	10		
MP-07	Media Use	a. [Selection (one): Restrict; Prohibit] the use of [Assignment: organization-defined types of system media] on [Assignment: organization-defined systems or system components] using [Assignment: organization-defined controls]; andb. Prohibit the	Functional	Intersects With	Media & Data Retention	DCH-18	Mechanisms exist to retain media and data in accordance with applicable statutory, regulatory and contractual obligations	5		MP-07
MP-07	Media Use	a. [Selection (one): Restrict; Prohibit] the use of [Assignment: organization-defined types of system media] on [Assignment: organization-defined systems or system components] using [Assignment: organization-defined controls]; andb. Prohibit the	Functional	Intersects With	Media Use	DCH-10	Mechanisms exist to restrict the use of types of digital media on systems or system components.	5		MP-07
MP-07	Media Use	a. [Selection (one): Restrict; Prohibit] the use of [Assignment: organization-defined types of system media] on [Assignment: organization-defined systems or system components] using [Assignment: organization-defined controls]; andb. Prohibit the	Functional	Intersects With	Prohibit Use Without Owner	DCH-10.2	Mechanisms exist to prohibit the use of portable storage devices in organizational systems when such	5		MP-07
MP-07(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
MP-07(02)	Media Use Prohibit Use of Sanitization-resistant Media	Prohibit the use of sanitization-resistant media in organizational systems.	Functional	No Relationship	N/A	N/A	N/A	0		
MP-08	Media Downgrading	a. Establish [Assignment: organization-defined system media downgrading process] that includes employing downgrading mechanisms with strength and integrity commensurate with the security category or classification of the information;b. Verify	Functional	Intersects With	Data Reclassification	DCH-11	Mechanisms exist to reclassify data, including associated Technology Assets, Applications and/or Services	5		
MP-08(01)	Media Downgrading Documentation of Process	Document system media downgrading actions.	Functional	No Relationship	N/A	N/A	N/A	0		
MP-08(02)	Media Downgrading Equipment Testing	Test downgrading equipment and procedures [Assignment: organization-defined frequency] to ensure that downgrading actions are being achieved.	Functional	No Relationship	N/A	N/A	N/A	0		
MP-08(03)	Media Downgrading Controlled Unclassified Information	Downgrade system media containing controlled unclassified information prior to public release.	Functional	Intersects With	Data Reclassification	DCH-11	Mechanisms exist to reclassify data, including associated Technology Assets, Applications and/or Services	5		
MP-08(04)	Media Downgrading Classified Information	Downgrade system media containing classified information prior to release to individuals without required access authorizations.	Functional	No Relationship	N/A	N/A	N/A	0		

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PE-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level physical and environmental protection policy]	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5		PE-01
PE-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level physical and environmental protection policy]	Functional	Subset Of	Physical & Environmental Protections	PES-01	Mechanisms exist to facilitate the operation of physical and environmental protection controls.	10		PE-01
PE-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level physical and environmental protection policy]	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies, standards and procedures, at least	5		PE-01
PE-02	Physical Access Authorizations	a. Develop, approve, and maintain a list of individuals with authorized access to the facility where the system resides;b. Issue authorization credentials for facility access;c. Review the access list detailing authorized facility access by individuals	Functional	Equal	Physical Access Authorizations	PES-02	Physical access control mechanisms exist to maintain a current list of personnel with authorized access to organizational facilities (except for	10		PE-02
PE-02(01)	Physical Access Authorizations Access by Position or Role	Authorize physical access to the facility where the system resides based on position or role.	Functional	Equal	Role-Based Physical Access	PES-02.1	Physical access control mechanisms exist to authorize physical access to facilities based on the position or role of the individual	10		
PE-02(02)	Physical Access Authorizations Two Forms of Identification	Require two forms of identification from the following forms of identification for visitor access to the facility where the system resides: [Assignment: organization-defined list of acceptable forms of identification]	Functional	Equal	Identification Requirement	PES-06.2	Physical access control mechanisms exist to requires at least one(1) form of government-issued or organization- issued photo identification to	10		
PE-02(03)	Physical Access Authorizations Restrict Unescorted Access	Restrict unescorted access to the facility where the system resides to personnel with [Selection (one or more): security clearances for all information contained within the system; formal access authorizations for all information contained	Functional	Equal	Restrict Unescorted Access	PES-06.3	Physical access control mechanisms exist to restrict unescorted access to facilities to personnel with required security clearances, formal access,	10		
PE-03	Physical Access Control	a. Enforce physical access authorizations at [Assignment: organization-defined entry and exit points to the facility where the system resides] by:1. Verifying individual access authorizations before granting access to the facility; and2.	Functional	Intersects With	Physical Access Control	PES-03	Physical access control mechanisms exist to enforce physical access authorizations for all physical access points (including designated entry/exit	5		PE-03
PE-03(01)	Physical Access Control System Access	Enforce physical access authorizations to the system in addition to the physical access controls for the facility at [Assignment: organization-defined physical spaces containing one or more components of the system]	Functional	Equal	Access To Critical Systems	PES-03.4	Physical access control mechanisms exist to enforce physical access to critical systems or sensitive and/or locations that the organization deems	10		
PE-03(02)	Physical Access Control Facility and Systems	Perform security checks [Assignment: organization-defined frequency] at the physical perimeter of the facility or system for exfiltration of information or removal of system components.	Functional	Intersects With	Physical Access Control	PES-03	Physical access control mechanisms exist to enforce physical access authorizations for all physical access points (including designated entry/exit	5		
PE-03(03)	Physical Access Control Continuous Guards	Employ guards to control [Assignment: organization-defined physical access points] to the facility where the system resides 24 hours per day, 7 days per week.	Functional	Intersects With	Physical Access Control	PES-03	Physical access control mechanisms exist to enforce physical access authorizations for all physical access points (including designated entry/exit	5		
PE-03(04)	Physical Access Control Lockable Casings	Use lockable physical casings to protect [Assignment: organization-defined system components] from unauthorized physical access.	Functional	Equal	Lockable Physical Casings	PES-03.2	Physical access control mechanisms exist to protect system components from unauthorized physical access (e.g., lockable physical casings).	10		
PE-03(05)	Physical Access Control Logical Tampering Protection	Employ [Assignment: organization-defined anti-tamper technologies] to [Selection (one or more): detect; prevent] physical tampering or alteration of [Assignment: organization- defined hardware components] within the system	Functional	Equal	Mobile Device Tampering	MDM-04	Mechanisms exist to protect mobile devices from tampering through inspecting devices returning from locations that the organization deems	10		
PE-03(06)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
PE-03(07)	Physical Access Control Physical Barriers	Limit access using physical barriers.	Functional	No Relationship	N/A	N/A	N/A	0		
PE-03(08)	Physical Access Control Access Control Vestibules	Employ access control vestibules at [Assignment: organization-defined locations within the facility].	Functional	No Relationship	N/A	N/A	N/A	0		
PE-04	Access Control for Transmission	Control physical access to [Assignment: organization-defined system distribution and transmission lines] within organizational facilities using [Assignment: organization-defined security controls]	Functional	Equal	Transmission Medium Security	PES-12.1	Physical security mechanisms exist to protect power and telecommunications cabling carrying data or supporting information	10		
PE-05	Access Control for Output Devices	Control physical access to output from [Assignment: organization-defined output devices] to prevent unauthorized individuals from obtaining the output.	Functional	Equal	Access Control for Output Devices	PES-12.2	Physical security mechanisms exist to restrict access to printers and other system output devices to prevent unauthorized individuals from	10		
PE-05(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
PE-05(02)	Access Control for Output Devices Link to Individual Identity	Link individual identity to receipt of output from output devices.	Functional	No Relationship	N/A	N/A	N/A	0		
PE-05(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
PE-06	Monitoring Physical Access	a. Monitor physical access to the facility where the system resides to detect and respond to physical security incidents;b. Review physical access logs [Assignment: organization-defined frequency] and upon occurrence of [Assignment: organization-	Functional	Equal	Monitoring Physical Access	PES-05	Physical access control mechanisms exist to monitor for, detect and respond to physical security incidents.	10		PE-06
PE-06(01)	Monitoring Physical Access Intrusion Alarms and Surveillance	Monitor physical access to the facility where the system resides using physical intrusion alarms and surveillance equipment.	Functional	Equal	Intrusion Alarms / Surveillance Equipment	PES-05.1	Physical access control mechanisms exist to monitor physical intrusion alarms and surveillance equipment.	10		
PE-06(02)	Monitoring Physical Access Automated Intrusion Recognition and Responses	Recognize [Assignment: organization-defined classes or types of intrusions] and initiate [Assignment: organization-defined response actions] using [Assignment: organization-defined automated mechanisms]	Functional	No Relationship	N/A	N/A	N/A	0		
PE-06(03)	Monitoring Physical Access Video Surveillance	a. Employ video surveillance of [Assignment: organization-defined operational areas];b. Review video recordings [Assignment: organization-defined frequency]; andc. Retain video recordings for [Assignment: organization-defined time	Functional	No Relationship	N/A	N/A	N/A	0		
PE-06(04)	Monitoring Physical Access Monitoring Physical Access to Systems	Monitor physical access to the system in addition to the physical access monitoring of the facility at [Assignment: organization-defined physical spaces containing one or more components of the system]	Functional	Equal	Monitoring Physical Access To Critical Systems	PES-05.2	Facility security mechanisms exist to monitor physical access to critical systems or sensitive and/or regulated data in addition to the physical	10		
PE-07		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
PE-08	Visitor Access Records	a. Maintain visitor access records to the facility where the system resides for [Assignment: organization-defined time period];b. Review visitor access records [Assignment: organization-defined frequency]; andc. Report anomalies in	Functional	Equal	Physical Access Logs	PES-03.3	Physical access control mechanisms generate a log entry for each access attempt through controlled ingress and egress points	10		PE-08
PE-08(01)	Visitor Access Records Automated Records Maintenance and Review	Maintain and review visitor access records using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Automated Records Management & Review	PES-06.4	Automated mechanisms exist to facilitate the maintenance and review of visitor access records.	10		
PE-08(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn

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PE-08(03)	Visitor Access Records Limit Personally Identifiable Information Elements	Limit personally identifiable information contained in visitor access records to the following elements identified in the privacy risk assessment: [Assignment: organization-defined elements]	Functional	Equal	Minimize Visitor Personal Data (PD)	PES-06.5	Mechanisms exist to minimize the collection of Personal Data (PD) contained in visitor access records.	10		
PE-09	Power Equipment and Cabling	Protect power equipment and power cabling for the system from damage and destruction.	Functional	Equal	Supporting Utilities	PES-07	Facility security mechanisms exist to protect power equipment and power cabling for the system from damage and destruction.	10		
PE-09(01)	Power Equipment and Cabling Redundant Cabling	Employ redundant power cabling paths that are physically separated by [Assignment: organization-defined distance].	Functional	Equal	Redundant Cabling	PES-07.7	Mechanisms exist to employ redundant power cabling paths that are physically separated to ensure that power continues to flow in the event of a power outage or disruption.	10		
PE-09(02)	Power Equipment and Cabling Automatic Voltage Controls	Employ automatic voltage controls for [Assignment: organization-defined critical system components].	Functional	Equal	Automatic Voltage Controls	PES-07.1	Facility security mechanisms exist to utilize automatic voltage controls for critical system components.	10		
PE-10	Emergency Shutoff	a. Provide the capability of shutting off power to [Assignment: organization-defined system or individual system components] in emergency situations;b. Place emergency shutoff switches or devices in [Assignment: organization-defined location by system]	Functional	Equal	Emergency Shutoff	PES-07.2	Facility security mechanisms exist to shut off power in emergency situations by: (1) Placing emergency shutoff	10		
PE-10(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
PE-11	Emergency Power	Provide an uninterruptible power supply to facilitate [Selection (one or more): an orderly shutdown of the system; transition of the system to long-term alternate power] in the event of a primary power source loss.	Functional	Intersects With	Emergency Power	PES-07.3	Facility security mechanisms exist to supply alternate power, capable of maintaining minimally-required operational capability, in the event of a power outage or disruption.	5		
PE-11(01)	Emergency Power Alternate Power Supply — Minimal Operational Capability	Provide an alternate power supply for the system that is activated [Selection (one): manually; automatically] and that can maintain minimally required operational capability in the event of an extended loss of the primary power source.	Functional	Intersects With	Emergency Power	PES-07.3	Facility security mechanisms exist to supply alternate power, capable of maintaining minimally-required operational capability, in the event of a power outage or disruption.	5		
PE-11(02)	Emergency Power Alternate Power Supply — Self-contained	Provide an alternate power supply for the system that is activated [Selection (one): manually; automatically] and that is:a. Self-contained;b. Not reliant on external power generation; and;c. Capable of maintaining [Selection (one): manually; automatically] emergency lighting for the system that activates in the event of a power outage or disruption and that covers emergency exits and evacuation routes within the facility.	Functional	Intersects With	Emergency Power	PES-07.3	Facility security mechanisms exist to supply alternate power, capable of maintaining minimally-required operational capability, in the event of a power outage or disruption.	5		
PE-12	Emergency Lighting	Employ and maintain automatic emergency lighting for the system that activates in the event of a power outage or disruption and that covers emergency exits and evacuation routes within the facility.	Functional	Equal	Emergency Lighting	PES-07.4	Facility security mechanisms exist to utilize and maintain automatic emergency lighting that activates in the event of a power outage or disruption.	10		PE-12
PE-12(01)	Emergency Lighting Essential Mission and Business Functions	Provide emergency lighting for all areas within the facility supporting essential mission and business functions.	Functional	No Relationship	N/A	N/A	N/A	0		
PE-13	Fire Protection	Employ and maintain fire detection and suppression systems that are supported by an independent energy source.	Functional	Equal	Fire Protection	PES-08	Facility security mechanisms exist to utilize and maintain fire suppression and detection devices/systems for the system that are supported by an independent energy source.	10		PE-13
PE-13(01)	Fire Protection Detection Systems — Automatic Activation and Notification	Employ fire detection systems that activate automatically and notify [Assignment: organization-defined personnel or roles] and [Assignment: organization-defined emergency responders] in the event of a fire.	Functional	Equal	Fire Detection Devices	PES-08.1	Facility security mechanisms exist to utilize and maintain fire detection devices/systems that activate automatically and notify [Assignment: organization-defined personnel or roles] and [Assignment: organization-defined emergency responders] in the event of a fire.	10		
PE-13(02)	Fire Protection Suppression Systems — Automatic Activation and Notification	a. Employ fire suppression systems that activate automatically and notify [Assignment: organization-defined personnel or roles] and [Assignment: organization-defined emergency responders]; andb. Employ an automatic fire suppression capability when the system resides at [Assignment: organization-defined acceptable levels] within the facility where the system resides at [Assignment: organization-defined acceptable levels] within the facility.	Functional	Intersects With	Automatic Fire Suppression	PES-08.3	Facility security mechanisms exist to employ an automatic fire suppression capability for critical systems when the system resides at [Assignment: organization-defined acceptable levels] within the facility where the system resides at [Assignment: organization-defined acceptable levels] within the facility.	5		
PE-13(02)	Fire Protection Suppression Systems — Automatic Activation and Notification	a. Employ fire suppression systems that activate automatically and notify [Assignment: organization-defined personnel or roles] and [Assignment: organization-defined emergency responders]; andb. Employ an automatic fire suppression capability when the system resides at [Assignment: organization-defined acceptable levels] within the facility where the system resides at [Assignment: organization-defined acceptable levels] within the facility.	Functional	Intersects With	Fire Suppression Devices	PES-08.2	Facility security mechanisms exist to utilize fire suppression devices/systems that provide automatic notification of any fire.	5		
PE-13(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
PE-13(04)	Fire Protection Inspections	Ensure that the facility undergoes [Assignment: organization-defined frequency] fire protection inspections by authorized and qualified inspectors and identified deficiencies are resolved within [Assignment: organization-defined time period].	Functional	No Relationship	N/A	N/A	N/A	0		
PE-14	Environmental Controls	a. Maintain [Selection (one or more): temperature; humidity; pressure; radiation; [Assignment: organization-defined environmental control]] levels within the facility where the system resides at [Assignment: organization-defined acceptable levels] within the facility where the system resides at [Assignment: organization-defined acceptable levels] within the facility.	Functional	Equal	Temperature & Humidity Controls	PES-09	Facility security mechanisms exist to maintain and monitor temperature and humidity levels within the facility.	10		PE-14
PE-14(01)	Environmental Controls Automatic Controls	Employ the following automatic environmental controls in the facility to prevent fluctuations potentially harmful to the system: [Assignment: organization-defined automatic environmental controls].	Functional	No Relationship	N/A	N/A	N/A	0		
PE-14(02)	Environmental Controls Monitoring with Alarms and Notifications	Employ environmental control monitoring that provides an alarm or notification of changes potentially harmful to personnel or equipment to [Assignment: organization-defined personnel or roles].	Functional	Equal	Monitoring with Alarms / Notifications	PES-09.1	Facility security mechanisms exist to trigger an alarm or notification of temperature and humidity changes that be potentially harmful to the system.	10		
PE-15	Water Damage Protection	Protect the system from damage resulting from water leakage by providing master shutoff or isolation valves that are accessible, working properly, and known to key personnel.	Functional	Equal	Water Damage Protection	PES-07.5	Facility security mechanisms exist to protect systems from damage resulting from water leakage by providing master shutoff valves that are accessible, working properly, and known to key personnel.	10		PE-15
PE-15(01)	Water Damage Protection Automation Support	Detect the presence of water near the system and alert [Assignment: organization-defined personnel or roles] using [Assignment: organization-defined automated mechanisms].	Functional	Equal	Automation Support for Water Damage Protection	PES-07.6	Facility security mechanisms exist to detect the presence of water in the vicinity of critical systems and alert [Assignment: organization-defined personnel or roles] using [Assignment: organization-defined automated mechanisms].	10		
PE-16	Delivery and Removal	a. Authorize and control [Assignment: organization-defined types of system components] entering and exiting the facility; andb. Maintain records of the system components.	Functional	Equal	Delivery & Removal	PES-10	Physical security mechanisms exist to isolate information processing facilities from points such as delivery and loading areas and other points in the facility.	10		PE-16
PE-17	Alternate Work Site	a. Determine and document the [Assignment: organization-defined alternate work sites] allowed for use by employees;b. Employ the following controls at alternate work sites: [Assignment: organization-defined controls].	Functional	Equal	Alternate Work Site	PES-11	Physical security mechanisms exist to utilize appropriate management, operational and technical controls at alternate work sites.	10		
PE-18	Location of System Components	Position system components within the facility to minimize potential damage from [Assignment: organization-defined physical and environmental hazards] and to minimize the opportunity for unauthorized access.	Functional	Intersects With	Equipment Siting & Protection	PES-12	Physical security mechanisms exist to locate system components within the facility to minimize potential damage from physical and environmental hazards.	5		
PE-18(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
PE-19	Information Leakage	Protect the system from information leakage due to electromagnetic signals emanations.	Functional	Equal	Information Leakage Due To Electromagnetic Signals Emanations	PES-13	Facility security mechanisms exist to protect the system from information leakage due to electromagnetic signals emanations.	10		
PE-19(01)	Information Leakage National Emissions Policies and Procedures	Protect system components, associated data communications, and networks in accordance with national Emissions Security policies and procedures based on the security category or classification of the information.	Functional	No Relationship	N/A	N/A	N/A	0		
PE-20	Asset Monitoring and Tracking	Employ [Assignment: organization-defined asset location technologies] to track and monitor the location and movement of [Assignment: organization-defined assets] within [Assignment: organization-defined controlled areas].	Functional	Equal	Asset Monitoring and Tracking	PES-14	Physical security mechanisms exist to employ asset location technologies that track and monitor the location and movement of organization-	10		

Secure Controls Framework (SCF)

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
PL-09	Central Management	Centrally manage [Assignment: organization-defined controls and related processes].	Functional	Intersects With	Centralized Management of Security, Compliance & Resilience Controls	SEA-01.1	Mechanisms exist to centrally-manage the organization-wide management and implementation of security, compliance and resilience controls.	5		
PL-09	Central Management	Centrally manage [Assignment: organization-defined controls and related processes].	Functional	Intersects With	Centralized Management of Flaw Remediation Processes	VPM-05.1	Mechanisms exist to centrally-manage the flaw remediation process.	5		
PL-09	Central Management	Centrally manage [Assignment: organization-defined controls and related processes].	Functional	Intersects With	Assigned Security, Compliance & Resilience Responsibilities	GOV-04	Mechanisms exist to assign one or more qualified individuals with the mission and resources to centrally-manage, coordinate, develop	5		
PL-09	Central Management	Centrally manage [Assignment: organization-defined controls and related processes].	Functional	Intersects With	Centralized Management of Antimalware Technologies	END-04.3	Mechanisms exist to centrally-manage antimalware technologies.	5		
PL-09	Central Management	Centrally manage [Assignment: organization-defined controls and related processes].	Functional	Intersects With	Phishing & Spam Protection Centralized Management	END-08.1	Mechanisms exist to centrally manage anti-phishing and spam protection technologies.	5		
PL-09	Central Management	Centrally manage [Assignment: organization-defined controls and related processes].	Functional	Intersects With	Centralized Management of Event Log Content	MON-03.6	Mechanisms exist to centrally manage and update the criteria to be captured in event logs generated by	5		
PL-10	Baseline Selection	Select a control baseline for the system.	Functional	Equal	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for	10		PL-10
PL-11	Baseline Tailoring	Tailor the selected control baseline by applying specified tailoring actions.	Functional	Equal	Baseline Tailoring	CFG-02.9	Mechanisms exist to allow baseline controls to be specialized or customized by applying a defined set of tailoring actions that are specific to	10		PL-11
PM-01	Information Security Program Plan	a. Develop and disseminate an organization-wide information security program plan that:1. Provides an overview of the requirements for the security program and a description of the security program management controls and common controls	Functional	Subset Of	Security, Compliance & Resilience Program (SCRPP)	GOV-01	Mechanisms exist to facilitate the implementation of security, compliance and resilience governance controls.	10		PM-01
PM-01	Information Security Program Plan	a. Develop and disseminate an organization-wide information security program plan that:1. Provides an overview of the requirements for the security program and a description of the security program management controls and common controls	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies,	5		PM-01
PM-01	Information Security Program Plan	a. Develop and disseminate an organization-wide information security program plan that:1. Provides an overview of the requirements for the security program and a description of the security program management controls and common controls	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5		PM-01
PM-02	Information Security Program Leadership Role	Appoint a senior agency information security officer with the mission and resources to coordinate, develop, implement, and maintain an organization-wide information security program.	Functional	Intersects With	Assigned Security, Compliance & Resilience Responsibilities	GOV-04	Mechanisms exist to assign one or more qualified individuals with the mission and resources to centrally-manage, coordinate, develop	5		PM-02
PM-03	Information Security and Privacy Resources	a. Include the resources needed to implement the information security and privacy programs in capital planning and investment requests and document all exceptions to this requirement;b. Prepare documentation required for addressing	Functional	Equal	Security, Compliance & Resilience Resource Management	PRM-02	Mechanisms exist to address all capital planning and investment requests, including the resources needed to implement the Security	10		PM-03
PM-04	Plan of Action and Milestones Process	a. Implement a process to ensure that plans of action and milestones for the information security, privacy, and supply chain risk management programs and associated organizational systems:1. Are developed and maintained;2. Document the	Functional	Intersects With	Vulnerability Remediation Process	VPM-02	Mechanisms exist to ensure that vulnerabilities are properly identified, tracked and remediated.	5		PM-04
PM-04	Plan of Action and Milestones Process	a. Implement a process to ensure that plans of action and milestones for the information security, privacy, and supply chain risk management programs and associated organizational systems:1. Are developed and maintained;2. Document the	Functional	Intersects With	Capabilities Deficiency Tracking	IAO-05	Mechanisms exist to govern identified deficiencies (e.g., Plan of Action and Milestones (POA&M) or similar methodology) that formally	5		PM-04
PM-05	System Inventory	Develop and update [Assignment: organization-defined frequency] an inventory of organizational systems.	Functional	Intersects With	Asset Governance	AST-01	Asset Management (ITAM) program to implement and manage asset management controls.	5		PM-05
PM-05	System Inventory	Develop and update [Assignment: organization-defined frequency] an inventory of organizational systems.	Functional	Intersects With	Asset Inventories	AST-02	Mechanisms exist to perform inventories of Technology Assets, Applications, Services and/or Data	5		PM-05
PM-05(01)	System Inventory Inventory of Personally Identifiable Information	Establish, maintain, and update [Assignment: organization-defined frequency] an inventory of all systems, applications, and projects that process personally identifiable information.	Functional	Intersects With	Inventory of Personal Data (PD)	PRI-05.5	Mechanisms exist to establish and maintain a current inventory of all Technology Assets, Applications	5		
PM-05(01)	System Inventory Inventory of Personally Identifiable Information	Establish, maintain, and update [Assignment: organization-defined frequency] an inventory of all systems, applications, and projects that process personally identifiable information.	Functional	Intersects With	Personal Data (PD) Inventory Automation Support	PRI-05.6	Automated mechanisms exist to determine if Personal Data (PD) is maintained in electronic form.	5		
PM-06	Measures of Performance	Develop, monitor, and report on the results of information security and privacy measures of performance.	Functional	Intersects With	Assigned Security, Compliance & Resilience Responsibilities	GOV-04	Mechanisms exist to assign one or more qualified individuals with the mission and resources to centrally-manage, coordinate, develop	5		PM-06
PM-06	Measures of Performance	Develop, monitor, and report on the results of information security and privacy measures of performance.	Functional	Intersects With	Measures of Performance	GOV-05	Mechanisms exist to develop, report and monitor Security, Compliance & Resilience Program (SCRPP) measures of performance.	5		PM-06
PM-07	Enterprise Architecture	Develop and maintain an enterprise architecture with consideration for information security, privacy, and the resulting risk to organizational operations and assets, individuals, other organizations, and the Nation.	Functional	Intersects With	Alignment With Enterprise Architecture	SEA-02	Mechanisms exist to develop an enterprise architecture, aligned with industry-recognized leading practices, with consideration for security	5		PM-07
PM-07(01)	Enterprise Architecture Offloading	Offload [Assignment: organization-defined non-essential functions or services] to other systems, system components, or an external provider.	Functional	Equal	Outsourcing Non-Essential Functions or Services	SEA-02.2	Mechanisms exist to identify non-essential functions or services that are capable of being outsourced to external service providers and align	10		
PM-08	Critical Infrastructure Plan	Address information security and privacy issues in the development, documentation, and updating of a critical infrastructure and key resources protection plan.	Functional	Intersects With	Business Continuity Management System (BCMS)	BCD-01	Mechanisms exist to facilitate the implementation of contingency planning controls to help ensure	5		PM-08
PM-08	Critical Infrastructure Plan	Address information security and privacy issues in the development, documentation, and updating of a critical infrastructure and key resources protection plan.	Functional	Intersects With	Statutory, Regulatory & Contractual Compliance	CPL-01	Mechanisms exist to facilitate the identification and implementation of relevant statutory, regulatory and contractual controls.	5		PM-08
PM-09	Risk Management Strategy	a. Develops a comprehensive strategy to manage:1. Security risk to organizational operations and assets, individuals, other organizations, and the Nation associated with the operation and use of organizational systems; and2. Privacy risk to individuals	Functional	Equal	Risk Management Program	RSK-01	Mechanisms exist to facilitate the implementation of strategic, operational and tactical risk management controls that are aligned	10		PM-09
PM-10	Authorization Process	a. Manage the security and privacy state of organizational systems and the environments in which those systems operate through authorization processes;b. Designate individuals to fulfill specific roles and responsibilities within the organizational	Functional	Equal	Information Assurance (IA) Operations	IAO-01	Mechanisms exist to facilitate the implementation of security, compliance and resilience assessment and authorization	10		PM-10

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
PM-11	Mission and Business Process Definition	a. Define organizational mission and business processes with consideration for information security and privacy and the resulting risk to organizational operations, organizational assets, individuals, other organizations, and the Nation; and b.	Functional	Equal	Business Process Definition	PRM-06	Mechanisms exist to define business processes with consideration for security, compliance and resilience that determines:	10		PM-11
PM-12	Insider Threat Program	Implement an insider threat program that includes a cross-discipline insider threat incident handling team.	Functional	Equal	Insider Threat Program	THR-04	Mechanisms exist to implement an insider threat program that includes a cross-discipline insider threat incident handling team.	10		PM-12
PM-13	Security and Privacy Workforce	Establish a security and privacy workforce development and improvement program.	Functional	Intersects With	Defined Roles & Responsibilities	HRS-03	Mechanisms exist to define cybersecurity roles & responsibilities for all personnel.	5		PM-13
PM-13	Security and Privacy Workforce	Establish a security and privacy workforce development and improvement program.	Functional	Intersects With	Security, Compliance & Resilience-Minded Workforce	SAT-01	Mechanisms exist to facilitate the implementation of security workforce development and awareness controls.	5		PM-13
PM-14	Testing, Training, and Monitoring	a. Implement a process for ensuring that organizational plans for conducting security and privacy testing, training, and monitoring activities associated with organizational systems:1.	Functional	Intersects With	Personal Data (PD) Control Testing, Training & Monitoring	PRI-08	Mechanisms exist to conduct testing, training and monitoring activities for Personal Data (PD) controls.	5		PM-14
PM-14	Testing, Training, and Monitoring	a. Implement a process for ensuring that organizational plans for conducting security and privacy testing, training, and monitoring activities associated with organizational systems:1.	Functional	Intersects With	Security, Compliance & Resilience Controls Oversight	CPL-02	Mechanisms exist to provide a security, compliance and resilience controls oversight function that reports to the organization's executive.	5		PM-14
PM-15	Security and Privacy Groups and Associations	Establish and institutionalize contact with selected groups and associations within the security and privacy communities:a. To facilitate ongoing security and privacy education and training for organizational personnel-h. To maintain currency with	Functional	Intersects With	Threat Intelligence Program	THR-01	Mechanisms exist to implement a threat intelligence program that includes a cross-organization information-sharing capability that	5		PM-15
PM-15	Security and Privacy Groups and Associations	Establish and institutionalize contact with selected groups and associations within the security and privacy communities:a. To facilitate ongoing security and privacy education and training for organizational personnel-h. To maintain currency with	Functional	Intersects With	Contacts With Groups & Associations	GOV-07	Mechanisms exist to establish contact with selected groups and associations within the security, compliance and resilience communities to	5		PM-15
PM-16	Threat Awareness Program	Implement a threat awareness program that includes a cross-organization information-sharing capability for threat intelligence.	Functional	Intersects With	Threat Intelligence Program	THR-01	Mechanisms exist to implement a threat intelligence program that includes a cross-organization information-sharing capability that	5		PM-16
PM-16(01)	Threat Awareness Program Automated Means for Sharing Threat Intelligence	Employ automated mechanisms to maximize the effectiveness of sharing threat intelligence information.	Functional	Intersects With	Threat Intelligence Feeds	THR-03	Mechanisms exist to maintain situational awareness of vulnerabilities and evolving threats by leveraging the knowledge of attacker	5		
PM-17	Protecting Controlled Unclassified Information on External Systems	a. Establish policy and procedures to ensure that requirements for the protection of controlled unclassified information that is processed, stored or transmitted on external systems, are implemented in accordance with applicable laws, executive	Functional	Equal	Protecting Sensitive / Regulated Data on External Technology Assets, Applications	DCH-13.3	Mechanisms exist to ensure that the requirements for the protection of sensitive and/or regulated data	10		PM-17
PM-18	Privacy Program Plan	a. Develop and disseminate an organization-wide privacy program plan that provides an overview of the agency's privacy program, and:1. Includes a description of the structure of the privacy program and the resources dedicated to the privacy	Functional	Equal	Data Privacy Program	PRI-01	Mechanisms exist to facilitate the implementation and operation of data protection controls throughout the data lifecycle and across	10		PM-18
PM-19	Privacy Program Leadership Role	Appoint a senior agency official for privacy with the authority, mission, accountability, and resources to coordinate, develop, and implement, applicable privacy requirements and manage privacy risks through the organization-wide privacy program	Functional	Equal	Chief Privacy Officer (CPO)	PRI-01.1	Mechanisms exist to appoints a Chief Privacy Officer (CPO) or similar role, with the authority, mission, accountability and resources to	10		PM-19
PM-20	Dissemination of Privacy Program Information	Maintain a central resource webpage on the organization's principal public website that serves as a central source of information about the organization's privacy program and that:a. Ensures that the public has access to information about	Functional	Equal	Dissemination of Data Privacy Program Information	PRI-01.3	Mechanisms exist to: (1) Ensure that the public has access to information about organizational data privacy activities and can	10		PM-20
PM-20(01)	Dissemination of Privacy Program Information Privacy Policies on Websites	Develop and post privacy policies on all external-facing websites, mobile applications, and other digital services, that:a. Are written in plain language and organized in a way that is easy to understand and navigate;b. Provide information needed by	Functional	Equal	Data Privacy Notice	PRI-02	Mechanisms exist to: (1) Make data privacy notice(s) available to individuals upon first interaction with organization's	10		PM-20(01)
PM-21	Accounting of Disclosures	a. Develop and maintain an accurate accounting of disclosures of personally identifiable information, including:1. Date, nature, and purpose of each disclosure; and2. Name and address, or other contact information of the individual or organization to	Functional	Equal	Accounting of Disclosures	PRI-14.1	Mechanisms exist to provide data subjects with an accounting of disclosures of their Personal Data (PD) collected by	10		PM-21
PM-22	Personally Identifiable Information Quality Management	Develop and document organization-wide policies and procedures for:a. Reviewing for the accuracy, relevance, timeliness, and completeness of personally identifiable information across the information life cycle;b. Correcting or	Functional	Intersects With	Data Quality Management	PRI-10	Mechanisms exist to manage the quality, utility, objectivity, integrity and impact determination and de-identification of sensitive and/or	5		PM-22
PM-22	Personally Identifiable Information Quality Management	Develop and document organization-wide policies and procedures for:a. Reviewing for the accuracy, relevance, timeliness, and completeness of personally identifiable information across the information life cycle;b. Correcting or	Functional	Intersects With	Data Quality Operations	DCH-22	Mechanisms exist to check for Redundant, Obsolete/Outdated, Toxic or Trivial (ROTT) data to ensure the accuracy, relevance, timeliness	5		PM-22
PM-23	Data Governance Body	Establish a Data Governance Body consisting of [Assignment: organization-defined roles] with [Assignment: organization-defined responsibilities].	Functional	Intersects With	Data Management Board	PRI-13	Mechanisms exist to establish a written charter for a Data Management Board (DMB) and assigned organization-defined roles to	5		PM-23
PM-23	Data Governance Body	Establish a Data Governance Body consisting of [Assignment: organization-defined roles] with [Assignment: organization-defined responsibilities].	Functional	Intersects With	Data Quality Management	PRI-10	Mechanisms exist to manage the quality, utility, objectivity, integrity and impact determination and de-identification of sensitive and/or	5		PM-23
PM-23	Data Governance Body	Establish a Data Governance Body consisting of [Assignment: organization-defined roles] with [Assignment: organization-defined responsibilities].	Functional	Intersects With	Data Governance	GOV-10	Mechanisms exist to facilitate data governance to oversee the organization's policies, standards and procedures so that sensitive and/or	5		PM-23
PM-24	Data Integrity Board	Establish a Data Integrity Board to:a. Review proposals to conduct or participate in a matching program; andb. Conduct an annual review of all matching programs in which the agency has participated	Functional	Intersects With	Data Governance	GOV-10	Mechanisms exist to facilitate data governance to oversee the organization's policies, standards and procedures so that sensitive and/or	5		PM-24
PM-24	Data Integrity Board	Establish a Data Integrity Board to:a. Review proposals to conduct or participate in a matching program; andb. Conduct an annual review of all matching programs in which the agency has participated	Functional	Intersects With	Data Management Board	PRI-13	Mechanisms exist to establish a written charter for a Data Management Board (DMB) and assigned organization-defined roles to	5		PM-24
PM-24	Data Integrity Board	Establish a Data Integrity Board to:a. Review proposals to conduct or participate in a matching program; andb. Conduct an annual review of all matching programs in which the agency has participated	Functional	Intersects With	Data Quality Management	PRI-10	Mechanisms exist to manage the quality, utility, objectivity, integrity and impact determination and de-identification of sensitive and/or	5		PM-24
PM-24	Data Integrity Board	Establish a Data Integrity Board to:a. Review proposals to conduct or participate in a matching program; andb. Conduct an annual review of all matching programs in which the agency has participated	Functional	Intersects With	Personal Data (PD) Accuracy & Integrity	PRI-05.2	Mechanisms exist to ensure the accuracy and relevance of Personal Data (PD) throughout the information lifecycle	5		PM-24
PM-24	Data Integrity Board	Establish a Data Integrity Board to:a. Review proposals to conduct or participate in a matching program; andb. Conduct an annual review of all matching programs in which the agency has participated	Functional	Intersects With	Computer Matching Agreements (CMA)	PRI-02.3	Mechanisms exist to publish Computer Matching Agreements (CMA) on the organization's public website	5		PM-24
PM-24	Data Integrity Board	Establish a Data Integrity Board to:a. Review proposals to conduct or participate in a matching program; andb. Conduct an annual review of all matching programs in which the agency has participated	Functional	Intersects With	Automated Data Management Processes	PRI-02.2	Automated mechanisms exist to adjust data that is able to be collected, received, processed, stored, transmitted, processed, and	5		PM-24

Secure Controls Framework (SCF)

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
PS-03(04)	Personnel Screening Citizenship Requirements	Verify that individuals accessing a system processing, storing, or transmitting [Assignment: organization-defined information types] meet [Assignment: organization-defined citizenship requirements].	Functional	Equal	Citizenship Requirements	HRS-04.3	Mechanisms exist to verify that individuals accessing a system processing, storing, or transmitting sensitive information meet applicable	10		
PS-04	Personnel Termination	Upon termination of individual employment:a. Disable system access within [Assignment: organization-defined time period];b. Terminate or revoke any authenticators and credentials associated with the individual;c. Conduct exit interviews that	Functional	Equal	Personnel Termination	HRS-09	Mechanisms exist to govern the termination of individual employment.	10		PS-04
PS-04(01)	Personnel Termination Post-employment Requirements	a. Notify terminated individuals of applicable, legally binding post-employment requirements for the protection of organizational information; andb. Require terminated individuals to sign an acknowledgment of post-employment	Functional	Equal	Post-Employment Requirements Notification	HRS-09.3	Mechanisms exist to govern former employee behavior by formally notifying terminated individuals of	10		
PS-04(02)	Personnel Termination Automated Actions	Use [Assignment: organization-defined automated mechanisms] to [Selection (one or more): notify [Assignment: organization-defined personnel or roles] of individual termination actions; disable access to system resources]	Functional	Equal	Automated Employment Status Notifications	HRS-09.4	Automated mechanisms exist to notify Identity and Access Management (IAM) personnel or roles upon termination of an individual	10		
PS-05	Personnel Transfer	a. Review and confirm ongoing operational need for current logical and physical access authorizations to systems and facilities when individuals are reassigned or transferred to other positions within the organization;b. Initiate [Assignment:	Functional	Equal	Personnel Transfer	HRS-08	Mechanisms exist to adjust logical and physical access authorizations to Technology Assets, Applications	10		PS-05
PS-06	Access Agreements	a. Develop and document access agreements for organizational systems;b. Review and update the access agreements [Assignment: organization-defined frequency]; andc. Verify that individuals require access to organizational information and	Functional	Intersects With	Confidentiality Agreements	HRS-06.1	Mechanisms exist to require Non-Disclosure Agreements (NDAs) or similar confidentiality agreements that reflect the needs to protect data and	5		PS-06
PS-06	Access Agreements	a. Develop and document access agreements for organizational systems;b. Review and update the access agreements [Assignment: organization-defined frequency]; andc. Verify that individuals require access to organizational information and	Functional	Intersects With	Access Agreements	HRS-06	Mechanisms exist to require internal and third-party users to sign appropriate access agreements prior to being granted access.	5		PS-06
PS-06(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
PS-06(02)	Access Agreements Classified Information Requiring Special Protection	Verify that access to classified information requiring special protection is granted only to individuals who:a. Have a valid access authorization that is demonstrated by assigned official government duties;b. Satisfy associated personnel security	Functional	Intersects With	Confidentiality Agreements	HRS-06.1	Mechanisms exist to require Non-Disclosure Agreements (NDAs) or similar confidentiality agreements that reflect the needs to protect data and	5		
PS-06(02)	Access Agreements Classified Information Requiring Special Protection	Verify that access to classified information requiring special protection is granted only to individuals who:a. Have a valid access authorization that is demonstrated by assigned official government duties;b. Satisfy associated personnel security	Functional	Intersects With	Access Agreements	HRS-06	Mechanisms exist to require internal and third-party users to sign appropriate access agreements prior to being granted access.	5		
PS-06(03)	Access Agreements Post-employment Requirements	a. Notify individuals of applicable, legally binding post-employment requirements for protection of organizational information; andb. Require individuals to sign an acknowledgment of these requirements, if applicable, as part of	Functional	Equal	Post-Employment Requirements Awareness	HRS-06.2	Mechanisms exist to notify individuals of their applicable, legally-binding post-employment requirements for	10		
PS-07	External Personnel Security	a. Establish personnel security requirements, including security roles and responsibilities for external providers;b. Require external providers to comply with personnel security policies and procedures established by the organization;c. Document	Functional	Equal	Third-Party Personnel	HRS-10	Mechanisms exist to govern third-party personnel by reviewing and monitoring third-party security, compliance and/or resilience roles	10		PS-07
PS-08	Personnel Sanctions	a. Employ a formal sanctions process for individuals failing to comply with established information security and privacy policies and procedures; andb. Notify [Assignment: organization-defined personnel or roles] within [Assignment: organization-	Functional	Equal	Personnel Sanctions	HRS-07	Mechanisms exist to sanction personnel failing to comply with established security policies, standards and procedures	10		PS-08
PS-09	Position Descriptions	Incorporate security and privacy roles and responsibilities into organizational position descriptions.	Functional	Equal	Defined Roles & Responsibilities	HRS-03	Mechanisms exist to define cybersecurity roles & responsibilities for all personnel.	10		PS-09
PT-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] personally identifiable information processing and	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5		
PT-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] personally identifiable information processing and	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCR), including policies,	5		
PT-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] personally identifiable information processing and	Functional	Subset Of	Data Privacy Program	PRI-01	Mechanisms exist to facilitate the implementation and operation of data protection controls throughout the	10		
PT-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] personally identifiable information processing and	Functional	Subset Of	Secure Engineering Principles	SEA-01	Mechanisms exist to facilitate the implementation of industry-recognized security, compliance and	10		
PT-02	Authority to Process Personally Identifiable Information	a. Determine and document the [Assignment: organization-defined authority] that permits the [Assignment: organization-defined processing] of personally identifiable information; andb. Restrict the [Assignment: organization-defined processing] of	Functional	Intersects With	Authority To Collect, Process, Store & Share Personal Data (PD)	PRI-04.1	Mechanisms exist to determine and document the legal authority that permits the organization to collect,	5		
PT-02	Authority to Process Personally Identifiable Information	a. Determine and document the [Assignment: organization-defined authority] that permits the [Assignment: organization-defined processing] of personally identifiable information; andb. Restrict the [Assignment: organization-defined processing] of	Functional	Intersects With	Internal Use of Personal Data (PD) For Testing, Training and Research	PRI-05.1	Mechanisms exist to address the use of Personal Data (PD) for internal testing, training and research that:	5		
PT-02	Authority to Process Personally Identifiable Information	a. Determine and document the [Assignment: organization-defined authority] that permits the [Assignment: organization-defined processing] of personally identifiable information; andb. Restrict the [Assignment: organization-defined processing] of	Functional	Intersects With	Usage Restrictions of Personal Data (PD)	PRI-05.4	Mechanisms exist to restrict collecting, receiving, processing, storing, transmitting, sharing and/or	5		
PT-02	Authority to Process Personally Identifiable Information	a. Determine and document the [Assignment: organization-defined authority] that permits the [Assignment: organization-defined processing] of personally identifiable information; andb. Restrict the [Assignment: organization-defined processing] of	Functional	Intersects With	Restrict Collection, Processing & Sharing To Identified Purpose	PRI-04	Mechanisms exist to minimize the collection, processing and/or sharing of Personal Data (PD) to only what is	5		
PT-02(01)	Authority to Process Personally Identifiable Information Data Tagging	Attach data tags containing [Assignment: organization-defined authorized processing] to [Assignment: organization-defined elements of personally identifiable information].	Functional	Equal	Data Tags	DCH-22.2	Mechanisms exist to utilize data tags to automate tracking of sensitive and/or regulated data across the information lifecycle	10		
PT-02(02)	Authority to Process Personally Identifiable Information Automation	Manage enforcement of the authorized processing of personally identifiable information using [Assignment: organization-defined automated mechanisms].	Functional	Intersects With	Automated Data Management Processes	PRI-02.2	Automated mechanisms exist to adjust data that is able to be collected, received, processed,	5		
PT-03	Personally Identifiable Information Processing Purposes	a. Identify and document the [Assignment: organization-defined purpose(s)] for processing personally identifiable information;b. Describe the purpose(s) in the public privacy notices and policies of the organization;c. Restrict the [Assignment:	Functional	Intersects With	Internal Use of Personal Data (PD) For Testing, Training and Research	PRI-05.1	Mechanisms exist to address the use of Personal Data (PD) for internal testing, training and research that:	5		
PT-03	Personally Identifiable Information Processing Purposes	a. Identify and document the [Assignment: organization-defined purpose(s)] for processing personally identifiable information;b. Describe the purpose(s) in the public privacy notices and policies of the organization;c. Restrict the [Assignment:	Functional	Intersects With	Purpose Specification	PRI-02.1	Mechanisms exist to ensure data privacy notices identify the purpose(s) for which Personal Data (PD) is	5		
PT-03(01)	Personally Identifiable Information Processing Purposes Data Tagging	Attach data tags containing the following purposes to [Assignment: organization-defined elements of personally identifiable information]: [Assignment: organization-defined processing purposes].	Functional	Intersects With	Data Tagging	PRI-11	Mechanisms exist to issue data modeling guidelines to support tagging of sensitive and/or regulated data.	5		

31 of 52

Secure Controls Framework (SCF)

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SA-03(01)	System Development Life Cycle Manage Preproduction Environment	Protect system preproduction environments commensurate with risk throughout the system development life cycle for the system, system component, or system service.	Functional	Intersects With	Technology Lifecycle Management	SEA-07.1	Mechanisms exist to manage the usable lifecycles of Technology Assets, Applications and/or Services (TAAS).	5		
SA-03(01)	System Development Life Cycle Manage Preproduction Environment	Protect system preproduction environments commensurate with risk throughout the system development life cycle for the system, system component, or system service.	Functional	Intersects With	Secure Development Life Cycle (SDLC) Management	PRM-07	Mechanisms exist to ensure changes to Technology Assets, Applications and/or Services (TAAS) within the system development life cycle.	5		
SA-03(01)	System Development Life Cycle Manage Preproduction Environment	Protect system preproduction environments commensurate with risk throughout the system development life cycle for the system, system component, or system service.	Functional	Intersects With	Secure Development Environments	TDA-07	Mechanisms exist to maintain a segmented development network to ensure a secure development environment.	5		
SA-03(02)	System Development Life Cycle Use of Live or Operational Data	a. Approve, document, and control the use of live data in preproduction environments for the system, system component, or system service; andb. Protect preproduction environments for the system, system component, or system service.	Functional	Equal	Use of Live Data	TDA-10	Mechanisms exist to approve, document and control the use of live data in development and test environments.	10		
SA-03(03)	System Development Life Cycle Technology Refresh	Plan for and implement a technology refresh schedule for the system throughout the system development life cycle.	Functional	Intersects With	Technology Lifecycle Management	SEA-07.1	Mechanisms exist to manage the usable lifecycles of Technology Assets, Applications and/or Services (TAAS).	5		
SA-03(03)	System Development Life Cycle Technology Refresh	Plan for and implement a technology refresh schedule for the system throughout the system development life cycle.	Functional	Intersects With	Refresh from Trusted Sources	SEA-08.1	Mechanisms exist to ensure that software and data needed for system component and service refreshes are accurate and reliable.	5		
SA-04	Acquisition Process	Include the following requirements, descriptions, and criteria, explicitly or by reference, using [Selection (one or more): standardized contract language; [Assignment: organization-defined contract language]] in the acquisition contract for the system, system component, or system service.	Functional	Intersects With	Minimum Viable Product (MVP) Security Requirements	TDA-02	Mechanisms exist to design, develop and produce Technology Assets, Applications and/or Services (TAAS) in accordance with the system, system component, or system service requirements.	5		SA-04
SA-04	Acquisition Process	Include the following requirements, descriptions, and criteria, explicitly or by reference, using [Selection (one or more): standardized contract language; [Assignment: organization-defined contract language]] in the acquisition contract for the system, system component, or system service.	Functional	Intersects With	Third-Party Management	TPM-01	Mechanisms exist to facilitate the implementation of third-party management controls.	5		SA-04
SA-04	Acquisition Process	Include the following requirements, descriptions, and criteria, explicitly or by reference, using [Selection (one or more): standardized contract language; [Assignment: organization-defined contract language]] in the acquisition contract for the system, system component, or system service.	Functional	Intersects With	Technology Development & Acquisition	TDA-01	Mechanisms exist to facilitate the implementation of tailored development and acquisition strategies, contract tools and processes.	5		SA-04
SA-04	Acquisition Process	Include the following requirements, descriptions, and criteria, explicitly or by reference, using [Selection (one or more): standardized contract language; [Assignment: organization-defined contract language]] in the acquisition contract for the system, system component, or system service.	Functional	Intersects With	Managing Changes To Third-Party Services	TPM-10	Mechanisms exist to control changes to services by suppliers, taking into account the criticality of business processes.	5		SA-04
SA-04(01)	Acquisition Process Functional Properties of Controls	Require the developer of the system, system component, or system service to provide a description of the functional properties of the controls to be implemented.	Functional	Intersects With	Functional Properties	TDA-04.1	Mechanisms exist to require software developers to provide information describing the functional properties of controls.	5		
SA-04(01)	Acquisition Process Functional Properties of Controls	Require the developer of the system, system component, or system service to provide a description of the functional properties of the controls to be implemented.	Functional	Intersects With	Network Diagrams & Data Flow Diagrams (DFDs)	AST-04	Mechanisms exist to maintain network architecture diagrams that: (1) Contain sufficient detail to assess the security of the network's architecture.	5		
SA-04(02)	Acquisition Process Design and Implementation Information for Controls	Require the developer of the system, system component, or system service to provide design and implementation information for the controls that includes: [Selection (one or more): security-relevant external system interfaces; high-level information for controls; and low-level information for controls]	Functional	Intersects With	Network Diagrams & Data Flow Diagrams (DFDs)	AST-04	Mechanisms exist to maintain network architecture diagrams that: (1) Contain sufficient detail to assess the security of the network's architecture.	5		
SA-04(02)	Acquisition Process Design and Implementation Information for Controls	Require the developer of the system, system component, or system service to provide design and implementation information for the controls that includes: [Selection (one or more): security-relevant external system interfaces; high-level information for controls; and low-level information for controls]	Functional	Intersects With	Access to Program Source Code	TDA-20	Mechanisms exist to limit privileges to change software resident within software libraries.	5		
SA-04(02)	Acquisition Process Design and Implementation Information for Controls	Require the developer of the system, system component, or system service to provide design and implementation information for the controls that includes: [Selection (one or more): security-relevant external system interfaces; high-level information for controls; and low-level information for controls]	Functional	Intersects With	Functional Properties	TDA-04.1	Mechanisms exist to require software developers to provide information describing the functional properties of controls.	5		
SA-04(03)	Acquisition Process Development Methods, Techniques, and Practices	Require the developer of the system, system component, or system service to demonstrate the use of a system development life cycle process that includes:a. [Assignment: organization-defined systems engineering methods];b. [Assignment: organization-defined development practices]; andc. [Assignment: organization-defined testing practices].	Functional	Intersects With	Development Methods, Techniques & Processes	TDA-02.3	Mechanisms exist to require software developers to ensure that their software development processes employ industry-recognized secure development practices.	5		
SA-04(03)	Acquisition Process Development Methods, Techniques, and Practices	Require the developer of the system, system component, or system service to demonstrate the use of a system development life cycle process that includes:a. [Assignment: organization-defined systems engineering methods];b. [Assignment: organization-defined development practices]; andc. [Assignment: organization-defined testing practices].	Functional	Intersects With	Secure Software Development Practices (SSDP)	TDA-06	Mechanisms exist to develop applications based on Secure Software Development Practices (SSDP).	5		
SA-04(04)	Acquisition Process Development Methods, Techniques, and Practices	Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-04(05)	Acquisition Process System, Component, and Service Configurations	Require the developer of the system, system component, or system service to:a. Deliver the system, component, or service with [Assignment: organization-defined security configurations] implemented; andb. Use the configurations as the default for all subsequent system, component, or service configurations.	Functional	Equal	Pre-Established Secure Configurations	TDA-02.4	Mechanisms exist to ensure vendors / manufacturers: (1) Deliver the Technology Asset, Application, and/or Service (TAAS) with configuration settings that meet the system, system component, or system service requirements.	10		
SA-04(06)	Acquisition Process Use of Information Assurance Products	Employ only government off-the-shelf or commercial off-the-shelf information assurance and information assurance-enabled information technology products that compose an NSA-approved solution to protect classified information when the use of commercially provided information assurance and information assurance-enabled information technology products to those products that have been successfully evaluated against a National Information Assurance partnership.	Functional	Equal	Commercial Off-The-Shelf (COTS) Security Solutions	TDA-03	Mechanisms exist to utilize only Commercial Off-the-Shelf (COTS) security products.	10		
SA-04(07)	Acquisition Process NIAP-approved Protection Profiles	Require the developer of the system, system component, or system service to produce a plan for continuous monitoring of control effectiveness that is consistent with the continuous monitoring program of the organization.	Functional	Intersects With	Information Assurance Enabled Products	TDA-02.2	Mechanisms exist to limit the use of commercially-provided Information Assurance (IA) and IA-enabled IT products to those products that have been successfully evaluated against a National Information Assurance partnership.	5		
SA-04(08)	Acquisition Process Continuous Monitoring Plan for Controls	Require the developer of the system, system component, or system service to produce a plan for continuous monitoring of control effectiveness that is consistent with the continuous monitoring program of the organization.	Functional	Equal	Continuous Monitoring Plan	TDA-09.1	Mechanisms exist to require the developers of Technology Assets, Applications and/or Services (TAAS) to implement continuous monitoring of control effectiveness.	10		
SA-04(09)	Acquisition Process Functions, Ports, Protocols, and Services in Use	Require the developer of the system, system component, or system service to identify the functions, ports, protocols, and services intended for organizational use.	Functional	Equal	Ports, Protocols & Services In Use	TDA-02.1	Mechanisms exist to require the developers of Technology Assets, Applications and/or Services (TAAS) to implement controls that ensure the security of the system, system component, or system service.	10		
SA-04(10)	Acquisition Process Use of Approved PIV Products	Employ only information technology products on the FIPS 201-approved products list for Personal Identity Verification (PIV) capability implemented within organizational systems.	Functional	Intersects With	Information Assurance Enabled Products	TDA-02.2	Mechanisms exist to limit the use of commercially-provided Information Assurance (IA) and IA-enabled IT products to those products that have been successfully evaluated against a National Information Assurance partnership.	5		SA-04(10)
SA-04(11)	Acquisition Process System of Records	Include [Assignment: organization-defined Privacy Act requirements] in the acquisition contract for the operation of a system of records on behalf of an organization to accomplish an organizational mission or function.	Functional	No Relationship	N/A	N/A	N/A	0		
SA-04(12)	Acquisition Process Data Ownership	a. Include organizational data ownership requirements in the acquisition contract; andb. Require all data to be removed from the contractor's system and returned to the organization within [Assignment: organization-defined time frame].	Functional	Intersects With	Personal Data (PD) Lineage	PRI-09	Mechanisms exist to maintain a process to document the lineage of Personal Data (PD) by recording how the organization collects, stores, uses, and disseminates the data.	5		SA-04(12)
SA-04(12)	Acquisition Process Data Ownership	a. Include organizational data ownership requirements in the acquisition contract; andb. Require all data to be removed from the contractor's system and returned to the organization within [Assignment: organization-defined time frame].	Functional	Intersects With	Data Stewardship	DCH-01.1	Mechanisms exist to ensure data stewardship is assigned, documented and communicated.	5		SA-04(12)

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SA-04(12)	Acquisition Process Data Ownership	a. Include organizational data ownership requirements in the acquisition contract; andb. Require all data to be removed from the contractor's system and returned to the organization within [Assignment: organization-defined time frame].	Functional	Intersects With	Asset Ownership Assignment	AST-03	Mechanisms exist to ensure asset ownership responsibilities are assigned, tracked and managed at a team, individual or responsible.	5		SA-04(12)
SA-05	System Documentation	a. Obtain or develop administrator documentation for the system, system component, or system service that describes:1. Secure configuration, installation, and operation of the system, component, or service;2. Effective use and maintenance of	Functional	Intersects With	Documentation Requirements	TDA-04	Mechanisms exist to obtain, protect and distribute administrator documentation for Technology Assets.	5		SA-05
SA-05	System Documentation	a. Obtain or develop administrator documentation for the system, system component, or system service that describes:1. Secure configuration, installation, and operation of the system, component, or service;2. Effective use and maintenance of	Functional	Intersects With	Asset Scope Classification	AST-04.1	Mechanisms exist to determine security, compliance and resilience control applicability by identifying, classifying and documenting the	5		SA-05
SA-05(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-05(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-05(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-05(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-05(05)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-06		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-07		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-08	Security and Privacy Engineering Principles	Apply the following systems security and privacy engineering principles in the specification, design, development, implementation, and modification of the system and system components: [Assignment: organization-defined systems	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for	5		SA-08
SA-08	Security and Privacy Engineering Principles	Apply the following systems security and privacy engineering principles in the specification, design, development, implementation, and modification of the system and system components: [Assignment: organization-defined systems	Functional	Intersects With	Secure Engineering Principles	SEA-01	Mechanisms exist to facilitate the implementation of industry-recognized security, compliance and resilience practices to the	5		SA-08
SA-08(01)	Security and Privacy Engineering Principles Clear Abstractions	Implement the security design principle of clear abstractions.	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(02)	Security and Privacy Engineering Principles Least Common Mechanism	Implement the security design principle of least common mechanism in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(03)	Security and Privacy Engineering Principles Modularity and Layering	Implement the security design principles of modularity and layering in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(04)	Security and Privacy Engineering Principles Partially Ordered Dependencies	Implement the security design principle of partially ordered dependencies in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(05)	Security and Privacy Engineering Principles Efficiently Mediated Access	Implement the security design principle of efficiently mediated access in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(06)	Security and Privacy Engineering Principles Minimized Sharing	Implement the security design principle of minimized sharing in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(07)	Security and Privacy Engineering Principles Reduced Complexity	Implement the security design principle of reduced complexity in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(08)	Security and Privacy Engineering Principles Secure Evolvability	Implement the security design principle of secure evolvability in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(09)	Security and Privacy Engineering Principles Trusted Components	Implement the security design principle of trusted components in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(10)	Security and Privacy Engineering Principles Hierarchical Trust	Implement the security design principle of hierarchical trust in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(11)	Security and Privacy Engineering Principles Inverse Modification Threshold	Implement the security design principle of inverse modification threshold in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(12)	Security and Privacy Engineering Principles Hierarchical Protection	Implement the security design principle of hierarchical protection in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(13)	Security and Privacy Engineering Principles Minimized Security Elements	Implement the security design principle of minimized security elements in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(14)	Security and Privacy Engineering Principles Least Privilege	Implement the security design principle of least privilege in [Assignment: organization-defined systems or system components].	Functional	Equal	Least Privilege	IAC-21	Mechanisms exist to utilize the concept of least privilege, allowing only authorized access to processes necessary to accomplish assigned	10		
SA-08(15)	Security and Privacy Engineering Principles Predicate Permission	Implement the security design principle of predicate permission in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(16)	Security and Privacy Engineering Principles Self-reliant Trustworthiness	Implement the security design principle of self-reliant trustworthiness in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(17)	Security and Privacy Engineering Principles Secure Distributed Composition	Implement the security design principle of secure distributed composition in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(18)	Security and Privacy Engineering Principles Trusted Communications	Implement the security design principle of trusted communications channels in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(19)	Security and Privacy Engineering Principles Continuous Protection	Implement the security design principle of continuous protection in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(20)	Security and Privacy Engineering Principles Secure Metadata Management	Implement the security design principle of secure metadata management in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SA-08(21)	Security and Privacy Engineering Principles Self-analysis	Implement the security design principle of self-analysis in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(22)	Security and Privacy Engineering Principles Accountability and Traceability	Implement the security design principle of accountability and traceability in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(23)	Security and Privacy Engineering Principles Secure Defaults	Implement the security design principle of secure defaults in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(24)	Security and Privacy Engineering Principles Secure Failure and Recovery	Implement the security design principle of secure failure and recovery in [Assignment: organization-defined systems or system components].	Functional	Equal	Fail Secure	SEA-07.2	Mechanisms exist to enable systems to fail to an organization-defined known-state for types of failures, preserving system state information in	10		
SA-08(25)	Security and Privacy Engineering Principles Economic Security	Implement the security design principle of economic security in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(26)	Security and Privacy Engineering Principles Performance Security	Implement the security design principle of performance security in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(27)	Security and Privacy Engineering Principles Human Factored Security	Implement the security design principle of human factored security in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(28)	Security and Privacy Engineering Principles Acceptable Security	Implement the security design principle of acceptable security in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(29)	Security and Privacy Engineering Principles Repeatable and Documented	Implement the security design principle of repeatable and documented procedures in [Assignment: organization-defined systems or system components].	Functional	No Relationship	N/A	N/A	N/A	0		
SA-08(30)	Security and Privacy Engineering Principles Procedural Rigor	Implement the security design principle of procedural rigor in [Assignment: organization-defined systems or system components].	Functional	Intersects With	Secure Development Life Cycle (SDLC) Management	PRM-07	Mechanisms exist to ensure changes to Technology Assets, Applications and/or Services (TAAS) within the	5		
SA-08(30)	Security and Privacy Engineering Principles Procedural Rigor	Implement the security design principle of procedural rigor in [Assignment: organization-defined systems or system components].	Functional	Intersects With	Technology Lifecycle Management	SEA-07.1	Mechanisms exist to manage the usable lifecycles of Technology Assets, Applications and/or Services	5		
SA-08(31)	Security and Privacy Engineering Principles Secure System Modification	Implement the security design principle of secure system modification in [Assignment: organization-defined systems or system components].	Functional	Intersects With	Configuration Change Control	CHG-02	Mechanisms exist to govern the technical configuration change control processes.	5		
SA-08(31)	Security and Privacy Engineering Principles Secure System Modification	Implement the security design principle of secure system modification in [Assignment: organization-defined systems or system components].	Functional	Intersects With	Control Functionality Verification	CHG-06	Mechanisms exist to verify the functionality of security, compliance and resilience controls following implemented changes to ensure	5		
SA-08(31)	Security and Privacy Engineering Principles Secure System Modification	Implement the security design principle of secure system modification in [Assignment: organization-defined systems or system components].	Functional	Intersects With	Test, Validate & Document Changes	CHG-02.2	Mechanisms exist to appropriately test and document proposed changes in a non-production environment before changes are implemented in a	5		
SA-08(32)	Security and Privacy Engineering Principles Sufficient Documentation	Implement the security design principle of sufficient documentation in [Assignment: organization-defined systems or system components].	Functional	Equal	Standardized Operating Procedures (SOP)	OPS-01.1	Mechanisms exist to identify and document Standardized Operating Procedures (SOP), or similar documentation, to enable the proper	10		
SA-08(33)	Security and Privacy Engineering Principles Minimization	Implement the privacy principle of minimization using [Assignment: organization-defined processes].	Functional	Intersects With	Collection Minimization	END-13.3	Mechanisms exist to utilize sensors that are configured to minimize the collection of information about individuals	5		
SA-08(33)	Security and Privacy Engineering Principles Minimization	Implement the privacy principle of minimization using [Assignment: organization-defined processes].	Functional	Intersects With	Limit Sensitive / Regulated Data In Testing, Training & Research	DCH-18.2	Mechanisms exist to minimize the use of sensitive and/or regulated data for research, testing, or training, in accordance with authorized	5		
SA-08(33)	Security and Privacy Engineering Principles Minimization	Implement the privacy principle of minimization using [Assignment: organization-defined processes].	Functional	Intersects With	Minimize Visitor Personal Data (PD)	PES-06.5	Mechanisms exist to minimize the collection of Personal Data (PD) contained in visitor access records.	5		
SA-09	External System Services	a. Require that providers of external system services comply with organizational security and privacy requirements and employ the following controls: [Assignment: organization-defined controls]. b. Define and document organizational security and privacy requirements for the acquisition or outsourcing of information security services; and b. Verify that the acquisition or outsourcing of dedicated information security services is approved by [Assignment: organization-defined external system services].	Functional	Equal	Third-Party Services	TPM-04	Mechanisms exist to mitigate the risks associated with third-party access to the organization's Technology Assets, Applications and/or Services (TAAS).	10		SA-09
SA-09(01)	External System Services Risk Assessments and Organizational Services	a. Conduct an organizational assessment of risk prior to the acquisition or outsourcing of information security services; and b. Verify that the acquisition or outsourcing of dedicated information security services is approved by [Assignment: organization-defined external system services].	Functional	Equal	Third-Party Risk Assessments & Approvals	TPM-04.1	Mechanisms exist to conduct a risk assessment prior to the acquisition or outsourcing of technology-related	10		
SA-09(02)	External System Services Identification of Functions, Ports, External System Services	Require providers of the following external system services to identify the functions, ports, protocols, and other services required for the use of such services: [Assignment: organization-defined security and privacy requirements, properties, factors]	Functional	Equal	External Connectivity Requirements - Identification of Ports, Protocols & Services	TPM-04.2	Mechanisms exist to require External Service Providers (ESPs) to identify and document the business need for	10		
SA-09(03)	External System Services Establish and Maintain Trust Relationship with	Establish, document, and maintain trust relationships with external service providers based on the following requirements, properties, factors, or conditions: [Assignment: organization-defined security and privacy requirements, properties, factors]	Functional	Intersects With	Supply Chain Risk Management (SCRM) Plan	RSK-09	Mechanisms exist to develop a plan for Supply Chain Risk Management (SCRM) associated with the	5		
SA-09(03)	External System Services Establish and Maintain Trust Relationship with	Establish, document, and maintain trust relationships with external service providers based on the following requirements, properties, factors, or conditions: [Assignment: organization-defined security and privacy requirements, properties, factors]	Functional	Intersects With	Conflict of Interests	TPM-04.3	Mechanisms exist to ensure that the interests of external service providers are consistent with and reflect	5		
SA-09(03)	External System Services Establish and Maintain Trust Relationship with	Establish, document, and maintain trust relationships with external service providers based on the following requirements, properties, factors, or conditions: [Assignment: organization-defined security and privacy requirements, properties, factors]	Functional	Intersects With	Third-Party Criticality Assessments	TPM-02	Mechanisms exist to identify, prioritize, and assess suppliers and partners of critical Technology Assets, Applications and/or Services (TAAS).	5		
SA-09(03)	External System Services Establish and Maintain Trust Relationship with	Establish, document, and maintain trust relationships with external service providers based on the following requirements, properties, factors, or conditions: [Assignment: organization-defined security and privacy requirements, properties, factors]	Functional	Intersects With	Supply Chain Risk Management (SCRM)	TPM-03	Mechanisms exist to: (1) Evaluate security risks and threats associated with Technology Assets, Applications and/or Services (TAAS).	5		
SA-09(03)	External System Services Establish and Maintain Trust Relationship with	Establish, document, and maintain trust relationships with external service providers based on the following requirements, properties, factors, or conditions: [Assignment: organization-defined security and privacy requirements, properties, factors]	Functional	Intersects With	Responsible, Accountable, Supportive, Consulted & Informed (RASCII)	TPM-05.4	Mechanisms exist to document and maintain a Responsible, Accountable, Supportive, Consulted & Informed (RASCII) matrix or similar	5		
SA-09(03)	External System Services Establish and Maintain Trust Relationship with	Establish, document, and maintain trust relationships with external service providers based on the following requirements, properties, factors, or conditions: [Assignment: organization-defined security and privacy requirements, properties, factors]	Functional	Intersects With	Break Clauses	TPM-05.7	Mechanisms exist to include "break clauses" within contracts for failure to meet contract criteria for security,	5		

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

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FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SA-09(04)	External System Services Consistent Interests of Consumers and External System	Take the following actions to verify that the interests of [Assignment: organization-defined external service providers] are consistent with and reflect organizational interests: [Assignment: organization-defined actions]	Functional	Equal	Conflict of Interests	TPM-04.3	Mechanisms exist to ensure that the interests of external service providers are consistent with and reflect organizational interests	10		
SA-09(05)	External System Services Processing, Storage, and Service Location	Restrict the location of [Selection (one or more): information processing; information or data; system services] to [Assignment: organization-defined locations] based on [Assignment: organization-defined requirements or conditions]	Functional	Intersects With	Geolocation Requirements for Processing, Storage and Service Locations	CLD-09	Mechanisms exist to control the location of cloud processing/storage based on business requirements that includes statutory, regulatory and	5		
SA-09(05)	External System Services Processing, Storage, and Service Location	Restrict the location of [Selection (one or more): information processing; information or data; system services] to [Assignment: organization-defined locations] based on [Assignment: organization-defined requirements or conditions]	Functional	Intersects With	Third-Party Processing, Storage and Service Locations	TPM-04.4	Mechanisms exist to restrict the location of information processing/storage based on business requirements	5		
SA-09(05)	External System Services Processing, Storage, and Service Location	Restrict the location of [Selection (one or more): information processing; information or data; system services] to [Assignment: organization-defined locations] based on [Assignment: organization-defined requirements or conditions]	Functional	Intersects With	Geographic Location of Data	DCH-19	Mechanisms exist to inventory, document and maintain data flows for data that is resident (permanently or temporarily) within a service's	5		
SA-09(06)	External System Services Organization controlled Cryptographic Keys	Maintain exclusive control of cryptographic keys for encrypted material stored or transmitted through an external system.	Functional	Equal	External System Cryptographic Key Control	CRY-09.7	Mechanisms exist to maintain control of cryptographic keys for encrypted material stored or transmitted through an external system.	10		
SA-09(07)	External System Services Organization controlled Integrity Checking	Provide the capability to check the integrity of information while it resides in the external system.	Functional	No Relationship	N/A	N/A	N/A	0		
SA-09(08)	External System Services Processing and Storage Location — U.S. Jurisdiction	Restrict the geographic location of information processing and data storage to facilities located within in the legal jurisdictional boundary of the United States.	Functional	Intersects With	Geographic Location of Data	DCH-19	Mechanisms exist to inventory, document and maintain data flows for data that is resident (permanently or temporarily) within a service's	5		
SA-09(08)	External System Services Processing and Storage Location — U.S. Jurisdiction	Restrict the geographic location of information processing and data storage to facilities located within in the legal jurisdictional boundary of the United States.	Functional	Intersects With	Geolocation Requirements for Processing, Storage and Service Locations	CLD-09	Mechanisms exist to control the location of cloud processing/storage based on business requirements that includes statutory, regulatory and	5		
SA-10	Developer Configuration Management	Require the developer of the system, system component, or system service to: a. Perform configuration management during system, component, or service [Selection (one or more): design; development; implementation; operation; disposal] b.	Functional	Equal	Developer Configuration Management	TDA-14	Mechanisms exist to require system developers and integrators to perform configuration management during system design, development	10		
SA-10(01)	Developer Configuration Management Software and	Require the developer of the system, system component, or system service to enable integrity verification of software and firmware components.	Functional	Equal	Software / Firmware Integrity Verification	TDA-14.1	Mechanisms exist to require developers of Technology Assets, Applications and/or Services (TAAS) to enable integrity verification of	10		
SA-10(02)	Developer Configuration Management Alternative Developer	Provide an alternate configuration management process using organizational personnel in the absence of a dedicated developer configuration management team.	Functional	No Relationship	N/A	N/A	N/A	0		
SA-10(03)	Developer Configuration Management Hardware Integrity	Require the developer of the system, system component, or system service to enable integrity verification of hardware components.	Functional	Equal	Hardware Integrity Verification	TDA-14.2	Mechanisms exist to require developers of Technology Assets, Applications and/or Services (TAAS) to enable integrity verification of	10		
SA-10(04)	Developer Configuration Management Trusted Generation	Require the developer of the system, system component, or system service to employ tools for comparing newly generated versions of security-relevant hardware descriptions, source code, and object code with previous versions.	Functional	No Relationship	N/A	N/A	N/A	0		
SA-10(05)	Developer Configuration Management Mapping Integrity for	Require the developer of the system, system component, or system service to maintain the integrity of the mapping between the master build data describing the current version of security-relevant hardware, software, and firmware and the on-site	Functional	No Relationship	N/A	N/A	N/A	0		
SA-10(06)	Developer Configuration Management Trusted Distribution	Require the developer of the system, system component, or system service to execute procedures for ensuring that security-relevant hardware, software, and firmware updates distributed to the organization are exactly as specified by the master	Functional	No Relationship	N/A	N/A	N/A	0		
SA-10(07)	Developer Configuration Management Security and Privacy	Require [Assignment: organization-defined security and privacy representatives] to be included in the [Assignment: organization-defined configuration change management and control process].	Functional	Equal	Security, Compliance & Resilience Representatives For Product Changes	TDA-02.7	Mechanisms exist to include appropriate security, compliance and resilience representatives in the product feature and/or functionality	10		
SA-11	Developer Testing and Evaluation	Require the developer of the system, system component, or system service, at all post-design stages of the system development life cycle, to: a. Develop and implement a plan for ongoing security and privacy control assessments b. Perform	Functional	Equal	Security, Compliance & Resilience Testing Throughout Development	TDA-09	Mechanisms exist to require system developers/integrators consult with security, compliance and/or resilience personnel to:	10		
SA-11(01)	Developer Testing and Evaluation Static Code Analysis	Require the developer of the system, system component, or system service to employ static code analysis tools to identify common flaws and document the results of the analysis.	Functional	Equal	Static Code Analysis	TDA-09.2	Mechanisms exist to require the developers of Technology Assets, Applications and/or Services (TAAS) to	10		
SA-11(02)	Developer Testing and Evaluation Threat Modeling and Vulnerability Analyses	Require the developer of the system, system component, or system service to perform threat modeling and vulnerability analyses during development and the subsequent testing and evaluation of the system, component, or service that: a. Uses	Functional	Intersects With	Threat Analysis & Flaw Remediation During Development	TDA-15	Mechanisms exist to require system developers and integrators to develop and implement an ongoing Security Testing and Evaluation (ST&E) plan or	5		
SA-11(02)	Developer Testing and Evaluation Threat Modeling and Vulnerability Analyses	Require the developer of the system, system component, or system service to perform threat modeling and vulnerability analyses during development and the subsequent testing and evaluation of the system, component, or service that: a. Uses	Functional	Intersects With	Threat Modeling	TDA-06.2	Mechanisms exist to perform threat modelling and other secure design techniques, to ensure that threats to software and solutions are identified	5		
SA-11(03)	Developer Testing and Evaluation Independent Verification of	Require an independent agent satisfying [Assignment: organization-defined independence criteria] to verify the correct implementation of the developer security and privacy assessment plans and the evidence produced during testing.	Functional	No Relationship	N/A	N/A	N/A	0		
SA-11(04)	Developer Testing and Evaluation Manual Code Reviews	Require the developer of the system, system component, or system service to perform a manual code review of [Assignment: organization-defined specific code] using the following processes, procedures, and/or techniques:	Functional	Equal	Manual Code Review	TDA-09.7	Mechanisms exist to require the developers of Technology Assets, Applications and/or Services (TAAS) to	10		
SA-11(05)	Developer Testing and Evaluation Penetration Testing	Require the developer of the system, system component, or system service to perform penetration testing: a. At the following level of rigor: [Assignment: organization-defined breadth and depth of testing]; and b. Under the following constraints:	Functional	Intersects With	Threat Analysis & Flaw Remediation During Development	IAO-04	Mechanisms exist to require system developers and integrators to create and execute a Security Testing and Evaluation (ST&E) plan or similar	5		
SA-11(05)	Developer Testing and Evaluation Penetration Testing	Require the developer of the system, system component, or system service to perform penetration testing: a. At the following level of rigor: [Assignment: organization-defined breadth and depth of testing]; and b. Under the following constraints:	Functional	Intersects With	Application Penetration Testing	TDA-09.5	Mechanisms exist to perform application-level penetration testing of custom-made Technology Assets, Applications and/or Services (TAAS)	5		
SA-11(05)	Developer Testing and Evaluation Penetration Testing	Require the developer of the system, system component, or system service to perform penetration testing: a. At the following level of rigor: [Assignment: organization-defined breadth and depth of testing]; and b. Under the following constraints:	Functional	Intersects With	Security, Compliance & Resilience Testing Throughout Development	TDA-09	Mechanisms exist to require system developers/integrators consult with security, compliance and/or resilience personnel to:	5		
SA-11(05)	Developer Testing and Evaluation Penetration Testing	Require the developer of the system, system component, or system service to perform penetration testing: a. At the following level of rigor: [Assignment: organization-defined breadth and depth of testing]; and b. Under the following constraints:	Functional	Intersects With	Specialized Assessments	IAO-02.2	Mechanisms exist to conduct specialized assessments for: (1) Statutory, regulatory and contractual compliance obligations;	5		
SA-11(05)	Developer Testing and Evaluation Penetration Testing	Require the developer of the system, system component, or system service to perform penetration testing: a. At the following level of rigor: [Assignment: organization-defined breadth and depth of testing]; and b. Under the following constraints:	Functional	Intersects With	Penetration Testing	VPM-07	Mechanisms exist to conduct penetration testing on Technology Assets, Applications and/or Services (TAAS)	5		

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SA-11(06)	Developer Testing and Evaluation Attack Surface Reviews	Require the developer of the system, system component, or system service to perform attack surface reviews.	Functional	Intersects With	Security, Compliance & Resilience Testing Throughout Development	TDA-09	Mechanisms exist to require system developers/integrators consult with security, compliance and/or resilience personnel to:	5		
SA-11(06)	Developer Testing and Evaluation Attack Surface Reviews	Require the developer of the system, system component, or system service to perform attack surface reviews.	Functional	Intersects With	Attack Surface Scope	VPM-01.1	Mechanisms exist to define and manage the scope for its attack surface management activities.	5		
SA-11(07)	Developer Testing and Evaluation Verify Scope of Testing and Evaluation	Require the developer of the system, system component, or system service to verify that the scope of testing and evaluation provides complete coverage of the required controls at the following level of rigor: [Assignment: organization-defined]	Functional	Intersects With	Attack Surface Scope	VPM-01.1	Mechanisms exist to define and manage the scope for its attack surface management activities.	5		
SA-11(07)	Developer Testing and Evaluation Verify Scope of Testing and Evaluation	Require the developer of the system, system component, or system service to verify that the scope of testing and evaluation provides complete coverage of the required controls at the following level of rigor: [Assignment: organization-defined]	Functional	Intersects With	Security, Compliance & Resilience Testing Throughout Development	TDA-09	Mechanisms exist to require system developers/integrators consult with security, compliance and/or resilience personnel to:	5		
SA-11(08)	Developer Testing and Evaluation Dynamic Code Analysis	Require the developer of the system, system component, or system service to employ dynamic code analysis tools to identify common flaws and document the results of the analysis.	Functional	Equal	Dynamic Code Analysis	TDA-09.3	Mechanisms exist to require the developers of Technology Assets, Applications and/or Services (TAAS) to	10		
SA-11(09)	Developer Testing and Evaluation Interactive Application Security Testing	Require the developer of the system, system component, or system service to employ interactive application security testing tools to identify flaws and document the results.	Functional	No Relationship	N/A	N/A	N/A	0		
SA-12		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-12(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-12(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-12(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-12(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-12(05)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-12(06)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-12(07)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-12(08)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-12(09)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-12(10)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-12(11)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-12(12)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-12(13)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-12(14)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-12(15)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-13		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-14		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-14(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-15	Development Process, Standards, and Tools	a. Require the developer of the system, system component, or system service to follow a documented development process that:1. Explicitly addresses security and privacy requirements;2. Identifies the standards and tools used in the development Require the developer of the system, system component, or system service to:a. Define quality metrics at the beginning of the development process; andb. Provide evidence of meeting the quality metrics [Selection (one or more): [Assignment: organization-defined]]	Functional	Equal	Secure Software Development Practices (SSDP)	TDA-06	Mechanisms exist to develop applications based on Secure Software Development Practices (SSDP)	10		
SA-15(01)	Development Process, Standards, and Tools Quality Metrics	Require the developer of the system, system component, or system service to:a. Define quality metrics at the beginning of the development process; andb. Provide evidence of meeting the quality metrics [Selection (one or more): [Assignment: organization-defined]]	Functional	No Relationship	N/A	N/A	N/A	0		
SA-15(02)	Development Process, Standards, and Tools Security and Privacy Tracking Tools	Require the developer of the system, system component, or system service to select and employ security and privacy tracking tools for use during the development process.	Functional	Equal	Capabilities Deficiency Tracking	IAO-05	Mechanisms exist to govern identified deficiencies (e.g., Plan of Action and Milestones (POA&M) or similar methodology) that formally	10		
SA-15(03)	Development Process, Standards, and Tools Criticality Analysis	Require the developer of the system, system component, or system service to perform a criticality analysis:a. At the following decision points in the system development life cycle: [Assignment: organization-defined decision points in the system	Functional	Equal	Criticality Analysis During Development	TDA-06.1	Mechanisms exist to require the developer of the Technology Asset, Application and/or Service (TAAS) to	10		
SA-15(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-15(05)	Development Process, Standards, and Tools Attack Surface Reduction	Require the developer of the system, system component, or system service to reduce attack surfaces to [Assignment: organization-defined thresholds].	Functional	Intersects With	Secure Baseline Configurations	CFG-02	Mechanisms exist to develop, document and maintain secure baseline configurations for	5		
SA-15(05)	Development Process, Standards, and Tools Attack Surface Reduction	Require the developer of the system, system component, or system service to reduce attack surfaces to [Assignment: organization-defined thresholds].	Functional	Intersects With	Secure Engineering Principles	SEA-01	Mechanisms exist to facilitate the implementation of industry-recognized security, compliance and	5		
SA-15(06)	Development Process, Standards, and Tools Continuous Improvement	Require the developer of the system, system component, or system service to implement an explicit process to continuously improve the development process.	Functional	No Relationship	N/A	N/A	N/A	0		
SA-15(07)	Development Process, Standards, and Tools Automated Vulnerability Analysis	Require the developer of the system, system component, or system service [Assignment: organization-defined frequency] to:a. Perform an automated vulnerability analysis using [Assignment: organization-defined tools];b. Determine the	Functional	No Relationship	N/A	N/A	N/A	0		
SA-15(08)	Development Process, Standards, and Tools Reuse of Threat and Vulnerability	Require the developer of the system, system component, or system service to use threat modeling and vulnerability analyses from similar systems, components, or services to inform the current development process.	Functional	Equal	Threat Modeling	TDA-06.2	Mechanisms exist to perform threat modelling and other secure design techniques, to ensure that threats to software and solutions are identified	10		
SA-15(09)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-15(10)	Development Process, Standards, and Tools Incident Response Plan	Require the developer of the system, system component, or system service to provide, implement, and test an incident response plan.	Functional	No Relationship	N/A	N/A	N/A	0		
SA-15(11)	Development Process, Standards, and Tools Archive System or Component	Require the developer of the system or system component to archive the system or component to be released or delivered together with the corresponding evidence supporting the final security and privacy review.	Functional	No Relationship	N/A	N/A	N/A	0		
SA-15(12)	Development Process, Standards, and Tools Minimize Personally Identifiable	Require the developer of the system or system component to minimize the use of personally identifiable information in development and test environments.	Functional	Equal	Limit Sensitive / Regulated Data In Testing, Training & Research	DCH-18.2	Mechanisms exist to minimize the use of sensitive and/or regulated data for research, testing, or training, in accordance with authorized.	10		
SA-15(13)	Development Process, Standards, and Tools Logging Syntax	Require the developer of the system, system component, or system service to use [Assignment: organization-defined secure logging format] to log [assignment: organization-defined event types] at [Assignment: organization-defined level of detail]	Functional	Equal	Logging Syntax	TDA-02.14	Mechanisms exist to require system developers to use an industry-defined secure logging format to generate	10		
SA-16	Developer-provided Training	Require the developer of the system, system component, or system service to provide the following training on the correct use and operation of the implemented security and privacy functions, controls, and/or mechanisms: [Assignment: organization-defined]	Functional	Equal	Developer-Provided Training	TDA-16	Mechanisms exist to require the developers of Technology Assets, Applications and/or Services (TAAS) to	10		
SA-17	Developer Security and Privacy Architecture and Design	Require the developer of the system, system component, or system service to produce a design specification and security and privacy architecture that:a. Is consistent with the organization's security and privacy architecture that is an	Functional	Equal	Developer Architecture & Design	TDA-05	Mechanisms exist to require the developers of Technology Assets, Applications and/or Services (TAAS) to	10		

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SA-17(01)	Developer Security and Privacy Architecture and Design Formal Policy	Require the developer of the system, system component, or system service to:a. Produce, as an integral part of the development process, a formal policy model describing the [Assignment: organization-defined elements of organizational	Functional	No Relationship	N/A	N/A	N/A	0		
SA-17(02)	Developer Security and Privacy Architecture and Design Security	Require the developer of the system, system component, or system service to:a. Define security-relevant hardware, software, and firmware; andb. Provide a rationale that the definition for security-relevant hardware, software, and	Functional	No Relationship	N/A	N/A	N/A	0		
SA-17(03)	Developer Security and Privacy Architecture and Design Formal Policy	Require the developer of the system, system component, or system service to:a. Produce, as an integral part of the development process, a formal top-level specification that specifies the interfaces to security-relevant hardware, software, and	Functional	No Relationship	N/A	N/A	N/A	0		
SA-17(04)	Developer Security and Privacy Architecture and Design Informal	Require the developer of the system, system component, or system service to:a. Produce, as an integral part of the development process, an informal descriptive top-level specification that specifies the interfaces to security-relevant	Functional	No Relationship	N/A	N/A	N/A	0		
SA-17(05)	Developer Security and Privacy Architecture and Design Conceptually	Require the developer of the system, system component, or system service to:a. Design and structure the security-relevant hardware, software, and firmware to use a complete, conceptually simple protection mechanism with precisely	Functional	No Relationship	N/A	N/A	N/A	0		
SA-17(06)	Developer Security and Privacy Architecture and Design Structure for	Require the developer of the system, system component, or system service to structure security-relevant hardware, software, and firmware to facilitate testing.	Functional	No Relationship	N/A	N/A	N/A	0		
SA-17(07)	Developer Security and Privacy Architecture and Design Structure for	Require the developer of the system, system component, or system service to structure security-relevant hardware, software, and firmware to facilitate controlling access with least privileges	Functional	No Relationship	N/A	N/A	N/A	0		
SA-17(08)	Developer Security and Privacy Architecture and Design Orchestration	Design [Assignment: organization-defined critical systems or system components] with coordinated behavior to implement the following capabilities: [Assignment: organization-defined capabilities, by system or component]	Functional	No Relationship	N/A	N/A	N/A	0		
SA-17(09)	Developer Security and Privacy Architecture and Design Design	Use different designs for [Assignment: organization-defined critical systems or system components] to satisfy a common set of requirements or to provide equivalent functionality.	Functional	No Relationship	N/A	N/A	N/A	0		
SA-18		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-18(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-18(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-19		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-19(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-19(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-19(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-19(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-20	Customized Development of Critical Components	Reimplement or custom develop the following critical system components: [Assignment: organization-defined critical system components].	Functional	Equal	Customized Development of Critical Components	TDA-12	Mechanisms exist to custom-develop critical system components, when Commercial Off The Shelf (COTS) solutions are unavailable.	10		
SA-21	Developer Screening	Require that the developer of [Assignment: organization-defined system, system component, or system service]:a. Has appropriate access authorizations as determined by assigned [Assignment: organization-defined official government duties];	Functional	Equal	Developer Screening	TDA-13	Mechanisms exist to ensure that the developers of Technology Assets, Applications and/or Services (TAAS)	10		
SA-21(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-22	Unsupported System Components	a. Replace system components when support for the components is no longer available from the developer, vendor, or manufacturer; orb. Provide the following options for alternative sources for continued support for unsupported	Functional	Intersects With	Unsupported Technology Assets, Applications and/or Services (TAAS)	TDA-17	Mechanisms exist to prevent unsupported Technology Assets, Applications and/or Services (TAAS)	5		SA-22
SA-22	Unsupported System Components	a. Replace system components when support for the components is no longer available from the developer, vendor, or manufacturer; orb. Provide the following options for alternative sources for continued support for unsupported	Functional	Intersects With	Alternate Sources for Continued Support	TDA-17.1	Mechanisms exist to provide in-house support or contract external providers for support with unsupported	5		SA-22
SA-22(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SA-23	Specialization	Employ [Selection (one or more): design; modification; augmentation; reconfiguration] on [Assignment: organization-defined systems or system components] supporting mission essential services or functions to increase the trustworthiness	Functional	Intersects With	Technology Development & Acquisition	TDA-01	Mechanisms exist to facilitate the implementation of tailored development and acquisition strategies, contract tools and	5		
SA-23	Specialization	Employ [Selection (one or more): design; modification; augmentation; reconfiguration] on [Assignment: organization-defined systems or system components] supporting mission essential services or functions to increase the trustworthiness	Functional	Intersects With	Product Management	TDA-01.1	Mechanisms exist to design and implement product management processes to proactively govern the	5		
SA-23	Specialization	Employ [Selection (one or more): design; modification; augmentation; reconfiguration] on [Assignment: organization-defined systems or system components] supporting mission essential services or functions to increase the trustworthiness	Functional	Intersects With	Customized Development of Critical Components	TDA-12	Mechanisms exist to custom-develop critical system components, when Commercial Off The Shelf (COTS) solutions are unavailable.	5		
SC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level system and communications protection policy	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5		SC-01
SC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level system and communications protection policy	Functional	Subset Of	Network Security Controls (NSC)	NET-01	Mechanisms exist to develop, govern & update procedures to facilitate the implementation of Network Security Controls (NSC)	10		SC-01
SC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level system and communications protection policy	Functional	Subset Of	Secure Engineering Principles	SEA-01	Mechanisms exist to facilitate the implementation of industry-recognized security, compliance and	10		SC-01
SC-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level system and communications protection policy	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies,	5		SC-01
SC-02	Separation of System and User Functionality	Separate user functionality, including user interface services, from system management functionality.	Functional	Equal	Application Partitioning	SEA-03.2	Mechanisms exist to separate user functionality from system management functionality.	10		
SC-02(01)	Separation of System and User Functionality Interfaces for Non-privileged Users	Prevent the presentation of system management functionality at interfaces to non-privileged users.	Functional	Equal	Application Partitioning	SEA-03.2	Mechanisms exist to separate user functionality from system management functionality.	10		
SC-02(02)	Separation of System and User Functionality Disassociability	Store state information from applications and software separately.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-03	Security Function Isolation	Isolate security functions from nonsecurity functions.	Functional	Intersects With	Restrict Access To Security Functions	END-16	Mechanisms exist to ensure security functions are restricted to authorized individuals and enforce least privilege control requirements for necessary	5		

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SC-03	Security Function Isolation	Isolate security functions from nonsecurity functions.	Functional	Intersects With	Security Function Isolation	SEA-04.1	Mechanisms exist to isolate security functions from non-security functions.	5		
SC-03(01)	Security Function Isolation Hardware Separation	Employ hardware separation mechanisms to implement security function isolation.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-03(02)	Security Function Isolation Access and Flow Control Functions	Isolate security functions enforcing access and information flow control from nonsecurity functions and from other security functions.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-03(03)	Security Function Isolation Minimize Nonsecurity Functionality	Minimize the number of nonsecurity functions included within the isolation boundary containing security functions.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-03(04)	Security Function Isolation Module Coupling and Cohesiveness	Implement security functions as largely independent modules that maximize internal cohesiveness within modules and minimize coupling between modules.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-03(05)	Security Function Isolation Layered Structures	Implement security functions as a layered structure minimizing interactions between layers of the design and avoiding any dependence by lower layers on the functionality or correctness of higher layers.	Functional	Equal	Defense-In-Depth (DiD) Architecture	SEA-03	Mechanisms exist to implement security functions as a layered structure minimizing interactions between layers of the design and	10		
SC-04	Information in Shared System Resources	Prevent unauthorized and unintended information transfer via shared system resources.	Functional	Equal	Information In Shared Resources	SEA-05	Mechanisms exist to prevent unauthorized and unintended information transfer via shared system resources.	10		
SC-04(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-04(02)	Information in Shared System Resources Multilevel or Periods Processing	Prevent unauthorized information transfer via shared resources in accordance with [Assignment: organization-defined procedures] when system processing explicitly switches between different information classification levels or security a. (Selection one): Protect against; Limit the effects of the following types of denial-of-service events: [Assignment: organization-defined types of denial-of-service events]; andb. Employ the following controls to achieve the denial-of-service a. (Selection one): Protect against; Limit the effects of the following types of denial-of-service events: [Assignment: organization-defined types of denial-of-service events]; andb. Employ the following controls to achieve the denial-of-service	Functional	No Relationship	N/A	N/A	N/A	0		
SC-05	Denial-of-service Protection	a. (Selection one): Protect against; Limit the effects of the following types of denial-of-service events: [Assignment: organization-defined types of denial-of-service events]; andb. Employ the following controls to achieve the denial-of-service	Functional	Intersects With	Resource Priority	CAP-02	Mechanisms exist to control resource utilization of Technology Assets, Applications and/or Services (TAAS)	5		SC-05
SC-05	Denial-of-service Protection	a. (Selection one): Protect against; Limit the effects of the following types of denial-of-service events: [Assignment: organization-defined types of denial-of-service events]; andb. Employ the following controls to achieve the denial-of-service	Functional	Intersects With	Capacity Planning	CAP-03	Mechanisms exist to conduct capacity planning so that necessary capacity for information processing, telecommunications and	5		SC-05
SC-05	Denial-of-service Protection	a. (Selection one): Protect against; Limit the effects of the following types of denial-of-service events: [Assignment: organization-defined types of denial-of-service events]; andb. Employ the following controls to achieve the denial-of-service	Functional	Intersects With	Capacity & Performance Management	CAP-01	Mechanisms exist to facilitate the implementation of capacity management controls to ensure optimal system performance to meet	5		SC-05
SC-05	Denial-of-service Protection	a. (Selection one): Protect against; Limit the effects of the following types of denial-of-service events: [Assignment: organization-defined types of denial-of-service events]; andb. Employ the following controls to achieve the denial-of-service	Functional	Intersects With	Denial of Service (DoS) Protection	NET-02.1	Automated mechanisms exist to protect against or limit the effects of denial of service attacks.	5		SC-05
SC-05(01)	Denial-of-service Protection Restrict Ability to Attack Other Systems	Restrict the ability of individuals to launch the following denial-of-service attacks against other systems: [Assignment: organization-defined denial-of-service attacks].	Functional	Intersects With	Resource Priority	CAP-02	Mechanisms exist to control resource utilization of Technology Assets, Applications and/or Services (TAAS)	5		
SC-05(02)	Denial-of-service Protection Capacity, Bandwidth, and Redundancy	Manage capacity, bandwidth, or other redundancy to limit the effects of information flooding denial-of-service attacks.	Functional	Intersects With	Resource Priority	CAP-02	Mechanisms exist to control resource utilization of Technology Assets, Applications and/or Services (TAAS)	5		
SC-05(02)	Denial-of-service Protection Capacity, Bandwidth, and Redundancy	Manage capacity, bandwidth, or other redundancy to limit the effects of information flooding denial-of-service attacks.	Functional	Intersects With	Capacity Planning	CAP-03	Mechanisms exist to conduct capacity planning so that necessary capacity for information processing, telecommunications and	5		
SC-05(03)	Denial-of-service Protection Detection and Monitoring	a. Employ the following monitoring tools to detect indicators of denial-of-service attacks against, or launched from, the system: [Assignment: organization-defined monitoring tools]; andb. Monitor the following system resources to determine if	Functional	Intersects With	Capacity & Performance Management	CAP-01	Mechanisms exist to facilitate the implementation of capacity management controls to ensure optimal system performance to meet	5		
SC-06	Resource Availability	Protect the availability of resources by allocating [Assignment: organization-defined resources] by [Selection (one or more): priority; quota; [Assignment: organization-defined controls]].	Functional	Intersects With	Resource Priority	CAP-02	Mechanisms exist to control resource utilization of Technology Assets, Applications and/or Services (TAAS)	5		
SC-07	Boundary Protection	a. Monitor and control communications at the external managed interfaces to the system and at key internal managed interfaces within the system;b. Implement subnetworks for publicly accessible system components that are [Selection (one):	Functional	Intersects With	Boundary Protection	NET-03	Mechanisms exist to monitor and control communications at the external network boundary and at key internal boundaries within the	5		SC-07
SC-07(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-07(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-07(03)	Boundary Protection Access Points	Limit the number of external network connections to the system.	Functional	Equal	Limit Network Connections	NET-03.1	Mechanisms exist to limit the number of concurrent external network connections to its Technology Assets, Applications and/or Services (TAAS)	10		
SC-07(04)	Boundary Protection External Telecommunications Services	a. Implement a managed interface for each external telecommunication service;b. Establish a traffic flow policy for each managed interface;c. Protect the confidentiality and integrity of the information being transmitted across each	Functional	Intersects With	External Telecommunications Services	NET-03.2	Mechanisms exist to maintain a managed interface for each external telecommunication service that protects the confidentiality and	5		
SC-07(05)	Boundary Protection Deny by Default—Allow by Exception	Deny network communications traffic by default and allow network communications traffic by exception [Selection (one or more): at managed interfaces; for [Assignment: organization-defined systems]]	Functional	Intersects With	Deny Traffic by Default & Allow Traffic by Exception	NET-04.1	Mechanisms exist to configure firewall and router configurations to deny network traffic by default and allow network traffic by exception (e.g., deny	5		
SC-07(06)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-07(07)	Boundary Protection Split Tunneling for Remote Devices	Prevent split tunneling for remote devices connecting to organizational systems unless the split tunnel is securely provisioned using [Assignment: organization-defined safeguards]	Functional	Equal	Split Tunneling	CFG-03.4	Mechanisms exist to prevent split tunneling for remote devices unless the split tunnel is securely provisioned using organization-defined	10		
SC-07(08)	Boundary Protection Route Traffic to Authenticated Proxy Servers	Route [Assignment: organization-defined internal communications traffic] to [Assignment: organization-defined external networks] through authenticated proxy servers at managed interfaces	Functional	Intersects With	Route Internal Traffic to Proxy Servers	NET-18.1	Mechanisms exist to route internal communications traffic to external networks through organization-approved proxy servers at managed	5		
SC-07(08)	Boundary Protection Route Traffic to Authenticated Proxy Servers	Route [Assignment: organization-defined internal communications traffic] to [Assignment: organization-defined external networks] through authenticated proxy servers at managed interfaces	Functional	Intersects With	DNS & Content Filtering	NET-18	Mechanisms exist to force Internet-bound network traffic through a proxy device (e.g., Policy Enforcement Point (PEP)) for URL content filtering and	5		
SC-07(09)	Boundary Protection Restrict Threatening Outgoing Communications	a. Detect and deny outgoing communications traffic posing a threat to external systems; andb. Audit the identity of internal users associated with denied communications.	Functional	Intersects With	Boundary Protection	NET-03	Mechanisms exist to monitor and control communications at the external network boundary and at key internal boundaries within the	5		
SC-07(09)	Boundary Protection Restrict Threatening Outgoing Communications	a. Detect and deny outgoing communications traffic posing a threat to external systems; andb. Audit the identity of internal users associated with denied communications.	Functional	Intersects With	External Telecommunications Services	NET-03.2	Mechanisms exist to maintain a managed interface for each external telecommunication service that protects the confidentiality and	5		

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SC-07(10)	Boundary Protection Prevent Exfiltration	a. Prevent the exfiltration of information; andb. Conduct exfiltration tests [Assignment: organization-defined frequency].	Functional	Intersects With	Prevent Unauthorized Exfiltration	NET-03.5	Automated mechanisms exist to prevent the unauthorized exfiltration of sensitive and/or regulated data across managed interfaces.	5		
SC-07(10)	Boundary Protection Prevent Exfiltration	a. Prevent the exfiltration of information; andb. Conduct exfiltration tests [Assignment: organization-defined frequency].	Functional	Intersects With	Data Loss Prevention (DLP)	NET-17	Automated mechanisms exist to implement Data Loss Prevention (DLP) to protect sensitive information as it is stored, transmitted and processed.	5		
SC-07(11)	Boundary Protection Restrict Incoming Communications Traffic	Only allow incoming communications from [Assignment: organization-defined authorized sources] to be routed to [Assignment: organization-defined authorized destinations].	Functional	Intersects With	Deny Traffic by Default & Allow Traffic by Exception	NET-04.1	Mechanisms exist to configure firewall and router configurations to deny network traffic by default and allow network traffic by exception (e.g., deny).	5		
SC-07(11)	Boundary Protection Restrict Incoming Communications Traffic	Only allow incoming communications from [Assignment: organization-defined authorized sources] to be routed to [Assignment: organization-defined authorized destinations].	Functional	Intersects With	Boundary Protection	NET-03	Mechanisms exist to monitor and control communications at the external network boundary and at key internal boundaries within the	5		
SC-07(12)	Boundary Protection Host-based Protection	Implement [Assignment: organization-defined host-based boundary protection mechanisms] at [Assignment: organization-defined system components].	Functional	Equal	Host-Based Security Function Isolation	END-16.1	Mechanisms exist to implement underlying software separation mechanisms to facilitate security function isolation.	10		
SC-07(13)	Boundary Protection Isolation of Security Tools, Mechanisms, and Support	Isolate [Assignment: organization-defined information security tools, mechanisms, and support components] from other internal system components by implementing physically separate subnetworks with managed interfaces to other	Functional	Intersects With	Security Management Subnets	NET-06.1	Mechanisms exist to implement security management subnets to isolate security tools and support components from other internal	5		
SC-07(14)	Boundary Protection Protect Against Unauthorized Physical Connections	Protect against unauthorized physical connections at [Assignment: organization-defined managed interfaces].	Functional	Intersects With	Equipment Siting & Protection	PES-12	Physical security mechanisms exist to locate system components within the facility to minimize potential damage from physical and environmental	5		
SC-07(14)	Boundary Protection Protect Against Unauthorized Physical Connections	Protect against unauthorized physical connections at [Assignment: organization-defined managed interfaces].	Functional	Intersects With	Lockable Physical Casings	PES-03.2	Physical access control mechanisms exist to protect system components from unauthorized physical access (e.g., lockable physical casings).	5		
SC-07(14)	Boundary Protection Protect Against Unauthorized Physical Connections	Protect against unauthorized physical connections at [Assignment: organization-defined managed interfaces].	Functional	Intersects With	Transmission Medium Security	PES-12.1	Physical security mechanisms exist to protect power and telecommunications cabling carrying data or supporting information.	5		
SC-07(15)	Boundary Protection Networked Privileged Accesses	Route networked, privileged accesses through a dedicated, managed interface for purposes of access control and auditing.	Functional	Equal	Route Privileged Network Access	NET-18.3	Automated mechanisms exist to route networked, privileged accesses through a dedicated, managed interface for purposes of access	10		
SC-07(16)	Boundary Protection Prevent Discovery of System Components	Prevent the discovery of specific system components that represent a managed interface.	Functional	Equal	Prevent Discovery of Internal Information	NET-03.3	Mechanisms exist to prevent the public disclosure of internal network information.	10		
SC-07(17)	Boundary Protection Automated Enforcement of Protocol Formats	Enforce adherence to protocol formats.	Functional	Equal	Web Application Firewall (WAF)	WEB-03	Mechanisms exist to deploy Web Application Firewalls (WAFs) to provide defense-in-depth protection for application-specific threats.	10		
SC-07(18)	Boundary Protection Fail Secure	Prevent systems from entering insecure states in the event of an operational failure of a boundary protection device.	Functional	Intersects With	Secure Engineering Principles	SEA-01	Mechanisms exist to facilitate the implementation of industry-recognized security, compliance and	5		
SC-07(19)	Boundary Protection Block Communication from Non-organizationally	Block inbound and outbound communications traffic between [Assignment: organization-defined communication clients] that are independently configured by end users and external service providers.	Functional	Intersects With	Network Access Control (NAC)	AST-02.5	Automated mechanisms exist to employ Network Access Control (NAC), or a similar technology, which is capable of detecting unauthorized	5		
SC-07(20)	Boundary Protection Dynamic Isolation and Segregation	Provide the capability to dynamically isolate [Assignment: organization-defined system components] from other system components.	Functional	Equal	Dynamic Isolation & Segregation (Sandboxing)	NET-03.6	Automated mechanisms exist to dynamically isolate (e.g., sandbox) untrusted components during runtime, where the component is	10		
SC-07(21)	Boundary Protection Isolation of System Components	Employ boundary protection mechanisms to isolate [Assignment: organization-defined system components] supporting [Assignment: organization-defined missions and/or business functions].	Functional	Equal	Isolation of System Components	NET-03.7	Mechanisms exist to employ boundary protections to isolate Technology Assets, Applications and/or Services (TAAS).	10		
SC-07(22)	Boundary Protection Separate Subnets for Connecting to Different Security Domains	Implement separate network addresses to connect to systems in different security domains.	Functional	Intersects With	Separate Subnet for Connecting to Different Security Domains	NET-03.8	Mechanisms exist to implement separate network addresses (e.g., different subnets) to connect to systems in different security domains.	5		
SC-07(23)	Boundary Protection Disable Sender Feedback on Protocol Validation Failure	Disable feedback to senders on protocol format validation failure.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-07(24)	Boundary Protection Personally Identifiable Information	For systems that process personally identifiable information:a. Apply the following processing rules to data elements of personally identifiable information: [Assignment: organization-defined processing rules];b. Monitor for permitted processing at	Functional	Equal	Personal Data (PD)	NET-03.4	Mechanisms exist to apply network-based processing rules to data elements of Personal Data (PD).	10		
SC-07(25)	Boundary Protection Unclassified National Security System Connections	Prohibit the direct connection of [Assignment: organization-defined unclassified national security system] to an external network without the use of [Assignment: organization-defined boundary protection device].	Functional	Intersects With	Interconnection Security Agreements (ISAs)	NET-05	Mechanisms exist to authorize connections from systems to other systems using Interconnection Security Agreements (ISAs) or similar	5		
SC-07(26)	Boundary Protection Classified National Security System Connections	Prohibit the direct connection of a classified national security system to an external network without the use of [Assignment: organization-defined boundary protection device].	Functional	Intersects With	Interconnection Security Agreements (ISAs)	NET-05	Mechanisms exist to authorize connections from systems to other systems using Interconnection Security Agreements (ISAs) or similar	5		
SC-07(27)	Boundary Protection Unclassified Non-national Security System Connections	Prohibit the direct connection of [Assignment: organization-defined unclassified non-national security system] to an external network without the use of [Assignment: organization-defined boundary protection device].	Functional	Equal	External System Connections	NET-05.1	Mechanisms exist to prohibit the direct connection of a sensitive system to an external network without the use of an organization-defined	10		
SC-07(28)	Boundary Protection Connections to Public Networks	Prohibit the direct connection of [Assignment: organization-defined system] to a public network.	Functional	Equal	Direct Internet Access Restrictions	NET-06.5	Mechanisms exist to prohibit, or strictly-control, Internet access from sensitive and/or regulated data enclaves (secure zones).	10		SC-07(28)
SC-07(29)	Boundary Protection Separate Subnets to Isolate Functions	Implement [Selection (one): physically, logically] separate subnetworks to isolate the following critical system components and functions: [Assignment: organization-defined critical system components and functions].	Functional	Intersects With	Cloud Infrastructure Security Subnet	CLD-03	Mechanisms exist to host security-specific technologies in a dedicated subnet.	5		SC-07(29)
SC-07(29)	Boundary Protection Separate Subnets to Isolate Functions	Implement [Selection (one): physically, logically] separate subnetworks to isolate the following critical system components and functions: [Assignment: organization-defined critical system components and functions].	Functional	Intersects With	Security Management Subnets	NET-06.1	Mechanisms exist to implement security management subnets to isolate security tools and support components from other internal	5		
SC-07(29)	Boundary Protection Separate Subnets to Isolate Functions	Implement [Selection (one): physically, logically] separate subnetworks to isolate the following critical system components and functions: [Assignment: organization-defined critical system components and functions].	Functional	Intersects With	Separate Subnet for Connecting to Different Security Domains	NET-03.8	Mechanisms exist to implement separate network addresses (e.g., different subnets) to connect to systems in different security domains.	5		
SC-08	Transmission Confidentiality and Integrity	Protect the [Selection (one or more): confidentiality; integrity] of transmitted information.	Functional	Intersects With	Transmission Confidentiality	CRY-03	Cryptographic mechanisms exist to protect the confidentiality of data being transmitted.	5		

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SC-08	Transmission Confidentiality and Integrity	Protect the [Selection (one or more): confidentiality; integrity] of transmitted information.	Functional	Intersects With	Transmission Integrity	CRY-04	Cryptographic mechanisms exist to protect the integrity of data being transmitted.	5		
SC-08(01)	Transmission Confidentiality and Integrity Confidentiality	Implement cryptographic mechanisms to [Selection (one or more): prevent unauthorized disclosure of information; detect changes to information] during transmission.	Functional	Intersects With	Alternate Physical Protection	CRY-01.1	Cryptographic mechanisms exist to prevent unauthorized disclosure of information as an alternative to physical safeguards.	5		
SC-08(01)	Transmission Confidentiality and Integrity Confidentiality	Implement cryptographic mechanisms to [Selection (one or more): prevent unauthorized disclosure of information; detect changes to information] during transmission.	Functional	Intersects With	Use of Cryptographic Controls	CRY-01	Mechanisms exist to facilitate the implementation of cryptographic protections controls using known public standards and trusted.	5		
SC-08(01)	Transmission Confidentiality and Integrity Confidentiality	Implement cryptographic mechanisms to [Selection (one or more): prevent unauthorized disclosure of information; detect changes to information] during transmission.	Functional	Intersects With	Transmission Confidentiality	CRY-03	Cryptographic mechanisms exist to protect the confidentiality of data being transmitted.	5		
SC-08(02)	Transmission Confidentiality and Integrity Pre- and Post-transmission	Maintain the [Selection (one or more): confidentiality; integrity] of information during preparation for transmission and during reception.	Functional	Intersects With	Pre/Post Transmission Handling	CRY-01.3	Cryptographic mechanisms exist to ensure the confidentiality and integrity of information during preparation for transmission and during reception.	5		
SC-08(02)	Transmission Confidentiality and Integrity Pre- and Post-transmission	Maintain the [Selection (one or more): confidentiality; integrity] of information during preparation for transmission and during reception.	Functional	Intersects With	Use of Cryptographic Controls	CRY-01	Mechanisms exist to facilitate the implementation of cryptographic protections controls using known public standards and trusted.	5		
SC-08(02)	Transmission Confidentiality and Integrity Pre- and Post-transmission	Maintain the [Selection (one or more): confidentiality; integrity] of information during preparation for transmission and during reception.	Functional	Intersects With	Media Use	DCH-10	Mechanisms exist to restrict the use of types of digital media on systems or system components.	5		
SC-08(03)	Transmission Confidentiality and Integrity Confidentiality	Implement cryptographic mechanisms to protect message externals unless otherwise protected by [Assignment: organization-defined alternative physical controls].	Functional	Equal	Electronic Messaging	NET-13	Mechanisms exist to protect the confidentiality, integrity and availability of electronic messaging communications.	10		
SC-08(04)	Transmission Confidentiality and Integrity Conceal or Randomize	Implement cryptographic mechanisms to conceal or randomize communication patterns unless otherwise protected by [Assignment: organization-defined alternative physical controls].	Functional	Equal	Conceal / Randomize Communications	CRY-01.4	Cryptographic mechanisms exist to conceal or randomize communication patterns.	10		
SC-08(05)	Transmission Confidentiality and Integrity Protected Distribution System	Implement [Assignment: organization-defined protected distribution system] to [Selection (one or more): prevent unauthorized disclosure of information; detect changes to information] during transmission.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-09		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-10	Network Disconnect	Terminate the network connection associated with a communications session at the end of the session or after [Assignment: organization-defined time period] of inactivity.	Functional	Equal	Network Connection Termination	NET-07	Mechanisms exist to terminate network connections at the end of a session or after an organization-defined time period of inactivity.	10		
SC-11	Trusted Path	a. Provide a [Selection (one): physically; logically] isolated trusted communications path for communications between the user and the trusted components of the system; andb. Permit users to invoke the trusted communications path for	Functional	Equal	Trusted Path	END-09	Mechanisms exist to establish a trusted communications path between the user and the security functions of the operating system.	10		
SC-11(01)	Trusted Path Irrefutable Communications Path	a. Provide a trusted communications path that is irrefutably distinguishable from other communications paths; andb. Initiate the trusted communications path for communications between the [Assignment: organization-defined security functions] of the	Functional	No Relationship	N/A	N/A	N/A	0		
SC-12	Cryptographic Key Establishment and Management	Establish and manage cryptographic keys when cryptography is employed within the system in accordance with the following key management requirements: [Assignment: organization-defined requirements for key generation, distribution, storage	Functional	Intersects With	Public Key Infrastructure (PKI)	CRY-08	Mechanisms exist to securely implement an internal Public Key Infrastructure (PKI) infrastructure or obtain PKI services from a reputable	5		SC-12
SC-12(01)	Cryptographic Key Establishment and Management Availability	Maintain availability of information in the event of the loss of cryptographic keys by users.	Functional	Equal	Cryptographic Key Loss or Change	CRY-09.3	Mechanisms exist to ensure the availability of information in the event of the loss of cryptographic keys by individual users.	10		
SC-12(02)	Cryptographic Key Establishment and Management Symmetric Keys	Produce, control, and distribute symmetric cryptographic keys using [Selection (one): NIST FIPS-validated; NSA-approved] key management technology and processes.	Functional	Equal	Symmetric Keys	CRY-09.1	Mechanisms exist to facilitate the production and management of symmetric cryptographic keys using Federal Information Processing	10		
SC-12(03)	Cryptographic Key Establishment and Management Asymmetric Keys	Produce, control, and distribute asymmetric cryptographic keys using [Selection (one): NSA-approved key management technology and processes; prepositioned keying material; DoD-approved or DoD-issued Medium Assurance PKI certificates;	Functional	Equal	Asymmetric Keys	CRY-09.2	Mechanisms exist to facilitate the production and management of asymmetric cryptographic keys using Federal Information Processing	10		
SC-12(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-12(05)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-12(06)	Cryptographic Key Establishment and Management Physical Control of	Maintain physical control of cryptographic keys when stored information is encrypted by external service providers.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-13	Cryptographic Protection	a. Determine the [Assignment: organization-defined cryptographic uses]; andb. Implement the following types of cryptography required for each specified cryptographic use: [Assignment: organization-defined types of cryptography for	Functional	Intersects With	Encrypting Data At Rest	CRY-05	Cryptographic mechanisms exist to prevent unauthorized disclosure of data at rest.	5		SC-13
SC-13	Cryptographic Protection	a. Determine the [Assignment: organization-defined cryptographic uses]; andb. Implement the following types of cryptography required for each specified cryptographic use: [Assignment: organization-defined types of cryptography for	Functional	Intersects With	Export-Controlled Cryptography	CRY-01.2	Mechanisms exist to address the exporting of cryptographic technologies in compliance with relevant statutory and regulatory	5		SC-13
SC-13	Cryptographic Protection	a. Determine the [Assignment: organization-defined cryptographic uses]; andb. Implement the following types of cryptography required for each specified cryptographic use: [Assignment: organization-defined types of cryptography for	Functional	Intersects With	Use of Cryptographic Controls	CRY-01	Mechanisms exist to facilitate the implementation of cryptographic protections controls using known public standards and trusted.	5		SC-13
SC-13(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-13(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-13(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-13(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-14		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-15	Collaborative Computing Devices and Applications	a. Prohibit remote activation of collaborative computing devices and applications with the following exceptions: [Assignment: organization-defined exceptions where remote activation is to be allowed]; andb. Provide an explicit indication of use to users	Functional	Intersects With	Collaborative Computing Devices	END-14	Mechanisms exist to unplug or prohibit the remote activation of collaborative computing devices with the following exceptions:	5		SC-15
SC-15(01)	Collaborative Computing Devices and Applications Physical or Logical	Provide [Selection (one or more): physical; logical] disconnect of collaborative computing devices in a manner that supports ease of use.	Functional	Intersects With	Collaborative Computing Devices	END-14	Mechanisms exist to unplug or prohibit the remote activation of collaborative computing devices with the following exceptions:	5		
SC-15(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-15(03)	Collaborative Computing Devices and Applications Disabling and Removal	Disable or remove collaborative computing devices and applications from [Assignment: organization-defined systems or system components] in [Assignment: organization-defined secure work areas]	Functional	Equal	Disabling / Removal In Secure Work Areas	END-14.1	Mechanisms exist to disable or remove collaborative computing devices from critical systems and	10		

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SC-15(04)	Collaborative Computing Devices and Applications Explicitly Indicate	Provide an explicit indication of current participants in [Assignment: organization-defined online meetings and teleconferences].	Functional	Equal	Explicitly Indicate Current Participants	END-14.2	Automated mechanisms exist to provide an explicit indication of current participants in online meetings and teleconferences.	10		
SC-16	Transmission of Security and Privacy Attributes	Associate [Assignment: organization-defined security and privacy attributes] with information exchanged between systems and between system components.	Functional	Intersects With	Transmission of Cybersecurity & Data Protection Attributes	CRY-10	Mechanisms exist to associate Technology Assets, Applications and/or Services (TAAS) security attributes with information exchanged.	5		
SC-16(01)	Transmission of Security and Privacy Attributes Integrity Verification	Verify the integrity of transmitted security and privacy attributes.	Functional	Intersects With	Transmission Integrity	CRY-04	Cryptographic mechanisms exist to protect the integrity of data being transmitted.	5		
SC-16(01)	Transmission of Security and Privacy Attributes Integrity Verification	Verify the integrity of transmitted security and privacy attributes.	Functional	Intersects With	Transmission of Cybersecurity & Data Protection Attributes	CRY-10	Mechanisms exist to associate Technology Assets, Applications and/or Services (TAAS) security attributes with information exchanged.	5		
SC-16(02)	Transmission of Security and Privacy Attributes Anti-spoofing Mechanisms	Implement anti-spoofing mechanisms to prevent adversaries from falsifying the security attributes indicating the successful application of the security process.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-16(03)	Transmission of Security and Privacy Attributes Cryptographic Binding	Implement [Assignment: organization-defined mechanisms or techniques] to bind security and privacy attributes to transmitted information.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-17	Public Key Infrastructure Certificates	a. Issue public key certificates under an [Assignment: organization-defined certificate policy] or obtain public key certificates from an approved service provider; andb. Include only approved trust anchors in trust stores or certificate stores.	Functional	Intersects With	Public Key Infrastructure (PKI)	CRY-08	Mechanisms exist to securely implement an internal Public Key Infrastructure (PKI) infrastructure or obtain PKI services from a reputable provider.	5		
SC-18	Mobile Code	a. Define acceptable and unacceptable mobile code and mobile code technologies; andb. Authorize, monitor, and control the use of mobile code within the system.	Functional	Intersects With	Mobile Code	END-10	Mechanisms exist to address mobile code / operating system-independent applications.	5		
SC-18(01)	Mobile Code Identify Unacceptable Code and Take Corrective Actions	Identify [Assignment: organization-defined unacceptable mobile code] and take [Assignment: organization-defined corrective actions].	Functional	Intersects With	Vulnerability Remediation Process	VPM-02	Mechanisms exist to ensure that vulnerabilities are properly identified, tracked and remediated.	5		
SC-18(01)	Mobile Code Identify Unacceptable Code and Take Corrective Actions	Identify [Assignment: organization-defined unacceptable mobile code] and take [Assignment: organization-defined corrective actions].	Functional	Intersects With	Mobile Code	END-10	Mechanisms exist to address mobile code / operating system-independent applications.	5		
SC-18(01)	Mobile Code Identify Unacceptable Code and Take Corrective Actions	Identify [Assignment: organization-defined unacceptable mobile code] and take [Assignment: organization-defined corrective actions].	Functional	Intersects With	Continuous Vulnerability Remediation Activities	VPM-04	Mechanisms exist to address new threats and vulnerabilities on an ongoing basis and ensure assets are protected against known attacks.	5		
SC-18(02)	Mobile Code Acquisition, Development, and Use	Verify that the acquisition, development, and use of mobile code to be deployed in the system meets [Assignment: organization-defined mobile code requirements].	Functional	Intersects With	Software Licensing Restrictions	AST-02.7	Mechanisms exist to protect Intellectual Property (IP) rights with software licensing restrictions.	5		
SC-18(02)	Mobile Code Acquisition, Development, and Use	Verify that the acquisition, development, and use of mobile code to be deployed in the system meets [Assignment: organization-defined mobile code requirements].	Functional	Intersects With	Mobile Code	END-10	Mechanisms exist to address mobile code / operating system-independent applications.	5		
SC-18(03)	Mobile Code Prevent Downloading and Execution	Prevent the download and execution of [Assignment: organization-defined unacceptable mobile code].	Functional	Intersects With	DNS & Content Filtering	NET-18	Mechanisms exist to force Internet-bound network traffic through a proxy device (e.g., Policy Enforcement Point (PEP) for URL content filtering and	5		
SC-18(03)	Mobile Code Prevent Downloading and Execution	Prevent the download and execution of [Assignment: organization-defined unacceptable mobile code].	Functional	Intersects With	Mobile Code	END-10	Mechanisms exist to address mobile code / operating system-independent applications.	5		
SC-18(04)	Mobile Code Prevent Automatic Execution	Prevent the automatic execution of mobile code in [Assignment: organization-defined software applications] and enforce [Assignment: organization-defined actions] prior to executing the code.	Functional	Intersects With	Mobile Code	END-10	Mechanisms exist to address mobile code / operating system-independent applications.	5		
SC-18(04)	Mobile Code Prevent Automatic Execution	Prevent the automatic execution of mobile code in [Assignment: organization-defined software applications] and enforce [Assignment: organization-defined actions] prior to executing the code.	Functional	Intersects With	Explicitly Allow / Deny Applications	CFG-03.3	Mechanisms exist to explicitly allow (allowlist / whitelist) and/or block (denylist / blacklist) applications that are authorized to execute on systems.	5		
SC-18(05)	Mobile Code Allow Execution Only in Confined Environments	Allow execution of permitted mobile code only in confined virtual machine environments.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-19		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-20	Secure Name/address Resolution Service (authoritative Source)	a. Provide additional data origin authentication and integrity verification artifacts along with the authoritative name resolution data the system returns in response to external name/address resolution queries; andb. Provide the means to	Functional	Intersects With	Domain Name Service (DNS) Resolution	NET-10	Mechanisms exist to ensure Domain Name Service (DNS) resolution is designed, implemented and managed to protect the security of name /	5		SC-20
SC-20(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-20(02)	Secure Name/address Resolution Service (authoritative Source) Data Origin and	Provide data origin and integrity protection artifacts for internal name/address resolution queries.	Functional	Intersects With	Domain Name Service (DNS) Resolution	NET-10	Mechanisms exist to ensure Domain Name Service (DNS) resolution is designed, implemented and managed to protect the security of name /	5		
SC-21	Secure Name/address Resolution Service (recursive or Caching Resolver)	Request and perform data origin authentication and data integrity verification on the name/address resolution responses the system receives from authoritative sources.	Functional	Equal	Secure Name / Address Resolution Service (Recursive or Caching Resolver)	NET-10.2	Mechanisms exist to perform data origin authentication and data integrity verification on the Domain Name Service (DNS) resolution responses	10		SC-21
SC-21(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-22	Architecture and Provisioning for Name/address Resolution Service	Ensure the systems that collectively provide name/address resolution service for an organization are fault-tolerant and implement internal and external role separation.	Functional	Equal	Architecture & Provisioning for Name / Address Resolution Service	NET-10.1	Mechanisms exist to ensure systems that collectively provide Domain Name Service (DNS) resolution service are fault-tolerant and	10		SC-22
SC-23	Session Authenticity	Protect the authenticity of communications sessions.	Functional	Equal	Session Integrity	NET-09	Mechanisms exist to protect the authenticity and integrity of communications sessions.	10		
SC-23(01)	Session Authenticity Invalidate Session Identifiers at Logout	Invalidate session identifiers upon user logout or other session termination.	Functional	Equal	Invalidate Session Identifiers at Logout	NET-09.1	Automated mechanisms exist to invalidate session identifiers upon user logout or other session termination	10		
SC-23(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-23(03)	Session Authenticity Unique System-generated Session Identifiers	Generate a unique session identifier for each session with [Assignment: organization-defined randomness requirements] and recognize only session identifiers that are system-generated.	Functional	Equal	Unique System-Generated Session Identifiers	NET-09.2	Automated mechanisms exist to generate and recognize unique session identifiers for each session.	10		
SC-23(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SC-23(05)	Session Authenticity Allowed Certificate Authorities	Only allow the use of [Assignment: organization-defined certificate authorities] for verification of the establishment of protected sessions.	Functional	Equal	Certificate Authorities	CRY-11	Automated mechanisms exist to enable the use of organization-defined Certificate Authorities (CAs) to facilitate the establishment of	10		
SC-24	Fail in Known State	Fail to a [Assignment: organization-defined known system state] for the following failures on the indicated components while preserving [Assignment: organization-defined system state information]: in failure: [Assignment: list of organization-defined	Functional	Intersects With	Fail Secure	SEA-07.2	Mechanisms exist to enable systems to fail to an organization-defined known-state for types of failures, preserving system state information in	5		
SC-25	Thin Nodes	Employ minimal functionality and information storage on the following system components: [Assignment: organization-defined system components].	Functional	Equal	Thin Nodes	END-11	Mechanisms exist to configure thin nodes to have minimal functionality and information storage.	10		
SC-26	Decoys	Include components within organizational systems specifically designed to be the target of malicious attacks for detecting, deflecting, and analyzing such attacks.	Functional	Equal	Honeypots	SEA-11	Mechanisms exist to utilize honeypots that are specifically designed to be the target of malicious attacks for the purpose of detecting, deflecting and	10		
SC-26(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-27	Platform-independent Applications	Include within organizational systems the following platform independent applications: [Assignment: organization-defined platform-independent applications].	Functional	Equal	Mobile Code	END-10	Mechanisms exist to address mobile code / operating system-independent applications.	10		
SC-28	Protection of Information at Rest	Protect the [Selection (one or more): confidentiality; integrity] of the following information at rest: [Assignment: organization-defined information at rest].	Functional	Intersects With	Endpoint Protection Measures	END-02	Mechanisms exist to protect the confidentiality, integrity, availability and safety of endpoint devices.	5		
SC-28	Protection of Information at Rest	Protect the [Selection (one or more): confidentiality; integrity] of the following information at rest: [Assignment: organization-defined information at rest].	Functional	Intersects With	Encrypting Data At Rest	CRY-05	Cryptographic mechanisms exist to prevent unauthorized disclosure of data at rest.	5		
SC-28(01)	Protection of Information at Rest Cryptographic	Implement cryptographic mechanisms to prevent unauthorized disclosure and modification of the following information at rest on [Assignment: organization-defined system components or media]: [Assignment: organization-defined information]	Functional	Intersects With	Cryptographic Protection	BCD-11.4	Cryptographic mechanisms exist to prevent the unauthorized disclosure and/or modification of backup information	5		
SC-28(01)	Protection of Information at Rest Cryptographic	Implement cryptographic mechanisms to prevent unauthorized disclosure and modification of the following information at rest on [Assignment: organization-defined system components or media]: [Assignment: organization-defined information]	Functional	Intersects With	Encrypting Data At Rest	CRY-05	Cryptographic mechanisms exist to prevent unauthorized disclosure of data at rest.	5		
SC-28(01)	Protection of Information at Rest Cryptographic	Implement cryptographic mechanisms to prevent unauthorized disclosure and modification of the following information at rest on [Assignment: organization-defined system components or media]: [Assignment: organization-defined information]	Functional	Intersects With	Transmission Integrity	CRY-04	Cryptographic mechanisms exist to protect the integrity of data being transmitted.	5		
SC-28(01)	Protection of Information at Rest Cryptographic	Implement cryptographic mechanisms to prevent unauthorized disclosure and modification of the following information at rest on [Assignment: organization-defined system components or media]: [Assignment: organization-defined information]	Functional	Intersects With	Encrypting Data In Storage Media	DCH-07.2	Cryptographic mechanisms exist to protect the confidentiality and integrity of information stored on digital media during transport outside	5		
SC-28(02)	Protection of Information at Rest Offline Storage	Remove the following information from online storage and store offline in a secure location: [Assignment: organization-defined information].	Functional	Intersects With	Offline Storage	CRY-05.2	Mechanisms exist to remove unused data from online storage and archive it off-line in a secure location until it can be disposed of according to data	5		
SC-28(02)	Protection of Information at Rest Offline Storage	Remove the following information from online storage and store offline in a secure location: [Assignment: organization-defined information].	Functional	Intersects With	Data Backups	BCD-11	Mechanisms exist to create recurring backups of data, software and/or system images, as well as verify the	5		
SC-28(03)	Protection of Information at Rest Cryptographic Keys	Provide protected storage for cryptographic keys [Selection (one): [Assignment: organization-defined safeguards]; hardware-protected key store].	Functional	Equal	Cryptographic Key Management	CRY-09	Mechanisms exist to facilitate cryptographic key management controls to protect the confidentiality, integrity and availability of keys	10		
SC-29	Heterogeneity	Employ a diverse set of information technologies for the following system components in the implementation of the system: [Assignment: organization-defined system components]	Functional	Equal	Heterogeneity	SEA-13	Mechanisms exist to utilize a diverse set of technologies for system components to reduce the impact of technical vulnerabilities from the	10		
SC-29(01)	Heterogeneity Virtualization Techniques	Employ virtualization techniques to support the deployment of a diversity of operating systems and applications that are changed [Assignment: organization-defined frequency].	Functional	Equal	Virtualization Techniques	SEA-13.1	Mechanisms exist to utilize virtualization techniques to support the employment of a diversity of operating systems and applications	10		
SC-30	Concealment and Misdirection	Employ the following concealment and misdirection techniques for [Assignment: organization-defined systems] at [Assignment: organization-defined time periods] to confuse and mislead adversaries: [Assignment: organization-defined concealment	Functional	Intersects With	Concealment & Misdirection	SEA-14	Mechanisms exist to utilize concealment and misdirection techniques for Technology Assets, Applications and/or Services (TAAPS)	5		
SC-30(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-30(02)	Concealment and Misdirection Randomness	Employ [Assignment: organization-defined techniques] to introduce randomness into organizational operations and assets.	Functional	Equal	Randomness	SEA-14.1	Automated mechanisms exist to introduce randomness into organizational operations and assets.	10		
SC-30(03)	Concealment and Misdirection Change Processing and Storage Locations	Change the location of [Assignment: organization-defined processing and/or storage] [Selection (one): [Assignment: organization-defined time frequency]; at random time intervals].	Functional	Equal	Change Processing & Storage Locations	SEA-14.2	Automated mechanisms exist to change the location of processing and/or storage at random time intervals	10		
SC-30(04)	Concealment and Misdirection Misleading Information	Employ realistic, but misleading information in [Assignment: organization-defined system components] about its security state or posture.	Functional	Intersects With	Concealment & Misdirection	SEA-14	Mechanisms exist to utilize concealment and misdirection techniques for Technology Assets, Applications and/or Services (TAAPS)	5		
SC-30(05)	Concealment and Misdirection Concealment of System Components	Employ the following techniques to hide or conceal [Assignment: organization-defined system components]: [Assignment: organization-defined techniques].	Functional	Intersects With	Concealment & Misdirection	SEA-14	Mechanisms exist to utilize concealment and misdirection techniques for Technology Assets, Applications and/or Services (TAAPS)	5		
SC-31	Covert Channel Analysis	a. Perform a covert channel analysis to identify those aspects of communications within the system that are potential avenues for covert [Selection (one or more): storage; timing] channels; and b. Estimate the maximum bandwidth of those channels	Functional	Equal	Covert Channel Analysis	MON-15	Mechanisms exist to conduct covert channel analysis to identify aspects of communications that are potential avenues for covert channels	10		
SC-31(01)	Covert Channel Analysis Test Covert Channels for Exploitability	Test a subset of the identified covert channels to determine the channels that are exploitable.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-31(02)	Covert Channel Analysis Maximum Bandwidth	Reduce the maximum bandwidth for identified covert [Selection (one or more): storage; timing] channels to [Assignment: organization-defined values].	Functional	No Relationship	N/A	N/A	N/A	0		
SC-31(03)	Covert Channel Analysis Measure Bandwidth in Operational	Measure the bandwidth of [Assignment: organization-defined subset of identified covert channels] in the operational environment of the system.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-32	System Partitioning	Partition the system into [Assignment: organization-defined system components] residing in separate [Selection (one): physical; logical] domains or environments based on [Assignment: organization-defined circumstances for physical or	Functional	Equal	System Partitioning	SEA-03.1	Mechanisms exist to partition systems so that partitions reside in separate physical domains or environments.	10		

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SC-32(01)	System Partitioning Separate Physical Domains for Privileged Functions	Partition privileged functions into separate physical domains.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-33		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-34	Non-modifiable Executable Programs	For [Assignment: organization-defined system components], load and execute: a. The operating environment from hardware-enforced, read-only media; and b. The following applications from hardware-enforced, read-only media: [Assignment: organization-defined system components]	Functional	Equal	Non-Modifiable Executable Programs	SEA-16	Mechanisms exist to utilize non-modifiable executable programs that load and execute the operating environment and applications from hardware-enforced, read-only media.	10		
SC-34(01)	Non-modifiable Executable Programs No Writable Storage	Employ [Assignment: organization-defined system components] with no writable storage that is persistent across component restart or power on/off.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-34(02)	Non-modifiable Executable Programs Integrity Protection on Read-only Media	Protect the integrity of information prior to storage on read-only media and control the media after such information has been recorded onto the media.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-34(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-35	External Malicious Code Identification	Include system components that proactively seek to identify network-based malicious code or malicious websites.	Functional	Equal	Honeyclients	SEA-12	Mechanisms exist to utilize honeyclients that proactively seek to identify malicious websites and/or web-based malicious code.	10		
SC-36	Distributed Processing and Storage	Distribute the following processing and storage components across multiple [Selection (one): physical locations; logical domains]: [Assignment: organization-defined processing and storage components]	Functional	Equal	Distributed Processing & Storage	SEA-15	Mechanisms exist to distribute processing and storage across multiple physical locations.	10		
SC-36(01)	Distributed Processing and Storage Polling Techniques	a. Employ polling techniques to identify potential faults, errors, or compromises to the following processing and storage components: [Assignment: organization-defined distributed processing and storage components]; and b. Take the following actions: [Assignment: organization-defined actions]	Functional	No Relationship	N/A	N/A	N/A	0		
SC-36(02)	Distributed Processing and Storage Synchronization	Synchronize the following duplicate systems or system components: [Assignment: organization-defined duplicate systems or system components]	Functional	No Relationship	N/A	N/A	N/A	0		
SC-37	Out-of-band Channels	Employ the following out-of-band channels for the physical delivery or electronic transmission of [Assignment: organization-defined information, system components, or devices] to [Assignment: organization-defined individuals or systems]: [Assignment: organization-defined controls] to ensure that only [Assignment: organization-defined individuals or systems] receive the following information, system components, or devices: [Assignment: organization-defined controls]	Functional	Intersects With	Out-of-Band Channels	NET-11	Mechanisms exist to utilize out-of-band channels for the electronic transmission of information and/or the physical shipment of system components.	5		
SC-37(01)	Out-of-band Channels Ensure Delivery and Transmission	Employ the following operations security controls to protect key organizational information throughout the system development life cycle: [Assignment: organization-defined operations security controls]	Functional	Intersects With	Out-of-Band Channels	NET-11	Mechanisms exist to utilize out-of-band channels for the electronic transmission of information and/or the physical shipment of system components.	5		
SC-38	Operations Security	Employ the following operations security controls to protect key organizational information throughout the system development life cycle: [Assignment: organization-defined operations security controls]	Functional	Intersects With	Security Operations Center (SOC)	OPS-04	Mechanisms exist to establish and maintain a Security Operations Center (SOC) that facilitates a 24x7 response capability.	5		
SC-38	Operations Security	Employ the following operations security controls to protect key organizational information throughout the system development life cycle: [Assignment: organization-defined operations security controls]	Functional	Intersects With	Operations Security	OPS-01	Mechanisms exist to facilitate the implementation of operational security controls.	5		
SC-39	Process Isolation	Maintain a separate execution domain for each executing system process.	Functional	Equal	Process Isolation	SEA-04	Mechanisms exist to implement a separate execution domain for each executing process.	10		SC-39
SC-39(01)	Process Isolation Hardware Separation	Implement hardware separation mechanisms to facilitate process isolation.	Functional	Equal	Hardware Separation	SEA-04.2	Mechanisms exist to implement underlying hardware separation mechanisms to facilitate process separation.	10		
SC-39(02)	Process Isolation Separate Execution Domain Per Thread	Maintain a separate execution domain for each thread in [Assignment: organization-defined multi-threaded processing].	Functional	Equal	Thread Separation	SEA-04.3	Mechanisms exist to maintain a separate execution domain for each thread in multi-threaded processing.	10		
SC-40	Wireless Link Protection	Protect external and internal [Assignment: organization-defined wireless links] from the following signal parameter attacks: [Assignment: organization-defined types of signal parameter attacks or references to sources for such attacks]	Functional	Intersects With	Wireless Link Protection	NET-12.1	Mechanisms exist to protect external and internal wireless links from signal parameter attacks through monitoring for unauthorized wireless.	5		
SC-40	Wireless Link Protection	Protect external and internal [Assignment: organization-defined wireless links] from the following signal parameter attacks: [Assignment: organization-defined types of signal parameter attacks or references to sources for such attacks]	Functional	Intersects With	Wireless Access Authentication & Encryption	CRY-07	Mechanisms exist to protect the confidentiality and integrity of wireless networking technologies by [Assignment: organization-defined controls]	5		
SC-40(01)	Wireless Link Protection Electromagnetic Interference	Implement cryptographic mechanisms that achieve [Assignment: organization-defined level of protection] against the effects of intentional electromagnetic interference.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-40(02)	Wireless Link Protection Reduce Detection Potential	Implement cryptographic mechanisms to reduce the detection potential of wireless links to [Assignment: organization-defined level of reduction].	Functional	No Relationship	N/A	N/A	N/A	0		
SC-40(03)	Wireless Link Protection Imitative or Manipulative Communications	Implement cryptographic mechanisms to identify and reject wireless transmissions that are deliberate attempts to achieve imitative or manipulative communications deception based on signal parameters.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-40(04)	Wireless Link Protection Signal Parameter Identification	Implement cryptographic mechanisms to prevent the identification of [Assignment: organization-defined wireless transmitters] by using the transmitter signal parameters.	Functional	No Relationship	N/A	N/A	N/A	0		
SC-41	Port and I/O Device Access	[Selection (one): Physically; Logically] disable or remove [Assignment: organization-defined connection ports or input/output devices] on the following systems or system components: [Assignment: organization-defined systems or components]	Functional	Equal	Port & Input / Output (I/O) Device Access	END-12	Mechanisms exist to physically disable or remove unnecessary connection ports or input/output devices from sensitive systems.	10		SC-41
SC-42	Sensor Capability and Data	a. Prohibit [Selection (one or more): the use of devices possessing [Assignment: organization-defined environmental sensing capabilities] in [Assignment: organization-defined facilities, areas, or systems]; the remote activation of [Assignment: organization-defined sensors]	Functional	Equal	Sensor Capability	END-13	Mechanisms exist to configure embedded sensors on systems to: (1) Prohibit the remote activation of [Assignment: organization-defined sensors]	10		
SC-42(01)	Sensor Capability and Data Reporting to Authorized Individuals or Roles	Verify that the system is configured so that data or information collected by the [Assignment: organization-defined sensors] is only reported to authorized individuals or roles.	Functional	Equal	Sensor Delivery Verification	END-13.4	Mechanisms exist to verify embedded technology sensors are configured so that data collected by the sensor(s) is only reported to authorized individuals.	10		
SC-42(02)	Sensor Capability and Data Authorized Use	Employ the following measures so that data or information collected by [Assignment: organization-defined sensors] is only used for authorized purposes: [Assignment: organization-defined measures]	Functional	Equal	Authorized Use	END-13.1	Mechanisms exist to utilize organization-defined measures so that data or information collected by sensors is only used for authorized purposes.	10		
SC-42(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SC-42(04)	Sensor Capability and Data Notice of Collection	Employ the following measures to facilitate an individual's awareness that personally identifiable information is being collected by [Assignment: organization-defined sensors]: [Assignment: organization-defined measures]	Functional	Equal	Notice of Collection	END-13.2	Mechanisms exist to notify individuals that Personal Data (PD) is collected by sensors.	10		

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

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FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SC-42(05)	Sensor Capability and Data Collection Minimization	Employ [Assignment: organization-defined sensors] that are configured to minimize the collection of information about individuals that is not needed.	Functional	Equal	Collection Minimization	END-13.3	Mechanisms exist to utilize sensors that are configured to minimize the collection of information about individuals.	10		
SC-43	Usage Restrictions	a. Establish usage restrictions and implementation guidelines for the following system components: [Assignment: organization-defined system components]; andb. Authorize, monitor, and control the use of such components within the system.	Functional	Equal	Usage Parameters	AST-14	Mechanisms exist to monitor and enforce usage parameters that limit the potential damage caused from the unauthorized or unintentional	10		
SC-44	Detonation Chambers	Employ a detonation chamber capability within [Assignment: organization-defined system, system component, or location].	Functional	Equal	Detonation Chambers (Sandboxes)	IRO-15	Mechanisms exist to utilize a detonation chamber capability to detect and/or block potentially-malicious files and email	10		
SC-45	System Time Synchronization	Synchronize system clocks within and between systems and system components.	Functional	Intersects With	Synchronization With Authoritative Time Source	MON-07.1	Mechanisms exist to synchronize internal system clocks with an authoritative time source.	5		SC-45
SC-45(01)	System Time Synchronization Synchronization with Authoritative Time Source	a. Compare the internal system clocks [Assignment: organization-defined frequency] with [Assignment: organization-defined authoritative time source]; andb. Synchronize the internal system clocks to the authoritative time source when the	Functional	Equal	Synchronization With Authoritative Time Source	MON-07.1	Mechanisms exist to synchronize internal system clocks with an authoritative time source.	10		
SC-45(02)	System Time Synchronization Secondary Authoritative Time Source	a. Identify a secondary authoritative time source that is in a different geographic region than the primary authoritative time source; andb. Synchronize the internal system clocks to the secondary authoritative time source if the primary authoritative	Functional	No Relationship	N/A	N/A	N/A	0		
SC-46	Cross Domain Policy Enforcement	Implement a policy enforcement mechanism [Selection (one): physically; logically] between the physical and/or network interfaces for the connecting security domains.	Functional	Equal	Cross Domain Solution (CDS)	NET-02.3	Mechanisms exist to implement a Cross Domain Solution (CDS) to mitigate the specific security risks of accessing or transferring information	10		
SC-47	Alternate Communications Channels	Establish [Assignment: organization-defined alternate communications paths] for system operations organizational command and control.	Functional	Equal	Alternate Communications Channels	BCD-10.4	Mechanisms exist to maintain command and control capabilities via alternate communications channels and designating alternative decision	10		
SC-48	Sensor Relocation	Relocate [Assignment: organization-defined sensors and monitoring capabilities] to [Assignment: organization-defined locations] under the following conditions or circumstances: [Assignment: organization-defined conditions or	Functional	Intersects With	Threat Hunting	THR-07	Mechanisms exist to perform cyber threat hunting that uses indicators of Compromise (IoC) to detect, track and disrupt threats that evade existing	5		
SC-48	Sensor Relocation	Relocate [Assignment: organization-defined sensors and monitoring capabilities] to [Assignment: organization-defined locations] under the following conditions or circumstances: [Assignment: organization-defined conditions or	Functional	Intersects With	Automated Tools for Real-Time Analysis	MON-01.2	Mechanisms exist to utilize a Security Incident Event Manager (SIEM), or similar automated tool, to support near real-time analysis and incident	5		
SC-48(01)	Sensor Relocation Dynamic Relocation of Sensors or Monitoring Capabilities	Dynamically relocate [Assignment: organization-defined sensors and monitoring capabilities] to [Assignment: organization-defined locations] under the following conditions or circumstances: [Assignment: organization-defined conditions	Functional	No Relationship	N/A	N/A	N/A	0		
SC-49	Hardware-enforced Separation and Policy Enforcement	Implement hardware-enforced separation and policy enforcement mechanisms between [Assignment: organization-defined security domains].	Functional	No Relationship	N/A	N/A	N/A	0		
SC-50	Software-enforced Separation and Policy Enforcement	Implement software-enforced separation and policy enforcement mechanisms between [Assignment: organization-defined security domains].	Functional	No Relationship	N/A	N/A	N/A	0		
SC-51	Hardware-based Protection	a. Employ hardware-based, write-protect for [Assignment: organization-defined system firmware components]; andb. Implement specific procedures for [Assignment: organization-defined authorized individuals] to manually disable hardware	Functional	No Relationship	N/A	N/A	N/A	0		
SI-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] system and information integrity policy that a	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCRPP), including policies,	5		SI-01
SI-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] system and information integrity policy that a	Functional	Subset Of	Secure Engineering Principles	SEA-01	Mechanisms exist to facilitate the implementation of industry-recognized security, compliance and	10		SI-01
SI-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]:1. [Selection (one or more): Organization-level; Mission/business process-level; System-level] system and information integrity policy that a	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5		SI-01
SI-02	Flaw Remediation	a. Identify, report, and correct system flaws;b. Test software and firmware updates related to flaw remediation for effectiveness and potential side effects before installation;c. Install security-relevant software and firmware updates within	Functional	Intersects With	Vulnerability & Patch Management Program (VPM)	VPM-01	Mechanisms exist to facilitate the implementation and monitoring of vulnerability management controls.	5		SI-02
SI-02	Flaw Remediation	a. Identify, report, and correct system flaws;b. Test software and firmware updates related to flaw remediation for effectiveness and potential side effects before installation;c. Install security-relevant software and firmware updates within	Functional	Intersects With	Software & Firmware Patching	VPM-05	Mechanisms exist to conduct software patching for all deployed Technology Assets, Applications and/or Services	5		SI-02
SI-02	Flaw Remediation	a. Identify, report, and correct system flaws;b. Test software and firmware updates related to flaw remediation for effectiveness and potential side effects before installation;c. Install security-relevant software and firmware updates within	Functional	Intersects With	Automatic Antimalware Signature Updates	END-04.1	Automated mechanisms exist to update antimalware technologies, including signature definitions.	5		SI-02
SI-02(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SI-02(02)	Flaw Remediation Automated Flaw Remediation Status	Determine if system components have applicable security-relevant software and firmware updates installed using [Assignment: organization-defined automated mechanisms] [Assignment: organization-defined frequency].	Functional	Intersects With	Automated Remediation Status	VPM-05.2	Automated mechanisms exist to determine the state of system components with regard to flaw remediation	5		
SI-02(03)	Flaw Remediation Time To Remediate Flaws and Benchmarks for	a. Measure the time between flaw identification and flaw remediation; andb. Establish the following benchmarks for taking corrective actions: [Assignment: organization-defined benchmarks]	Functional	Equal	Time To Remediate / Benchmarks For Corrective Action	VPM-05.3	Mechanisms exist to track the effectiveness of remediation operations through metrics reporting.	10		
SI-02(04)	Flaw Remediation Automated Patch Management Tools	Employ automated patch management tools to facilitate flaw remediation to the following system components: [Assignment: organization-defined system components].	Functional	Intersects With	Automated Remediation Status	VPM-05.2	Automated mechanisms exist to determine the state of system components with regard to flaw remediation	5		
SI-02(04)	Flaw Remediation Automated Patch Management Tools	Employ automated patch management tools to facilitate flaw remediation to the following system components: [Assignment: organization-defined system components].	Functional	Intersects With	Automated Software & Firmware Updates	VPM-05.4	Automated mechanisms exist to install the latest stable versions of security-relevant software and firmware updates.	5		
SI-02(04)	Flaw Remediation Automated Patch Management Tools	Employ automated patch management tools to facilitate flaw remediation to the following system components: [Assignment: organization-defined system components].	Functional	Intersects With	Centralized Management of Flaw Remediation Processes	VPM-05.1	Mechanisms exist to centrally-manage the flaw remediation process.	5		
SI-02(04)	Flaw Remediation Automated Patch Management Tools	Employ automated patch management tools to facilitate flaw remediation to the following system components: [Assignment: organization-defined system components].	Functional	Intersects With	Software & Firmware Patching	VPM-05	Mechanisms exist to conduct software patching for all deployed Technology Assets, Applications and/or Services	5		

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SI-02(05)	Flaw Remediation Automatic Software and Firmware Updates	Install [Assignment: organization-defined security-relevant software and firmware updates] automatically to [Assignment: organization-defined system components].	Functional	Intersects With	Automated Software & Firmware Updates	VPM-05.4	Automated mechanisms exist to install the latest stable versions of security-relevant software and firmware updates.	5		
SI-02(06)	Flaw Remediation Removal of Previous Versions of Software and Firmware	Remove previous versions of [Assignment: organization-defined software and firmware components] after updated versions have been installed.	Functional	Equal	Removal of Previous Versions	VPM-05.5	Mechanisms exist to remove old versions of software and firmware components after updated versions have been installed.	10		
SI-02(07)	Flaw Remediation Root Cause Analysis	a. Conduct root cause analysis to identify the underlying causes of issues or failures; b. Develop actions to address the root cause of the issue or failure.	Functional	Equal	Software Design Root Cause Analysis	TDA-06.6	Mechanisms exist to assess software design processes that includes: (1) Conducting Root Cause Analysis (RCA) to identify the underlying causes of issues or failures.	10		
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code; b. Automatically update malicious code protection.	Functional	Intersects With	Software & Firmware Patching	VPM-05	Mechanisms exist to conduct software patching for all deployed Technology Assets, Applications and/or Services (TAAS).	5		SI-03
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code; b. Automatically update malicious code protection.	Functional	Intersects With	Vulnerability & Patch Management Program (VPMP)	VPM-01	Mechanisms exist to facilitate the implementation and monitoring of vulnerability management controls.	5		SI-03
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code; b. Automatically update malicious code protection.	Functional	Intersects With	Malicious Code Protection (Anti-Malware)	END-04	Mechanisms exist to utilize antimicrobial technologies to detect and eradicate malicious code.	5		SI-03
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code; b. Automatically update malicious code protection.	Functional	Intersects With	Heuristic / Nonsignature-Based Detection	END-04.4	Mechanisms exist to utilize heuristic / nonsignature-based antimicrobial detection capabilities.	5		SI-03
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code; b. Automatically update malicious code protection.	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during transmission.	5		SI-03
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code; b. Automatically update malicious code protection.	Functional	Intersects With	Automatic Antimalware Signature Updates	END-04.1	Automated mechanisms exist to update antimicrobial technologies, including signature definitions.	5		SI-03
SI-03	Malicious Code Protection	a. Implement [Selection (one or more): signature based; non-signature based] malicious code protection mechanisms at system entry and exit points to detect and eradicate malicious code; b. Automatically update malicious code protection.	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5		SI-03
SI-03(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SI-03(02)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SI-03(03)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SI-03(04)	Malicious Code Protection Updates Only by Privileged Users	Update malicious code protection mechanisms only when directed by a privileged user.	Functional	No Relationship	N/A	N/A	N/A	0		
SI-03(05)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SI-03(06)	Malicious Code Protection Testing and Verification	a. Test malicious code protection mechanisms [Assignment: organization-defined frequency] by introducing known benign code into the system; and b. Verify that the detection of the code and the associated incident response occur.	Functional	Equal	Malware Protection Mechanism Testing	END-04.5	Mechanisms exist to test antimicrobial technologies by introducing a known benign, non-spreading test case into the system and subsequently verifying the detection and incident response.	10		
SI-03(07)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SI-03(08)	Malicious Code Protection Detect Unauthorized Commands	a. Detect the following unauthorized operating system commands through the kernel application programming interface on [Assignment: organization-defined system hardware components]: [Assignment: organization-defined commands]; and b. [Assignment: organization-defined actions to take when unauthorized commands are detected].	Functional	No Relationship	N/A	N/A	N/A	0		
SI-03(09)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SI-03(10)	Malicious Code Protection Malicious Code Analysis	a. Employ the following tools and techniques to analyze the characteristics and behavior of malicious code: [Assignment: organization-defined tools and techniques]; and b. Incorporate the results from malicious code analysis into organizational security posture.	Functional	No Relationship	N/A	N/A	N/A	0		
SI-04	System Monitoring	a. Monitor the system to detect: 1. Attacks and indicators of potential attacks in accordance with the following monitoring objectives: [Assignment: organization-defined monitoring objectives]; and 2. Unauthorized local, network, and remote access to the system.	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5		SI-04
SI-04	System Monitoring	a. Monitor the system to detect: 1. Attacks and indicators of potential attacks in accordance with the following monitoring objectives: [Assignment: organization-defined monitoring objectives]; and 2. Unauthorized local, network, and remote access to the system.	Functional	Intersects With	Centralized Collection of Security Event Logs	MON-02	Mechanisms exist to utilize a Security Incident Event Manager (SIEM), or similar automated tool, to support the centralized collection of security event logs.	5		SI-04
SI-04	System Monitoring	a. Monitor the system to detect: 1. Attacks and indicators of potential attacks in accordance with the following monitoring objectives: [Assignment: organization-defined monitoring objectives]; and 2. Unauthorized local, network, and remote access to the system.	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during transmission.	5		SI-04
SI-04	System Monitoring	a. Monitor the system to detect: 1. Attacks and indicators of potential attacks in accordance with the following monitoring objectives: [Assignment: organization-defined monitoring objectives]; and 2. Unauthorized local, network, and remote access to the system.	Functional	Intersects With	Continuous Monitoring	MON-01	Mechanisms exist to facilitate the implementation of enterprise-wide monitoring controls.	5		SI-04
SI-04(01)	System Monitoring System-wide Intrusion Detection System	Connect and configure individual intrusion detection tools into a system-wide intrusion detection system.	Functional	Equal	Intrusion Detection & Prevention Systems (IDS & IPS)	MON-01.1	Mechanisms exist to implement Intrusion Detection / Prevention Systems (IDS / IPS) technologies on critical systems, key network components, and other systems.	10		
SI-04(02)	System Monitoring Automated Tools and Mechanisms for Real-time Analysis	Employ automated tools and mechanisms to support near real-time analysis of events.	Functional	Equal	Automated Tools for Real-Time Analysis	MON-01.2	Mechanisms exist to utilize a Security Incident Event Manager (SIEM), or similar automated tool, to support near real-time analysis and incident response.	10		
SI-04(03)	System Monitoring Automated Tool and Mechanism Integration	Employ automated tools and mechanisms to integrate intrusion detection tools and mechanisms into access control and flow control mechanisms.	Functional	No Relationship	N/A	N/A	N/A	0		
SI-04(04)	System Monitoring Inbound and Outbound Communications	a. Determine criteria for unusual or unauthorized activities or conditions for inbound and outbound communications traffic; b. Monitor inbound and outbound communications traffic [Assignment: organization-defined frequency] for [Assignment: organization-defined indicators of compromise or unauthorized activities]; and c. Alert [Assignment: organization-defined personnel or roles] when the following system-generated indications of compromise or potential compromise occur: [Assignment: organization-defined compromise indicators].	Functional	Equal	Inbound & Outbound Communications Traffic	MON-01.3	Mechanisms exist to continuously monitor inbound and outbound communications traffic for unusual or unauthorized activities or conditions.	10		
SI-04(05)	System Monitoring System-generated Alerts	Alert [Assignment: organization-defined personnel or roles] when the following system-generated indications of compromise or potential compromise occur: [Assignment: organization-defined compromise indicators].	Functional	Equal	System Generated Alerts	MON-01.4	Mechanisms exist to generate, monitor, correlate and respond to alerts from physical, cybersecurity, and other systems.	10		
SI-04(06)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SI-04(07)	System Monitoring Automated Response to Suspicious Events	a. Notify [Assignment: organization-defined incident response personnel (identified by name and/or by role)] of detected suspicious events; and b. Take the following actions upon detection: [Assignment: organization-defined least-disruptive actions to take upon detection of suspicious events].	Functional	Intersects With	Automated Response to Suspicious Events	MON-01.11	Automated mechanisms exist to implement pre-determined corrective actions in response to detected suspicious events.	5		
SI-04(07)	System Monitoring Automated Response to Suspicious Events	a. Notify [Assignment: organization-defined incident response personnel (identified by name and/or by role)] of detected suspicious events; and b. Take the following actions upon detection: [Assignment: organization-defined least-disruptive actions to take upon detection of suspicious events].	Functional	Intersects With	Automated Incident Handling Processes	IRO-02.1	Automated mechanisms exist to support the incident handling process.	5		
SI-04(08)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn

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SI-04(09)	System Monitoring Testing of Monitoring Tools and Mechanisms	Test intrusion-monitoring tools and mechanisms [Assignment: organization-defined frequency].	Functional	Intersects With	Incident Response Testing	IRO-06	Mechanisms exist to formally test incident response capabilities through realistic exercises to determine the operational effectiveness of those mechanisms.	5		
SI-04(10)	System Monitoring Visibility of Encrypted Communications	Make provisions so that [Assignment: organization-defined encrypted communications traffic] is visible to [Assignment: organization-defined system monitoring tools and mechanisms].	Functional	Equal	Visibility of Encrypted Communications	NET-18.2	Mechanisms exist to configure the proxy to make encrypted communications traffic visible to monitoring tools and mechanisms.	10		
SI-04(11)	System Monitoring Analyze Communications Traffic Anomalies	Analyze outbound communications traffic at the external interfaces to the system and selected [Assignment: organization-defined interior points within the system] to discover anomalies.	Functional	Equal	Anomalous Behavior	MON-16	Mechanisms exist to utilize User & Entity Behavior Analytics (UEBA) and/or User Activity Monitoring (UAM) to detect and identify anomalies.	10		
SI-04(12)	System Monitoring Automated Organization-generated Alerts	Alert [Assignment: organization-defined personnel or roles] using [Assignment: organization-defined automated mechanisms] when the following indications of inappropriate or unusual activities with security or privacy implications occur:	Functional	Intersects With	Automated Alerts	MON-01.12	Mechanisms exist to automatically alert incident response personnel to inappropriate or anomalous activities that have potential security incident.	5		
SI-04(12)	System Monitoring Automated Organization-generated Alerts	Alert [Assignment: organization-defined personnel or roles] using [Assignment: organization-defined automated mechanisms] when the following indications of inappropriate or unusual activities with security or privacy implications occur:	Functional	Intersects With	Real-Time Alerts of Event Logging Failure	MON-05.1	Mechanisms exist to provide 24x7x365 near real-time alerting capability when an event log processing failure occurs.	5		
SI-04(13)	System Monitoring Analyze Traffic and Event Patterns	a. Analyze communications traffic and event patterns for the system;b. Develop profiles representing common traffic and event patterns; andc. Use the traffic and event profiles in tuning system monitoring devices.	Functional	Equal	Alert Threshold Tuning	MON-01.13	Mechanisms exist to "tune" event monitoring technologies through analyzing communications traffic/event patterns and developing network segments.	10		
SI-04(14)	System Monitoring Wireless Intrusion Detection	Employ a wireless intrusion detection system to identify rogue wireless devices and to detect attack attempts and potential compromises or breaches to the system.	Functional	Intersects With	Wireless Network Monitoring	MON-01.5	Mechanisms exist to monitor wireless network segments for: (1) Rogue wireless devices; and (2) Anomalous and/or hostile communications.	5		
SI-04(15)	System Monitoring Wireless to Wireline Communications	Employ an intrusion detection system to monitor wireless communications traffic as the traffic passes from wireless to wireline networks.	Functional	Intersects With	Wireless Intrusion Detection / Prevention Systems (WIDS / WIPS) Deployment	NET-08.2	Mechanisms exist to utilize Wireless Intrusion Detection / Protection Systems (WIDS / WIPS) on wireless network segments.	5		
SI-04(16)	System Monitoring Correlate Monitoring Information	Correlate information from monitoring tools and mechanisms employed throughout the system.	Functional	Equal	Correlate Monitoring Information	MON-02.1	Automated mechanisms exist to correlate both technical and non-technical information from across the enterprise by a Security Incident Event.	10		
SI-04(17)	System Monitoring Integrated Situational Awareness	Correlate information from monitoring physical, cyber, and supply chain activities to achieve integrated, organization-wide situational awareness.	Functional	Equal	Integration of Scanning & Other Monitoring Information	MON-02.3	Automated mechanisms exist to integrate the analysis of audit records with analysis of vulnerability scanners, network performance system.	10		
SI-04(18)	System Monitoring Analyze Traffic and Covert Exfiltration	Analyze outbound communications traffic at external interfaces to the system and at the following interior points to detect covert exfiltration of information: [Assignment: organization-defined interior points within the system].	Functional	Intersects With	Data Loss Prevention (DLP)	NET-17	Automated mechanisms exist to implement Data Loss Prevention (DLP) to protect sensitive information as it is stored, transmitted and processed.	5		
SI-04(18)	System Monitoring Analyze Traffic and Covert Exfiltration	Analyze outbound communications traffic at external interfaces to the system and at the following interior points to detect covert exfiltration of information: [Assignment: organization-defined interior points within the system].	Functional	Intersects With	Analyze Traffic for Covert Exfiltration	MON-11.1	Automated mechanisms exist to analyze network traffic to detect covert data exfiltration.	5		
SI-04(19)	System Monitoring Risk for Individuals	Implement [Assignment: organization-defined additional monitoring] of individuals who have been identified by [Assignment: organization-defined sources] as posing an increased level of risk.	Functional	Equal	Individuals Posing Greater Risk	MON-01.14	Mechanisms exist to implement enhanced activity monitoring for individuals who have been identified as posing an increased level of risk.	10		
SI-04(20)	System Monitoring Privileged Users	Implement the following additional monitoring of privileged users: [Assignment: organization-defined additional monitoring].	Functional	Equal	Privileged User Oversight	MON-01.15	Mechanisms exist to implement enhanced activity monitoring for privileged users.	10		
SI-04(21)	System Monitoring Probationary Periods	Implement the following additional monitoring of individuals during [Assignment: organization-defined probationary period]: [Assignment: organization-defined additional monitoring].	Functional	Equal	Probationary Periods	HRS-02.2	Mechanisms exist to identify newly onboarded personnel for enhanced monitoring during their probationary period.	10		
SI-04(22)	System Monitoring Unauthorized Network Services	a. Detect network services that have not been authorized or approved by [Assignment: organization-defined authorization or approval processes]; andb. [Selection (one or more): Audit; Alert [Assignment: organization-defined personnel or roles] when	Functional	Equal	Unauthorized Network Services	MON-11.2	Automated mechanisms exist to detect unauthorized network services and alert incident response personnel.	10		
SI-04(23)	System Monitoring Host-based Devices	Implement the following host-based monitoring mechanisms at [Assignment: organization-defined system components]: [Assignment: organization-defined host-based monitoring mechanisms].	Functional	Equal	Host-Based Devices	MON-01.6	Mechanisms exist to utilize Host-based Intrusion Detection / Prevention Systems (HIDS / HIPs) to actively alert on or block unwanted activities and	10		
SI-04(24)	System Monitoring Indicators of Compromise	Discover, collect, and distribute to [Assignment: organization-defined personnel or roles], indicators of compromise provided by [Assignment: organization-defined sources].	Functional	Intersects With	Monitoring for Indicators of Compromise (IOC)	MON-11.3	Automated mechanisms exist to identify and alert on Indicators of Compromise (IoC).	5		
SI-04(24)	System Monitoring Indicators of Compromise	Discover, collect, and distribute to [Assignment: organization-defined personnel or roles], indicators of compromise provided by [Assignment: organization-defined sources].	Functional	Intersects With	File Integrity Monitoring (FIM)	MON-01.7	Mechanisms exist to utilize a File Integrity Monitor (FIM), or similar change-detection technology, on critical systems, key network	5		
SI-04(25)	System Monitoring Optimize Network Traffic Analysis	Provide visibility into network traffic at external and key internal system interfaces to optimize the effectiveness of monitoring devices.	Functional	Intersects With	Limit Network Connections	NET-03.1	Mechanisms exist to limit the number of concurrent external network connections to its Technology Assets, and to ensure that connections are made to authorized sources.	5		
SI-04(25)	System Monitoring Optimize Network Traffic Analysis	Provide visibility into network traffic at external and key internal system interfaces to optimize the effectiveness of monitoring devices.	Functional	Intersects With	Intrusion Detection & Prevention Systems (IDS & IPS)	MON-01.1	Mechanisms exist to implement Intrusion Detection / Prevention Systems (IDS / IPS) technologies on critical systems, key network	5		
SI-05	Security Alerts, Advisories, and Directives	a. Receive system security alerts, advisories, and directives from [Assignment: organization-defined external organizations] on an ongoing basis;b. Generate internal security alerts, advisories, and directives as deemed necessary;c. Disseminate	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5		SI-05
SI-05	Security Alerts, Advisories, and Directives	a. Receive system security alerts, advisories, and directives from [Assignment: organization-defined external organizations] on an ongoing basis;b. Generate internal security alerts, advisories, and directives as deemed necessary;c. Disseminate	Functional	Intersects With	Threat Intelligence Feeds	THR-03	Mechanisms exist to maintain situational awareness of vulnerabilities and evolving threats by leveraging the knowledge of attacker	5		SI-05
SI-05	Security Alerts, Advisories, and Directives	a. Receive system security alerts, advisories, and directives from [Assignment: organization-defined external organizations] on an ongoing basis;b. Generate internal security alerts, advisories, and directives as deemed necessary;c. Disseminate	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5		SI-05
SI-05(01)	Security Alerts, Advisories, and Directives Automated Alerts and Advisories	Broadcast security alert and advisory information throughout the organization using [Assignment: organization-defined automated mechanisms].	Functional	Intersects With	Threat Intelligence Feeds	THR-03	Mechanisms exist to maintain situational awareness of vulnerabilities and evolving threats by leveraging the knowledge of attacker	5		
SI-06	Security and Privacy Function Verification	a. Verify the correct operation of [Assignment: organization-defined security and privacy functions];b. Perform the verification of the functions specified in SI-06a [Selection (one or more)]: [Assignment: organization-defined system transitional	Functional	Intersects With	Control Functionality Verification	CHG-06	Mechanisms exist to verify the functionality of security, compliance and resilience controls following implemented changes to ensure	5		
SI-06(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SI-06(02)	Security and Privacy Function Verification Automation Support for Distributed Testing	Implement automated mechanisms to support the management of distributed security and privacy function testing.	Functional	No Relationship	N/A	N/A	N/A	0		
SI-06(03)	Security and Privacy Function Verification Report Verification Results	Report the results of security and privacy function verification to [Assignment: organization-defined personnel or roles].	Functional	Equal	Report Verification Results	CHG-06.1	Mechanisms exist to report the results of security, compliance and resilience capability verification to appropriate organizational management.	10		
SI-07	Software, Firmware, and Information Integrity	a. Employ integrity verification tools to detect unauthorized changes to the following software, firmware, and information: [Assignment: organization-defined software, firmware, and information]; andb. Take the following actions when	Functional	Intersects With	Endpoint File Integrity Monitoring (FIM)	END-06	Mechanisms exist to utilize File Integrity Monitor (FIM), or similar technologies, to detect and report on unauthorized changes to selected	5		
SI-07	Software, Firmware, and Information Integrity	a. Employ integrity verification tools to detect unauthorized changes to the following software, firmware, and information: [Assignment: organization-defined software, firmware, and information]; andb. Take the following actions when	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5		
SI-07	Software, Firmware, and Information Integrity	a. Employ integrity verification tools to detect unauthorized changes to the following software, firmware, and information: [Assignment: organization-defined software, firmware, and information]; andb. Take the following actions when	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5		
SI-07(01)	Software, Firmware, and Information Integrity Integrity Checks	Perform an integrity check of [Assignment: organization-defined software, firmware, and information] [Selection (one or more): at startup; at [Assignment: organization-defined transitional states or security-relevant events]; [Assignment: organization-defined	Functional	Equal	Integrity Checks	END-06.1	Mechanisms exist to validate configurations through integrity checking of software and firmware.	10		
SI-07(02)	Software, Firmware, and Information Integrity Automated Notifications of	Employ automated tools that provide notification to [Assignment: organization-defined personnel or roles] upon discovering discrepancies during integrity verification.	Functional	Equal	Automated Notifications of Integrity Violations	END-06.3	Automated mechanisms exist to alert incident response personnel upon discovering discrepancies during integrity verification.	10		
SI-07(03)	Software, Firmware, and Information Integrity Centrally Managed Integrity	Employ centrally managed integrity verification tools.	Functional	No Relationship	N/A	N/A	N/A	0		
SI-07(04)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SI-07(05)	Software, Firmware, and Information Integrity Automated Response to Integrity	Automatically [Selection (one or more): shut the system down; restart the system; implement [Assignment: organization-defined controls]] when integrity violations are discovered.	Functional	Equal	Automated Response to Integrity Violations	END-06.4	Automated mechanisms exist to implement remediation actions when integrity violations are discovered.	10		
SI-07(06)	Software, Firmware, and Information Integrity Cryptographic	Implement cryptographic mechanisms to detect unauthorized changes to software, firmware, and information.	Functional	Equal	Use of Cryptographic Controls	CRY-01	Mechanisms exist to facilitate the implementation of cryptographic protections controls using known public standards and trusted	10		
SI-07(07)	Software, Firmware, and Information Integrity Integration of Detection and	Incorporate the detection of the following unauthorized changes into the organizational incident response capability: [Assignment: organization-defined security-relevant changes to the system].	Functional	Equal	Endpoint Detection & Response (EDR)	END-06.2	Mechanisms exist to detect and respond to unauthorized configuration changes as cybersecurity incidents.	10		
SI-07(08)	Software, Firmware, and Information Integrity Auditing Capability for	Upon detection of a potential integrity violation, provide the capability to audit the event and initiate the following actions: [Selection (one or more): generate an audit record; alert current user; alert [Assignment: organization-defined personnel or	Functional	No Relationship	N/A	N/A	N/A	0		
SI-07(09)	Software, Firmware, and Information Integrity Verify Boot Process	Verify the integrity of the boot process of the following system components: [Assignment: organization-defined system components].	Functional	Equal	Boot Process Integrity	END-06.5	Automated mechanisms exist to verify the integrity of the boot process of systems.	10		
SI-07(10)	Software, Firmware, and Information Integrity Protection of Boot Firmware	Implement the following mechanisms to protect the integrity of boot firmware in [Assignment: organization-defined system components]: [Assignment: organization-defined mechanisms].	Functional	Equal	Protection of Boot Firmware	END-06.6	Automated mechanisms exist to protect the integrity of boot firmware in systems.	10		
SI-07(11)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SI-07(12)	Software, Firmware, and Information Integrity Integrity Verification	Require that the integrity of the following user-installed software be verified prior to execution: [Assignment: organization-defined user-installed software].	Functional	No Relationship	N/A	N/A	N/A	0		
SI-07(13)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SI-07(14)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SI-07(15)	Software, Firmware, and Information Integrity Code Authentication	Implement cryptographic mechanisms to authenticate the following software or firmware components prior to installation: [Assignment: organization-defined software or firmware components].	Functional	Intersects With	Signed Components	CHG-04.2	Mechanisms exist to prevent the installation of software and firmware components without verification that the component has been digitally	5		
SI-07(16)	Software, Firmware, and Information Integrity Time Limit on Process Execution	Prohibit processes from executing without supervision for more than [Assignment: organization-defined time period].	Functional	No Relationship	N/A	N/A	N/A	0		
SI-07(17)	Software, Firmware, and Information Integrity Runtime Application Self-	Implement [Assignment: organization-defined controls] for application self-protection at runtime.	Functional	No Relationship	N/A	N/A	N/A	0		
SI-08	Spam Protection	a. Employ spam protection mechanisms at system entry and exit points to detect and act on unsolicited messages; andb. Update spam protection mechanisms when new releases are available in accordance with organizational configuration	Functional	Equal	Phishing & Spam Protection	END-08	Mechanisms exist to utilize anti-phishing and spam protection technologies to detect and take action on unsolicited messages transported	10		
SI-08(01)		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SI-08(02)	Spam Protection Automatic Updates	Automatically update spam protection mechanisms [Assignment: organization-defined frequency].	Functional	Equal	Automatic Spam and Phishing Protection Updates	END-08.2	Mechanisms exist to automatically update anti-phishing and spam protection technologies when new releases are available in accordance	10		
SI-08(03)	Spam Protection Continuous Learning Capability	Implement spam protection mechanisms with a learning capability to more effectively identify legitimate communications traffic.	Functional	No Relationship	N/A	N/A	N/A	0		
SI-09		Withdrawn	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SI-10	Information Input Validation	Check the validity of the following information inputs: [Assignment: organization-defined information inputs to the system].	Functional	Intersects With	Safeguarding Data Over Open Networks	NET-12	Cryptographic mechanisms exist to implement strong cryptography and security protocols to safeguard sensitive and/or regulated data during	5		
SI-10	Information Input Validation	Check the validity of the following information inputs: [Assignment: organization-defined information inputs to the system].	Functional	Intersects With	Input Data Validation	TDA-18	Mechanisms exist to check the validity of information inputs.	5		
SI-10(01)	Information Input Validation Manual Override Capability	a. Provide a manual override capability for input validation of the following information inputs: [Assignment: organization-defined inputs defined in the base control (SI-10)];b. Restrict the use of the manual override capability to only [Assignment: organization	Functional	No Relationship	N/A	N/A	N/A	0		
SI-10(02)	Information Input Validation Review and Resolve Errors	Review and resolve input validation errors within [Assignment: organization-defined time period].	Functional	No Relationship	N/A	N/A	N/A	0		

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SI-10(03)	Information Input Validation Predictable Behavior	Verify that the system behaves in a predictable and documented manner when invalid inputs are received.	Functional	No Relationship	N/A	N/A	N/A	0		
SI-10(04)	Information Input Validation Timing Interactions	Account for timing interactions among system components in determining appropriate responses for invalid inputs.	Functional	No Relationship	N/A	N/A	N/A	0		
SI-10(05)	Information Input Validation Restrict Inputs to Trusted Sources and Approved	Restrict the use of information inputs to [Assignment: organization-defined trusted sources] and/or [Assignment: organization-defined formats].	Functional	No Relationship	N/A	N/A	N/A	0		
SI-10(06)	Information Input Validation Injection Prevention	Prevent untrusted data injections.	Functional	No Relationship	N/A	N/A	N/A	0		
SI-11	Error Handling	a. Generate error messages that provide information necessary for corrective actions without revealing information that could be exploited; andb. Reveal error messages only to [Assignment: organization-defined personnel or roles].	Functional	Equal	Error Handling	TDA-19	Mechanisms exist to handle error conditions by: (1) Identifying potentially security-relevant error conditions.	10		
SI-12	Information Management and Retention	Manage and retain information within the system and information output from the system in accordance with applicable laws, executive orders, directives, regulations, policies, standards, guidelines and operational requirements.	Functional	Intersects With	Media & Data Retention	DCH-18	Mechanisms exist to retain media and data in accordance with applicable statutory, regulatory and contractual obligations.	5		
SI-12	Information Management and Retention	Manage and retain information within the system and information output from the system in accordance with applicable laws, executive orders, directives, regulations, policies, standards, guidelines and operational requirements.	Functional	Intersects With	Personal Data (PD) Retention & Disposal	PRI-05	Mechanisms exist to: (1) Retain Personal Data (PD), including metadata, for an	5		SI-12
SI-12(01)	Information Management and Retention Limit Personally Identifiable Information	Limit personally identifiable information being processed in the information life cycle to the following elements of personally identifiable information: [Assignment: organization-defined elements of personally identifiable information].	Functional	Intersects With	Internal Use of Personal Data (PD) For Testing, Training and Research	PRI-05.1	Mechanisms exist to address the use of Personal Data (PD) for internal testing, training and research that:	5		
SI-12(01)	Information Management and Retention Limit Personally Identifiable Information	Limit personally identifiable information being processed in the information life cycle to the following elements of personally identifiable information: [Assignment: organization-defined elements of personally identifiable information].	Functional	Intersects With	Minimize Sensitive / Regulated Data	DCH-18.1	Mechanisms exist to minimize sensitive and/or regulated data that is collected, received, processed, stored	5		
SI-12(02)	Information Management and Retention Minimize Personally Identifiable Information	Use the following techniques to minimize the use of personally identifiable information for research, testing, or training: [Assignment: organization-defined techniques].	Functional	Intersects With	Limit Sensitive / Regulated Data In Testing, Training & Research	DCH-18.2	Mechanisms exist to minimize the use of sensitive and/or regulated data for research, testing, or training, in accordance with authorized	5		
SI-12(02)	Information Management and Retention Minimize Personally Identifiable Information	Use the following techniques to minimize the use of personally identifiable information for research, testing, or training: [Assignment: organization-defined techniques].	Functional	Intersects With	Internal Use of Personal Data (PD) For Testing, Training and Research	PRI-05.1	Mechanisms exist to address the use of Personal Data (PD) for internal testing, training and research that:	5		
SI-12(03)	Information Management and Retention Information Disposal	Use the following techniques to dispose of, destroy, or erase information following the retention period: [Assignment: organization-defined techniques].	Functional	Intersects With	Personal Data (PD) Retention & Disposal	PRI-05	Mechanisms exist to: (1) Retain Personal Data (PD), including metadata, for an	5		
SI-12(03)	Information Management and Retention Information Disposal	Use the following techniques to dispose of, destroy, or erase information following the retention period: [Assignment: organization-defined techniques].	Functional	Intersects With	Information Disposal	DCH-21	Mechanisms exist to securely dispose of, destroy or erase information.	5		
SI-13	Predictable Failure Prevention	a. Determine mean time to failure (MTTF) for the following system components in specific environments of operation: [Assignment: organization-defined system components]; andb. Provide substitute system components and a means to	Functional	Intersects With	Failover Capability	BCD-12.2	Mechanisms exist to implement real-time or near-real-time failover capability to maintain availability of	5		
SI-13	Predictable Failure Prevention	a. Determine mean time to failure (MTTF) for the following system components in specific environments of operation: [Assignment: organization-defined system components]; andb. Provide substitute system components and a means to	Functional	Intersects With	Predictable Failure Analysis	SEA-07	Mechanisms exist to determine the Mean Time to Failure (MTTF) for system components in specific environments of operation	5		
SI-13(01)	Predictable Failure Prevention Transferring Component	Take system components out of service by transferring component responsibilities to substitute components no later than [Assignment: organization-defined fraction or percentage] of mean time to failure.	Functional	No Relationship	N/A	N/A	N/A	0		
SI-13(02)	Predictable Failure Prevention Manual Transfer Between Components	Manually initiate transfers between active and standby system components when the use of the active component reaches [Assignment: organization-defined percentage] of the mean time to failure.	Functional	No Relationship	N/A	N/A	N/A	0		Withdrawn
SI-13(03)	Predictable Failure Prevention Standby Component	If system component failures are detected:a. Ensure that the standby components are successfully and transparently installed within [Assignment: organization-defined time period]; andb. [Selection (one or more): Activate [Assignment:	Functional	No Relationship	N/A	N/A	N/A	0		
SI-13(05)	Predictable Failure Prevention Failover Capability	Provide [Selection (one): real-time; near real-time] [Assignment: organization-defined failover capability] for the system.	Functional	No Relationship	N/A	N/A	N/A	0		
SI-14	Non-persistence	Implement non-persistent [Assignment: organization-defined system components and services] that are initiated in a known state and terminated [Selection (one or more): upon end of session of use; periodically at [Assignment: organization-	Functional	Equal	Non-Persistence	SEA-08	Mechanisms exist to implement non-persistent system components and services that are initiated in a known state and terminated upon the end of	10		
SI-14(01)	Non-persistence Refresh from Trusted Sources	Obtain software and data employed during system component and service refreshes from the following trusted sources: [Assignment: organization-defined trusted sources].	Functional	Equal	Refresh from Trusted Sources	SEA-08.1	Mechanisms exist to ensure that software and data needed for system component and service refreshes are obtained from trusted sources	10		
SI-14(02)	Non-persistence Non-persistent Information	a. [Selection (one): Refresh [Assignment: organization-defined information] [Assignment: organization-defined frequency]; Generate [Assignment: organization-defined information] on demand; andb. Delete information when no longer needed.	Functional	No Relationship	N/A	N/A	N/A	0		
SI-14(03)	Non-persistence Non-persistent Connectivity	Establish connections to the system on demand and terminate connections after [Selection (one): completion of a request; a period of non-use].	Functional	No Relationship	N/A	N/A	N/A	0		
SI-15	Information Output Filtering	Validate information output from the following software programs and/or applications to ensure that the information is consistent with the expected content: [Assignment: organization-defined software programs and/or applications].	Functional	Equal	Information Output Filtering	SEA-09	Mechanisms exist to validate information output from software programs and/or applications to ensure that the information is	10		
SI-16	Memory Protection	Implement the following controls to protect the system memory from unauthorized code execution: [Assignment: organization-defined controls].	Functional	Equal	Memory Protection	SEA-10	Mechanisms exist to implement security safeguards to protect system memory from unauthorized code execution.	10		
SI-17	Fail-safe Procedures	Implement the indicated fail-safe procedures when the indicated failures occur: [Assignment: organization-defined list of failure conditions and associated fail-safe procedures].	Functional	Equal	Fail Safe	SEA-07.3	Mechanisms exist to implement fail-safe procedures when failure conditions occur.	10		SI-17

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FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SI-18	Personally Identifiable Information Quality Operations	a. Check the accuracy, relevance, timeliness, and completeness of personally identifiable information across the information life cycle [Assignment: organization-defined frequency]; andb. Correct or delete inaccurate or outdated	Functional	Intersects With	Data Quality Operations	DCH-22	Mechanisms exist to check for Redundant, Obsolete/Outdated, Toxic or Trivial (ROTT) data to ensure the accuracy, relevance, timeliness	5		
SI-18(01)	Personally Identifiable Information Quality Operations Automation Support	Correct or delete personally identifiable information that is inaccurate or outdated, incorrectly determined regarding impact, or incorrectly de-identified using [Assignment: organization-defined automated mechanisms]	Functional	Intersects With	Data Quality Operations	DCH-22	Mechanisms exist to check for Redundant, Obsolete/Outdated, Toxic or Trivial (ROTT) data to ensure the accuracy, relevance, timeliness	5		
SI-18(02)	Personally Identifiable Information Quality Operations Data Tags	Employ data tags to automate the correction or deletion of personally identifiable information across the information life cycle within organizational systems.	Functional	Equal	Data Tags	DCH-22.2	Mechanisms exist to utilize data tags to automate tracking of sensitive and/or regulated data across the information lifecycle	10		
SI-18(03)	Personally Identifiable Information Quality Operations Collection	Collect personally identifiable information directly from the individual.	Functional	Equal	Primary Source Personal Data (PD) Collection	DCH-22.3	Mechanisms exist to collect Personal Data (PD) directly from the individual.	10		
SI-18(04)	Personally Identifiable Information Quality Operations Individual Requests	Correct or delete personally identifiable information upon request by individuals or their designated representatives.	Functional	Intersects With	Correcting Inaccurate Personal Data (PD)	PRI-06.1	Mechanisms exist to maintain a process for: (1) Data subjects to have inaccurate Personal Data (PD) maintained by the	5		
SI-18(04)	Personally Identifiable Information Quality Operations Individual Requests	Correct or delete personally identifiable information upon request by individuals or their designated representatives.	Functional	Intersects With	Updating & Correcting Personal Data (PD)	DCH-22.1	Mechanisms exist to utilize technical controls to correct Personal Data (PD) that is inaccurate or outdated, incorrectly determined regarding	5		
SI-18(04)	Personally Identifiable Information Quality Operations Individual Requests	Correct or delete personally identifiable information upon request by individuals or their designated representatives.	Functional	Intersects With	Data Subject Empowerment	PRI-06	Mechanisms exist to provide authenticated data subjects the ability to:	5		
SI-18(05)	Personally Identifiable Information Quality Operations Notice of Correction or Deletion	Notify [Assignment: organization-defined recipients of personally identifiable information] and individuals that the personally identifiable information has been corrected or deleted	Functional	Intersects With	Updating & Correcting Personal Data (PD)	DCH-22.1	Mechanisms exist to utilize technical controls to correct Personal Data (PD) that is inaccurate or outdated, incorrectly determined regarding	5		
SI-18(05)	Personally Identifiable Information Quality Operations Notice of Correction or Deletion	Notify [Assignment: organization-defined recipients of personally identifiable information] and individuals that the personally identifiable information has been corrected or deleted	Functional	Intersects With	Correcting Inaccurate Personal Data (PD)	PRI-06.1	Mechanisms exist to maintain a process for: (1) Data subjects to have inaccurate Personal Data (PD) maintained by the	5		
SI-18(05)	Personally Identifiable Information Quality Operations Notice of Correction or Deletion	Notify [Assignment: organization-defined recipients of personally identifiable information] and individuals that the personally identifiable information has been corrected or deleted	Functional	Intersects With	Notice of Correction or Processing Change	PRI-06.2	Mechanisms exist to notify affected data subjects if their Personal Data (PD) has been corrected, amended or deleted	5		
SI-19	De-identification	a. Remove the following elements of personally identifiable information from datasets: [Assignment: organization-defined elements of personally identifiable information]; andb. Evaluate [Assignment: organization-defined frequency] for effectiveness	Functional	Equal	De-Identification (Anonymization)	DCH-23	Mechanisms exist to anonymize data by removing Personal Data (PD) from datasets.	10		
SI-19(01)	De-identification Collection	De-identify the dataset upon collection by not collecting personally identifiable information.	Functional	Intersects With	Primary Source Personal Data (PD) Collection	DCH-22.3	Mechanisms exist to collect Personal Data (PD) directly from the individual.	5		
SI-19(01)	De-identification Collection	De-identify the dataset upon collection by not collecting personally identifiable information.	Functional	Intersects With	De-Identify Dataset Upon Collection	DCH-23.1	Mechanisms exist to de-identify the dataset upon collection by not collecting Personal Data (PD).	5		
SI-19(02)	De-identification Archiving	Prohibit archiving of personally identifiable information elements if those elements in a dataset will not be needed after the dataset is archived.	Functional	Equal	Archiving	DCH-23.2	Mechanisms exist to refrain from archiving Personal Data (PD) elements if those elements in a dataset will not be needed after the dataset is	10		
SI-19(03)	De-identification Release	Remove personally identifiable information elements from a dataset prior to its release if those elements in the dataset do not need to be part of the data release.	Functional	Equal	Release	DCH-23.3	Mechanisms exist to remove Personal Data (PD) elements from a dataset prior to its release if those elements in the dataset do not need to be part of	10		
SI-19(04)	De-identification Removal, Masking, Encryption, Hashing, or Replacement of	Remove, mask, encrypt, hash, or replace direct identifiers in a dataset.	Functional	Intersects With	Data Anonymization	PRI-05.3	Mechanisms exist to mask sensitive and/or regulated data through data anonymization, pseudonymization, redaction and/or de-identification	5		
SI-19(04)	De-identification Removal, Masking, Encryption, Hashing, or Replacement of	Remove, mask, encrypt, hash, or replace direct identifiers in a dataset.	Functional	Intersects With	Removal, Masking, Encryption, Hashing or Replacement of Direct Identifiers	DCH-23.4	Mechanisms exist to remove, mask, encrypt, hash or replace direct identifiers in a dataset.	5		
SI-19(05)	De-identification Statistical Disclosure Control	Manipulate numerical data, contingency tables, and statistical findings so that no individual or organization is identifiable in the results of the analysis.	Functional	Equal	Statistical Disclosure Control	DCH-23.5	Mechanisms exist to manipulate numerical data, contingency tables and statistical findings so that no person or organization is identifiable in	10		
SI-19(06)	De-identification Differential Privacy	Prevent disclosure of personally identifiable information by adding non-deterministic noise to the results of mathematical operations before the results are reported.	Functional	Equal	Differential Data Privacy	DCH-23.6	Mechanisms exist to prevent disclosure of Personal Data (PD) by adding non-deterministic noise to the results of mathematical operations	10		
SI-19(07)	De-identification Validated Algorithms and Software	Perform de-identification using validated algorithms and software that is validated to implement the algorithms.	Functional	Equal	Automated De-Identification of Sensitive Data	DCH-23.7	Mechanisms exist to perform de-identification of sensitive and/or regulated data, using validated algorithms and software to implement	10		
SI-19(08)	De-identification Motivated Intruder	Perform a motivated intruder test on the de-identified dataset to determine if the identified data remains or if the de-identified data can be re-identified.	Functional	Equal	Motivated Intruder	DCH-23.8	Mechanisms exist to perform a motivated intruder test on the de-identified dataset to determine if the identified data remains or if the de-	10		
SI-20	Tainting	Embed data or capabilities in the following systems or system components to determine if organizational data has been exfiltrated or improperly removed from the organization: [Assignment: organization-defined systems or system	Functional	Equal	Tainting	THR-08	Mechanisms exist to embed false data or steganographic data in files to enable the organization to determine if data has been exfiltrated and provide	10		
SI-21	Information Refresh	Refresh [Assignment: organization-defined information] at [Assignment: organization-defined frequencies] or generate the information on demand and delete the information when no longer needed	Functional	No Relationship	N/A	N/A	N/A	0		
SI-22	Information Diversity	a. Identify the following alternative sources of information for [Assignment: organization-defined essential functions and services]; [Assignment: organization-defined alternative information sources]; andb. Use an alternative information	Functional	No Relationship	N/A	N/A	N/A	0		
SI-23	Information Fragmentation	Based on [Assignment: organization-defined circumstances];a. Fragment the following information: [Assignment: organization-defined information]; andb. Distribute the fragmented information across the following systems or system	Functional	No Relationship	N/A	N/A	N/A	0		
SR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level supply chain risk management policy that a	Functional	Intersects With	Periodic Review & Update of Security, Compliance & Resilience Program	GOV-03	Mechanisms exist to review the Security, Compliance & Resilience Program (SCR), including policies, standards and procedures necessary for secure, compliant and resilient	5		SR-01
SR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles];1. [Selection (one or more): Organization-level; Mission/business process-level; System-level supply chain risk management policy that a	Functional	Intersects With	Publishing Security, Compliance & Resilience Documentation	GOV-02	Mechanisms exist to establish, maintain and disseminate policies, standards and procedures necessary for secure, compliant and resilient	5		SR-01

NIST IR 8477-Based Set Theory Relationship Mapping (STRM)

Reference Document: Secure Controls Framework (SCF) version 2026.2

STRM Guidance: <https://securecontrolsframework.com/start-here/set-theory-relationship-mapping-strm/>

Focal Document: **NIST SP 800 - 82 R3 - Guide to Operational Technology (OT) Security - Low OT Overlay**

Focal Document URL: <https://csrc.nist.gov/pubs/sp/800/82/r3/final>

Published STRM URL: <https://content.securecontrolsframework.com/strm/scf-strm-general-nist-800-82-r3.pdf>

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SR-01	Policy and Procedures	a. Develop, document, and disseminate to [Assignment: organization-defined personnel or roles]: 1. [Selection (one or more): Organization-level; Mission/business process-level; System-level supply chain risk management policy that a	Functional	Subset Of	Third-Party Management	TPM-01	Mechanisms exist to facilitate the implementation of third-party management controls.	10		SR-01
SR-02	Supply Chain Risk Management Plan	a. Develop a plan for managing supply chain risks associated with the research and development, design, manufacturing, acquisition, delivery, integration, operations and maintenance, and disposal of the following systems, system components or	Functional	Intersects With	Supply Chain Risk Management (SCRM) Plan	RSK-09	Mechanisms exist to develop a plan for Supply Chain Risk Management (SCRM) associated with the	5		SR-02
SR-02	Supply Chain Risk Management Plan	a. Develop a plan for managing supply chain risks associated with the research and development, design, manufacturing, acquisition, delivery, integration, operations and maintenance, and disposal of the following systems, system components or	Functional	Intersects With	Supply Chain Risk Management (SCRM)	TPM-03	Mechanisms exist to: (1) Evaluate security risks and threats associated with Technology Assets, Applications and/or Services (TAAS)	5		SR-02
SR-02(01)	Supply Chain Risk Management Plan Establish SCRM Team	Establish a supply chain risk management team consisting of [Assignment: organization-defined personnel, roles, and responsibilities] to lead and support the following SCRM activities: [Assignment: organization-defined supply chain risk	Functional	Intersects With	Supply Chain Risk Management (SCRM)	TPM-03	Mechanisms exist to: (1) Evaluate security risks and threats associated with Technology Assets, Applications and/or Services (TAAS)	5		SR-02(01)
SR-03	Supply Chain Controls and Processes	a. Establish a process or processes to identify and address weaknesses or deficiencies in the supply chain elements and processes of [Assignment: organization-defined system or system component] in coordination with [Assignment:	Functional	Equal	Processes To Address Weaknesses or Deficiencies	TPM-03.3	Mechanisms exist to address identified weaknesses or deficiencies in the security of the supply chain	10		SR-03
SR-03(01)	Supply Chain Controls and Processes Diverse Supply Base	Employ a diverse set of sources for the following system components and services: [Assignment: organization-defined system components and services].	Functional	Intersects With	Development Methods, Techniques & Processes	TDA-02.3	Mechanisms exist to require software developers to ensure that their software development processes employ industry-recognized secure	5		
SR-03(01)	Supply Chain Controls and Processes Diverse Supply Base	Employ a diverse set of sources for the following system components and services: [Assignment: organization-defined system components and services].	Functional	Intersects With	Supplier Diversity	TDA-03.1	Mechanisms exist to obtain security, compliance and resilience technologies from different suppliers to minimize supply chain risk	5		
SR-03(01)	Supply Chain Controls and Processes Diverse Supply Base	Employ a diverse set of sources for the following system components and services: [Assignment: organization-defined system components and services].	Functional	Intersects With	Acquisition Strategies, Tools & Methods	TPM-03.1	Mechanisms exist to utilize tailored acquisition strategies, contract tools and procurement methods for the	5		
SR-03(02)	Supply Chain Controls and Processes Limitation of Harm	Employ the following controls to limit harm from potential adversaries identifying and targeting the organizational supply chain: [Assignment: organization-defined controls].	Functional	Equal	Limit Potential Harm	TPM-03.2	Mechanisms exist to utilize security safeguards to limit harm from potential adversaries who identify and target the organization's supply chain	10		
SR-03(03)	Supply Chain Controls and Processes Sub-tier Flow Down	Ensure that the controls included in the contracts of prime contractors are also included in the contracts of subcontractors.	Functional	Intersects With	Third-Party Contract Requirements	TPM-05	Mechanisms exist to require contractual requirements for applicable security, compliance and	5		
SR-03(03)	Supply Chain Controls and Processes Sub-tier Flow Down	Ensure that the controls included in the contracts of prime contractors are also included in the contracts of subcontractors.	Functional	Intersects With	Contract Flow-Down Requirements	TPM-05.2	Mechanisms exist to ensure applicable security, compliance and resilience requirements are included in contracts that flow-down to	5		
SR-04	Provenance	Document, monitor, and maintain valid provenance of the following systems, system components, and associated data: [Assignment: organization-defined systems, system components, and associated data]	Functional	Intersects With	Provenance	AST-03.2	Mechanisms exist to track the origin, development, ownership, location and changes to Technology Assets, Applications and/or Services (TAAS)	5		
SR-04(01)	Provenance Identity	Establish and maintain unique identification of the following supply chain elements, processes, and personnel associated with the identified system and critical system components: [Assignment: organization-defined supply chain elements]	Functional	Intersects With	Provenance	AST-03.2	Mechanisms exist to track the origin, development, ownership, location and changes to Technology Assets, Applications and/or Services (TAAS)	5		
SR-04(02)	Provenance Track and Trace	Establish and maintain unique identification of the following systems and critical system components for tracking through the supply chain: [Assignment: organization-defined systems and critical system components]	Functional	Intersects With	Provenance	AST-03.2	Mechanisms exist to track the origin, development, ownership, location and changes to Technology Assets, Applications and/or Services (TAAS)	5		
SR-04(03)	Provenance Validate as Genuine and Not Altered	Employ the following controls to validate that the system or system component received is genuine and has not been altered: [Assignment: organization-defined controls].	Functional	Intersects With	Product Tampering and Counterfeiting (PTC)	TDA-11	Mechanisms exist to maintain awareness of component authenticity by developing and implementing Product Tampering and Counterfeiting	5		
SR-04(04)	Provenance Supply Chain Integrity — Pedigree	Employ [Assignment: organization-defined controls] and conduct [Assignment: organization-defined analysis] to ensure the integrity of the system and system components by validating the internal composition and provenance of critical or mission-	Functional	Intersects With	Product Tampering and Counterfeiting (PTC)	TDA-11	Mechanisms exist to maintain awareness of component authenticity by developing and implementing Product Tampering and Counterfeiting	5		
SR-05	Acquisition Strategies, Tools, and Methods	Employ the following acquisition strategies, contract tools, and procurement methods to protect against, identify, and mitigate supply chain risks: [Assignment: organization-defined acquisition strategies, contract tools, and procurement	Functional	Intersects With	Acquisition Strategies, Tools & Methods	TPM-03.1	Mechanisms exist to utilize tailored acquisition strategies, contract tools and procurement methods for the	5		SR-05
SR-05(01)	Acquisition Strategies, Tools, and Methods Adequate Supply	Employ the following controls to ensure an adequate supply of [Assignment: organization-defined critical system components]: [Assignment: organization-defined controls].	Functional	Equal	Adequate Supply	TPM-03.4	Mechanisms exist to develop and implement a spare parts strategy to ensure that an adequate supply of critical components is available to	10		
SR-05(02)	Acquisition Strategies, Tools, and Methods Assessments Prior to Selection	Assess the system, system component, or system service prior to selection, acceptance, modification, or update.	Functional	No Relationship	N/A	N/A	N/A	0		
SR-06	Supplier Assessments and Reviews	Assess and review the supply chain-related risks associated with suppliers or contractors and the system, system component, or system service they provide [Assignment: organization-defined frequency]	Functional	Intersects With	Review of Third-Party Services	TPM-08	Mechanisms exist to monitor, regularly review and assess External Service Providers (ESPs) for compliance with established	5		
SR-06(01)	Supplier Assessments and Reviews Testing and Analysis	Employ [Selection (one or more): organizational analysis; independent third-party analysis; organizational testing; independent third-party testing] of the following supply chain elements, processes, and data associated with the system	Functional	Intersects With	Review of Third-Party Services	TPM-08	Mechanisms exist to monitor, regularly review and assess External Service Providers (ESPs) for compliance with established	5		
SR-07	Supply Chain Operations Security	Employ the following Operations Security (OPSEC) controls to protect supply chain-related information for the system, system component, or system service: [Assignment: organization- defined Operations Security (OPSEC) controls]	Functional	Intersects With	Supply Chain Risk Management (SCRM) Plan	RSK-09	Mechanisms exist to develop a plan for Supply Chain Risk Management (SCRM) associated with the	5		
SR-07	Supply Chain Operations Security	Employ the following Operations Security (OPSEC) controls to protect supply chain-related information for the system, system component, or system service: [Assignment: organization- defined Operations Security (OPSEC) controls]	Functional	Intersects With	Operations Security	OPS-01	Mechanisms exist to facilitate the implementation of operational security controls.	5		
SR-08	Notification Agreements	Establish agreements and procedures with entities involved in the supply chain for the system, system component, or system service for the [Selection (one or more): notification of supply chain compromises; results of assessments or audits;]	Functional	Equal	Security Compromise Notification Agreements	TPM-05.1	Mechanisms exist to compel External Service Providers (ESPs) to provide notification of actual or potential	10		SR-08
SR-09	Tamper Resistance and Detection	Implement a tamper protection program for the system, system component, or system service.	Functional	Intersects With	Logical Tampering Protection	AST-15	Mechanisms exist to assess the integrity of critical Technology Assets, Applications and/or Services (TAAS) to	5		
SR-09(01)	Tamper Resistance and Detection Multiple Stages of System Development	Employ anti-tamper technologies, tools, and techniques throughout the system development life cycle.	Functional	Intersects With	Logical Tampering Protection	AST-15	Mechanisms exist to assess the integrity of critical Technology Assets, Applications and/or Services (TAAS) to	5		
SR-10	Inspection of Systems or Components	Inspect the following systems or system components [Selection (one or more): at random; at [Assignment: organization-defined frequency], upon [Assignment: organization-defined indications of need for inspection] to detect tampering: [Assignment:]	Functional	Intersects With	Product Tampering and Counterfeiting (PTC)	TDA-11	Mechanisms exist to maintain awareness of component authenticity by developing and implementing Product Tampering and Counterfeiting	5		SR-10

FDE #	FDE Name	Focal Document Element (FDE) Description	STRM Rationale	STRM Relationship	SCF Control	SCF #	Secure Controls Framework (SCF) Control Description	Strength of Relationship	Notes	NIST SP 800-82 Low Overlay
SR-10	Inspection of Systems or Components	Inspect the following systems or system components [Selection (one or more): at random; at [Assignment: organization-defined frequency], upon [Assignment: organization-defined indications of need for inspection]] to detect tampering: [Assignment: a. Develop and implement anti-counterfeit policy and procedures that include the means to detect and prevent counterfeit components from entering the system; and b. Report counterfeit system components to [Selection (one or more):	Functional	Intersects With	Technology Asset Inspections	AST-15.1	Mechanisms exist to physically and logically inspect critical technology assets to detect evidence of tampering.	5		SR-10
SR-11	Component Authenticity		Functional	Intersects With	Product Tampering and Counterfeiting (PTC)	TDA-11	Mechanisms exist to maintain awareness of component authenticity by developing and implementing Product Tampering and Counterfeiting	5		SR-11
SR-11(01)	Component Authenticity Anti-counterfeit Training	Train [Assignment: organization-defined personnel or roles] to detect counterfeit system components (including hardware, software, and firmware).	Functional	Equal	Anti-Counterfeit Training	TDA-11.1	Mechanisms exist to train personnel to detect counterfeit system components, including hardware, software and firmware.	10		SR-11(01)
SR-11(02)	Component Authenticity Configuration Control for Component	Maintain configuration control over the following system components awaiting service or repair and serviced or repaired components awaiting return to service: [Assignment: organization-defined system components].	Functional	Equal	Maintain Configuration Control During Maintenance	MNT-07	Mechanisms exist to maintain proper physical security and configuration control over technology assets awaiting service or repair.	10		SR-11(02)
SR-11(03)	Component Authenticity Anti-counterfeit Scanning	Scan for counterfeit system components [Assignment: organization-defined frequency].	Functional	Intersects With	Product Tampering and Counterfeiting (PTC)	TDA-11	Mechanisms exist to maintain awareness of component authenticity by developing and implementing Product Tampering and Counterfeiting	5		
SR-12	Component Disposal	Dispose of [Assignment: organization-defined data, documentation, tools, or system components] using the following techniques and methods: [Assignment: organization-defined techniques and methods].	Functional	Intersects With	Secure Disposal, Destruction or Re-Use of Equipment	AST-09	Mechanisms exist to securely dispose of, destroy or repurpose system components using organization-defined techniques and methods to	5		SR-12